

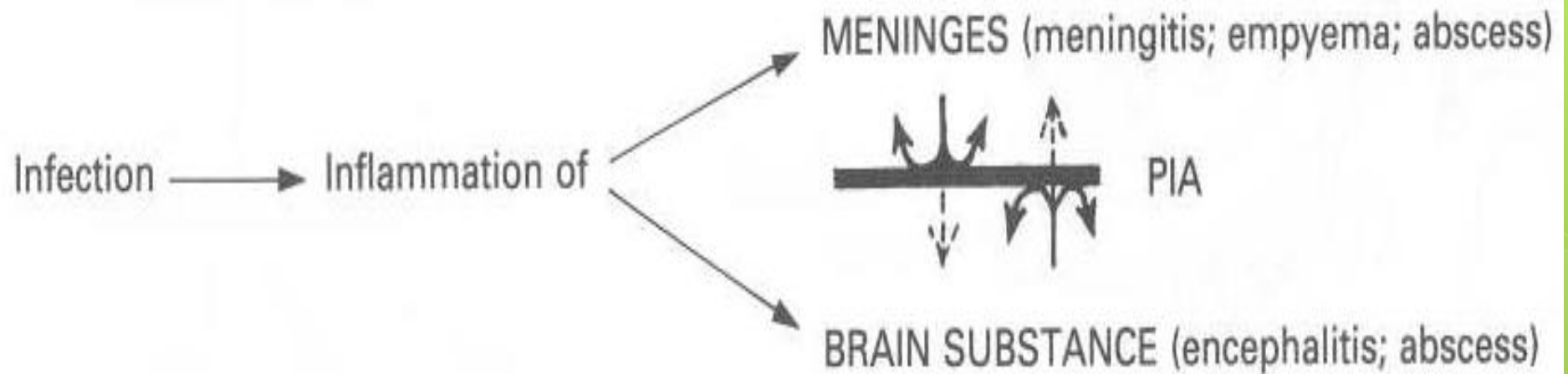


CNS Infections

Infections

- Principal routes in which microbes enter the nervous system
 - Hematogenous spread – the most common
 - Direct implantation – usually traumatic
 - Local extension from infection in air sinuses , infected tooth
 - Peripheral nervous system eg rabies, herpes zoster

- Injury to brain occurs
 - Direct effect of the agent
 - Toxic effect of elaborated toxin, Inflammatory response, or immune mediated mechanisms



Infections will be considered in three broad aetiological groups:

(1) Bacterial, (2) Viral and (3) Miscellaneous types.

Meningitis

- An inflammatory process of the leptomeninges & CSF in subarachnoid space
- Meningoencephalitis refers to inflammation of meninges & brain parenchyma
- Meningitis is usually caused by infections but other causes such as chemical irritant, cancer infiltration can cause meningitis

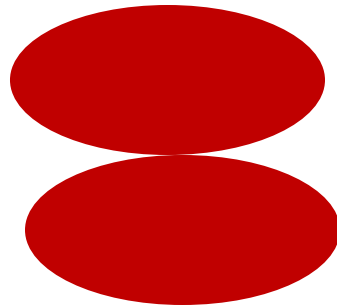
- Infectious meningitis can be classified based on the basis of inflammatory exudate on CSF examination & the clinical evolution of the illness
 - ❖ Acute pyogenic meningitis (usually bacterial meningitis)
 - ❖ Aseptic meningitis (usually acute viral meningitis)
 - ❖ Chronic meningitis (TB, neurosyphilis)

Acute pyogenic bacterial meningitis

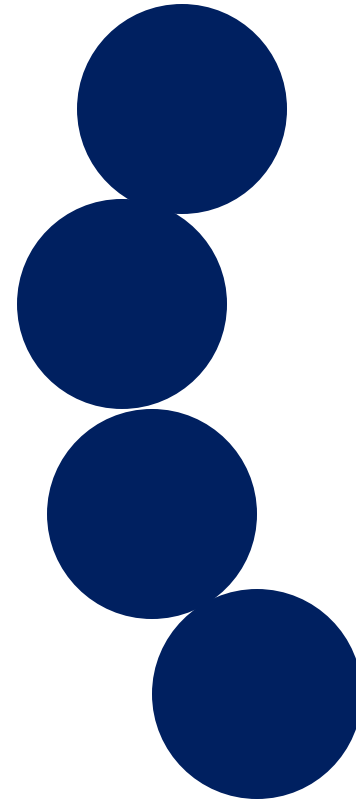
Causes

- *Streptococci pneumoniae*
- *Neisseria meningitidis*
- *Haemophilis influenzae*

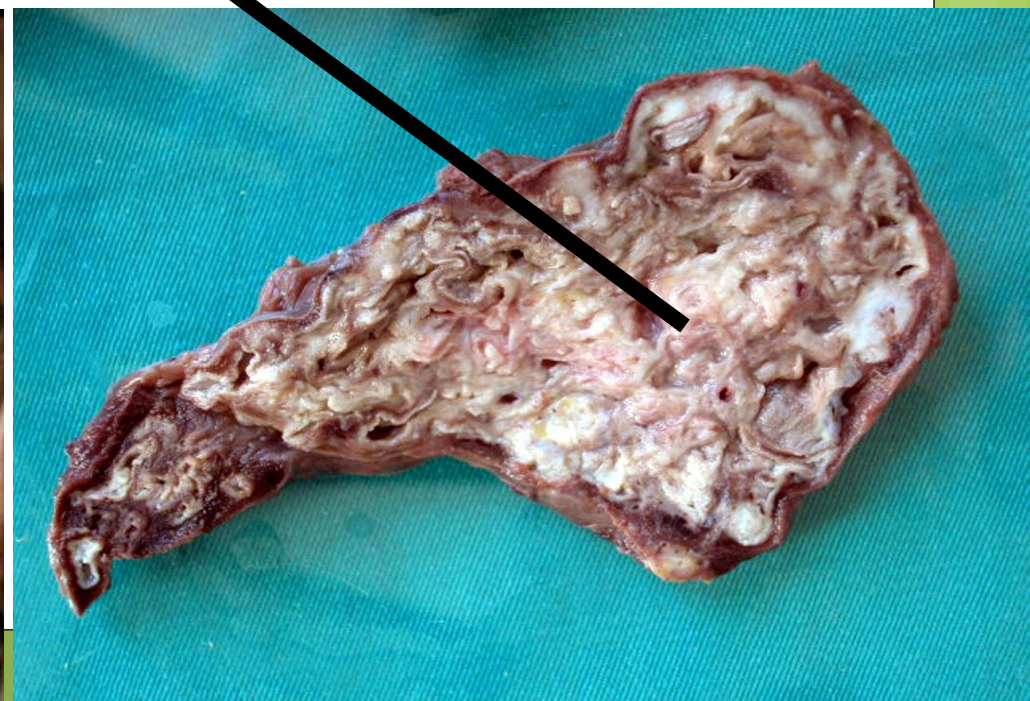
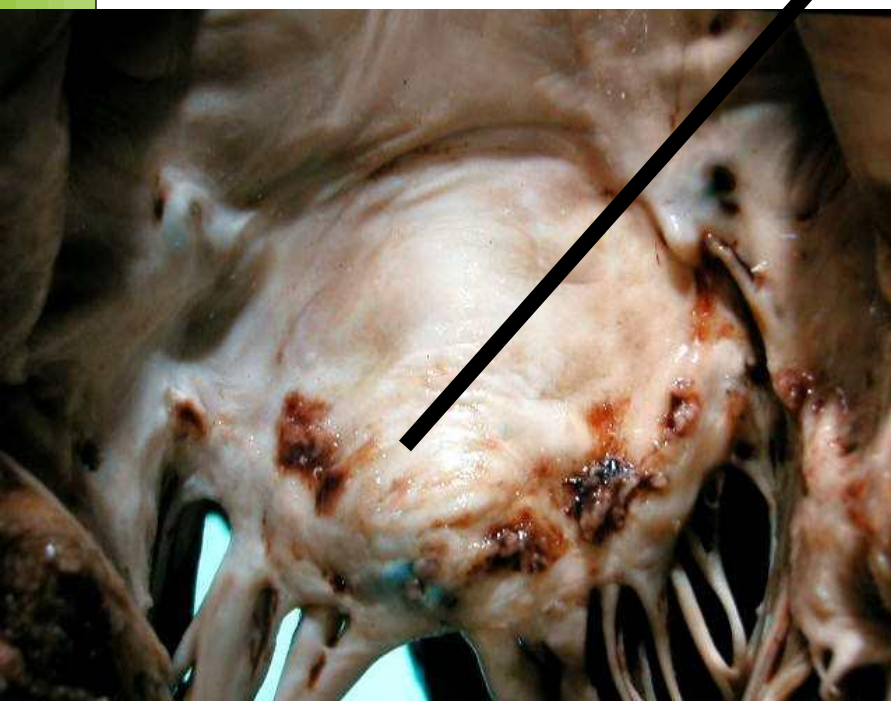
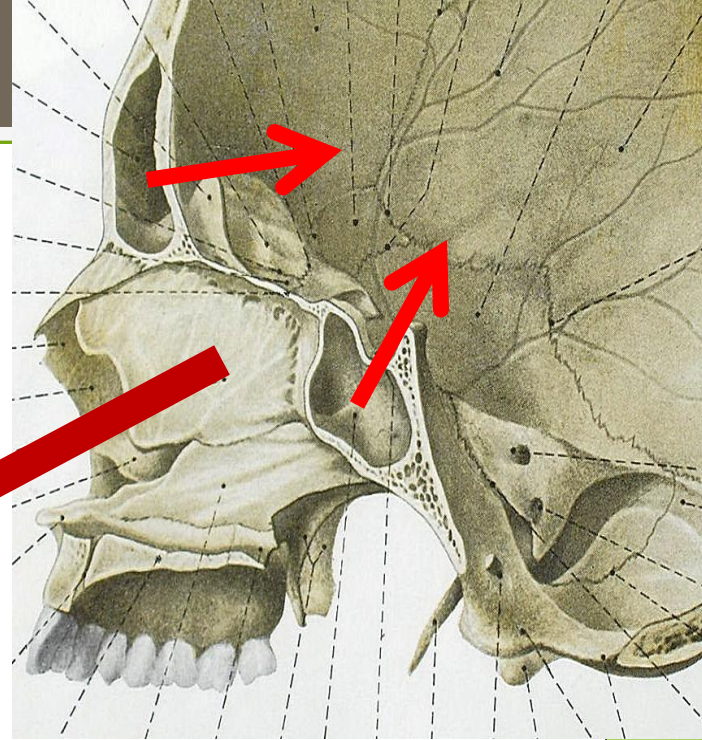
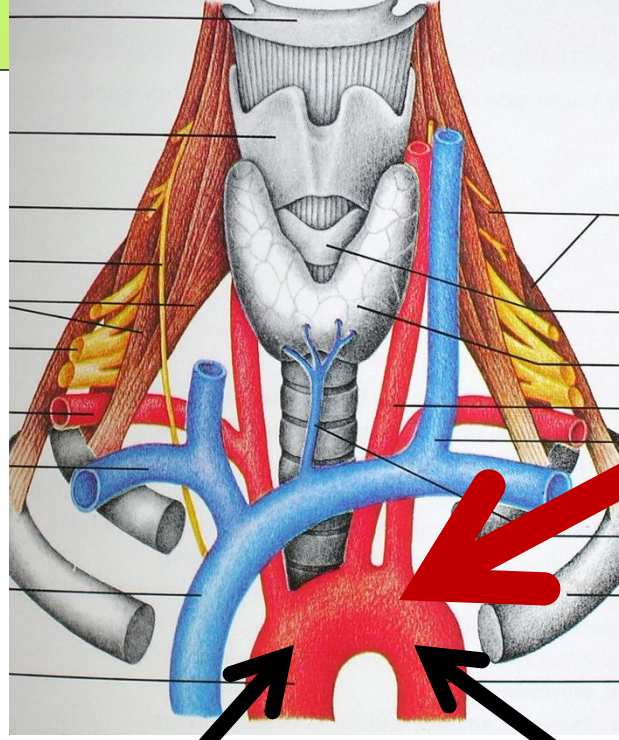
- The microorganisms that cause meningitis vary with the age of patients
 - Neonates – Escherichia coli, group B streptococci
 - Elderly – S. pneumoniae & Listeria monocytogens
 - Adolescents & young adults – Neisseria meningitidis, (H. Influenza)



Neisseria meningitidis
Gram –negative
„Meningococcus“



Streptococcus
pneumoniae
Gram-positive
„Pneumococcus“





Colonization
Carriage

Risk

Concomitant respiratory
infection
smoking (active +
passive)

Crowded living conditions
College



Military base: 492 recruits



13 initially seronegative



subsequently become colonized



5 / 13 invasive disease

Neisseria meningitidis

Type IV pilus

Nasal cavity

Nasal cavity

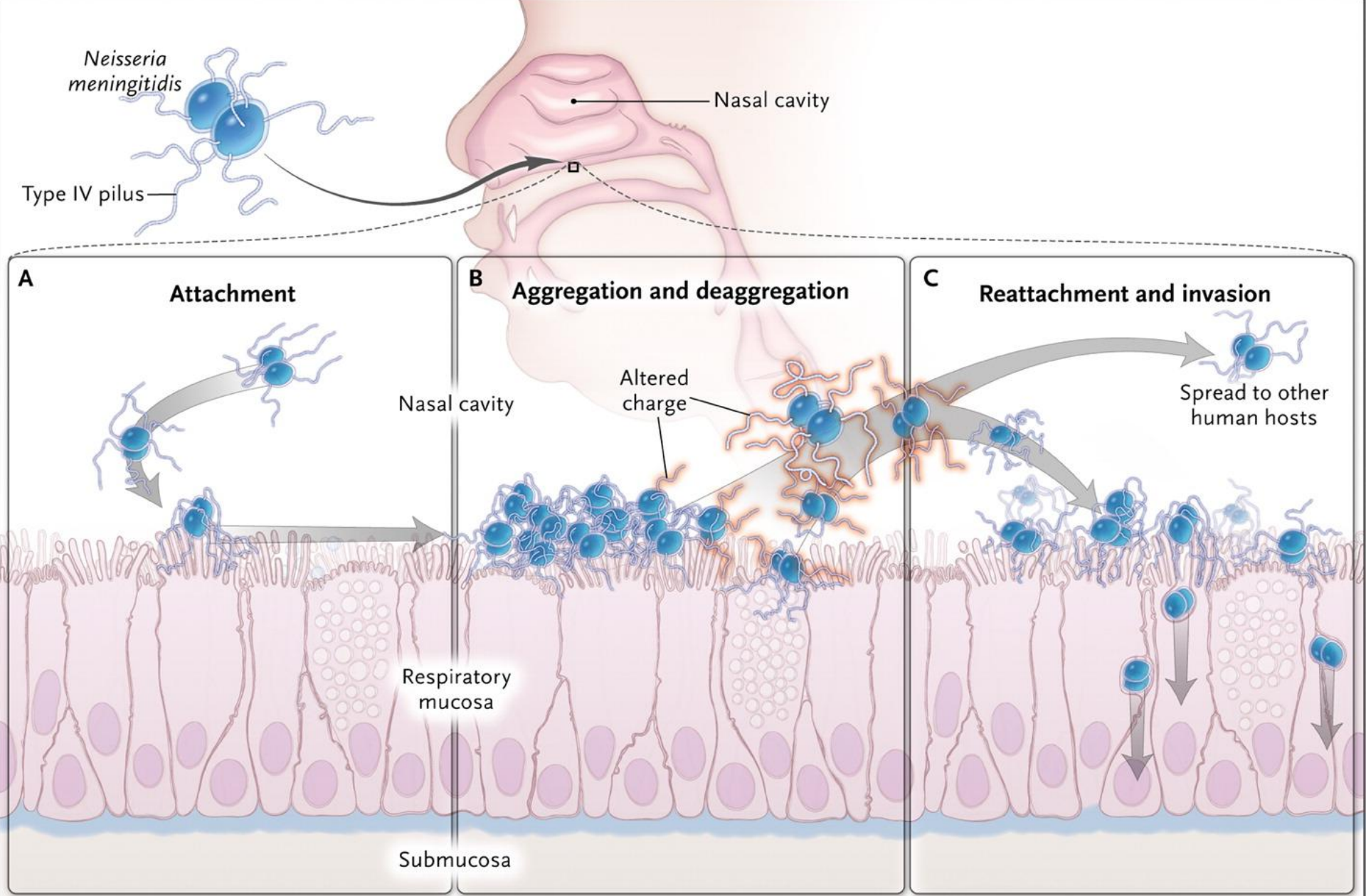
Altered charge

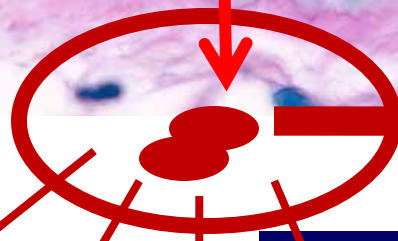
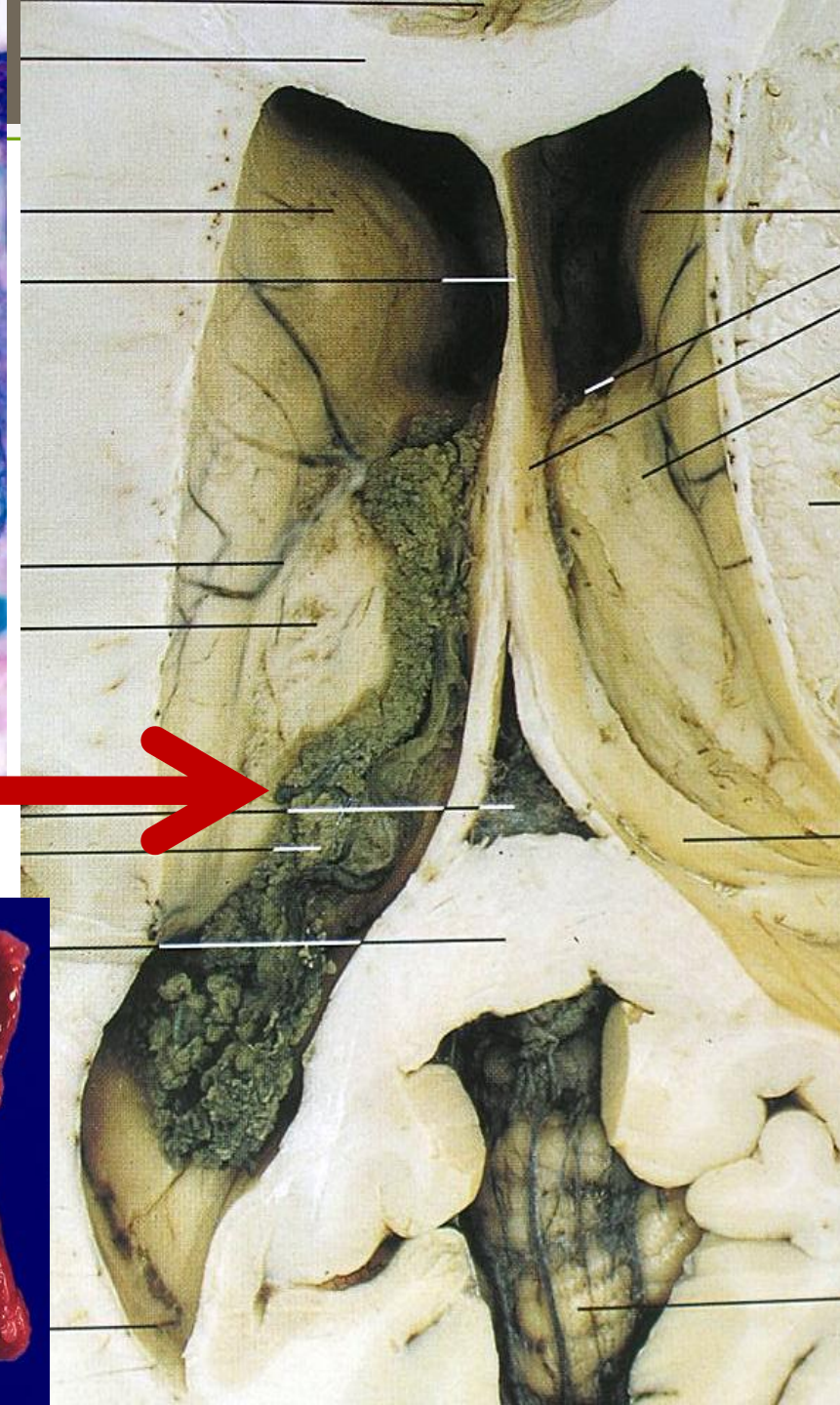
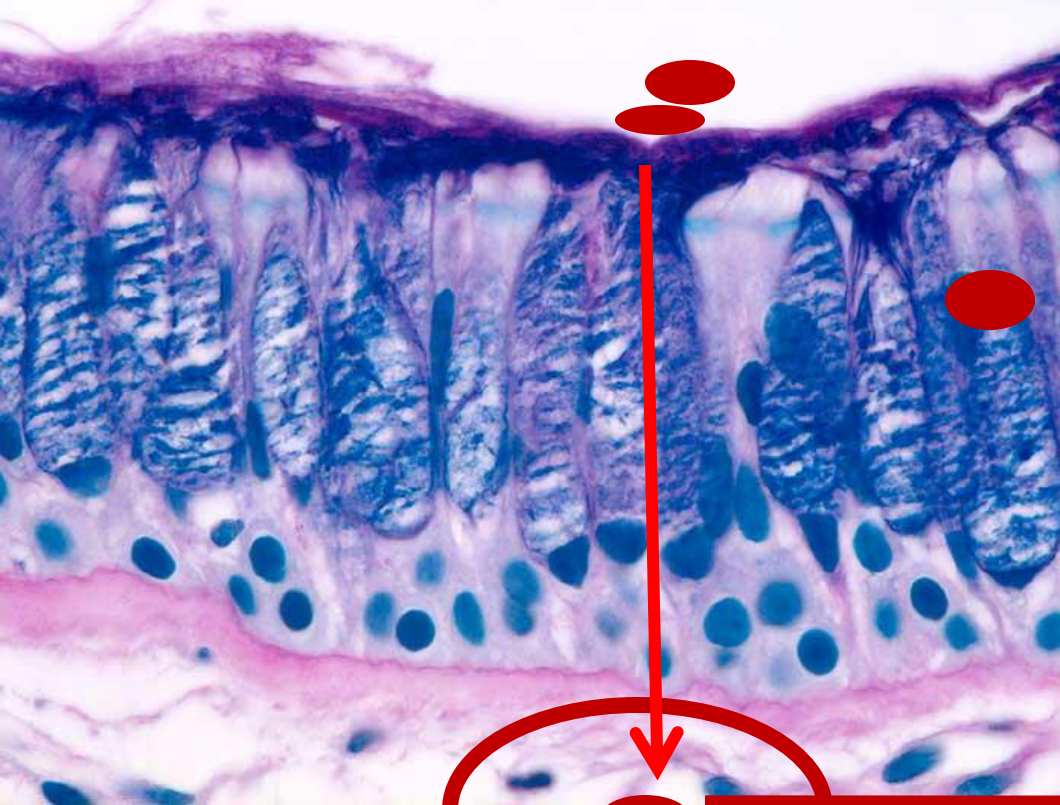
Respiratory mucosa

Submucosa

Reattachment and invasion

Spread to other human hosts





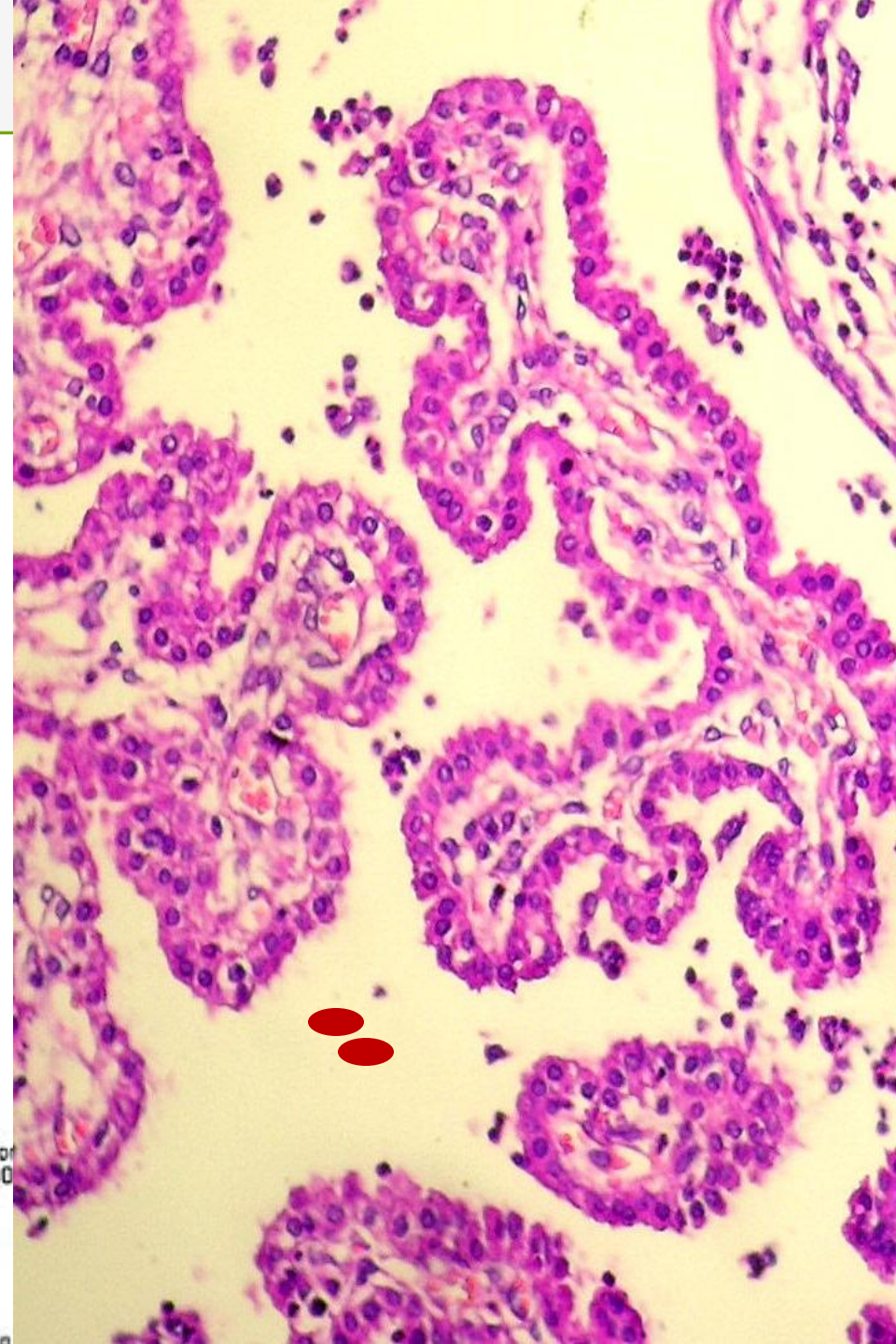
systemic

Gram-negative
=
lipooligosaccharide
e
WATERHOUSE-
FRIEDERICHSEN

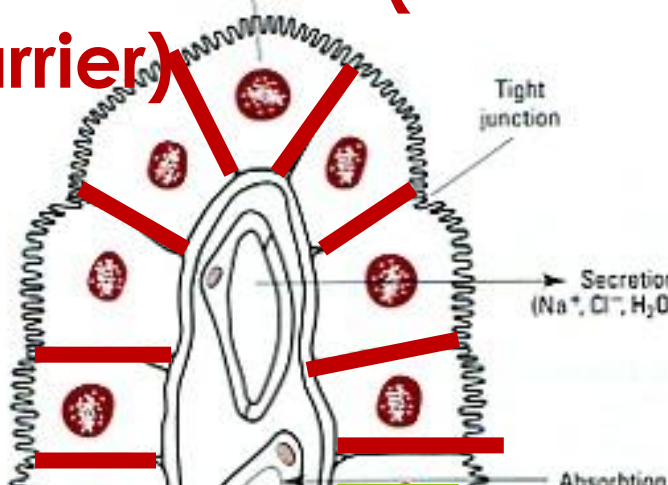


Waterhouse-Frederichsen syndrome

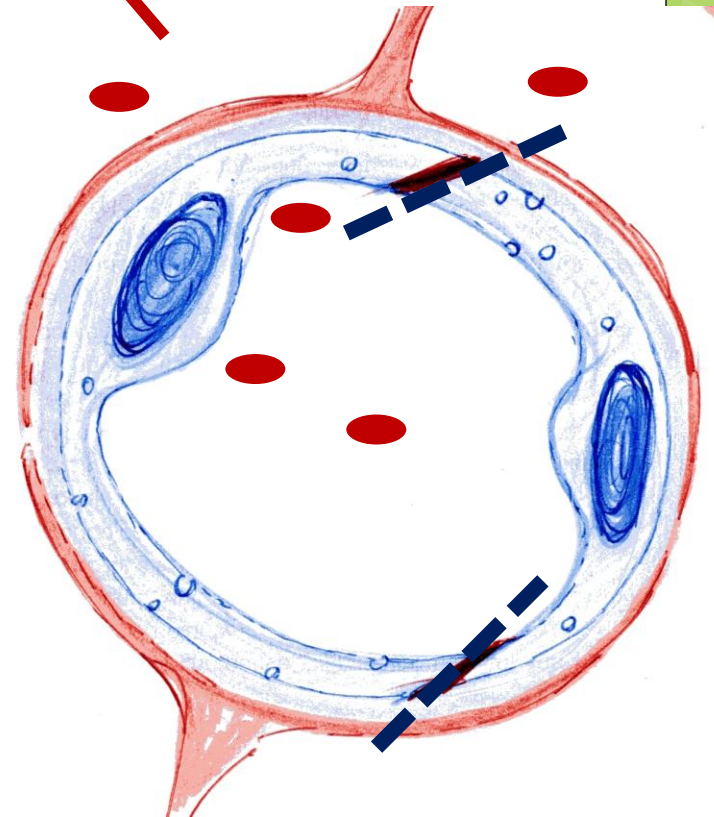
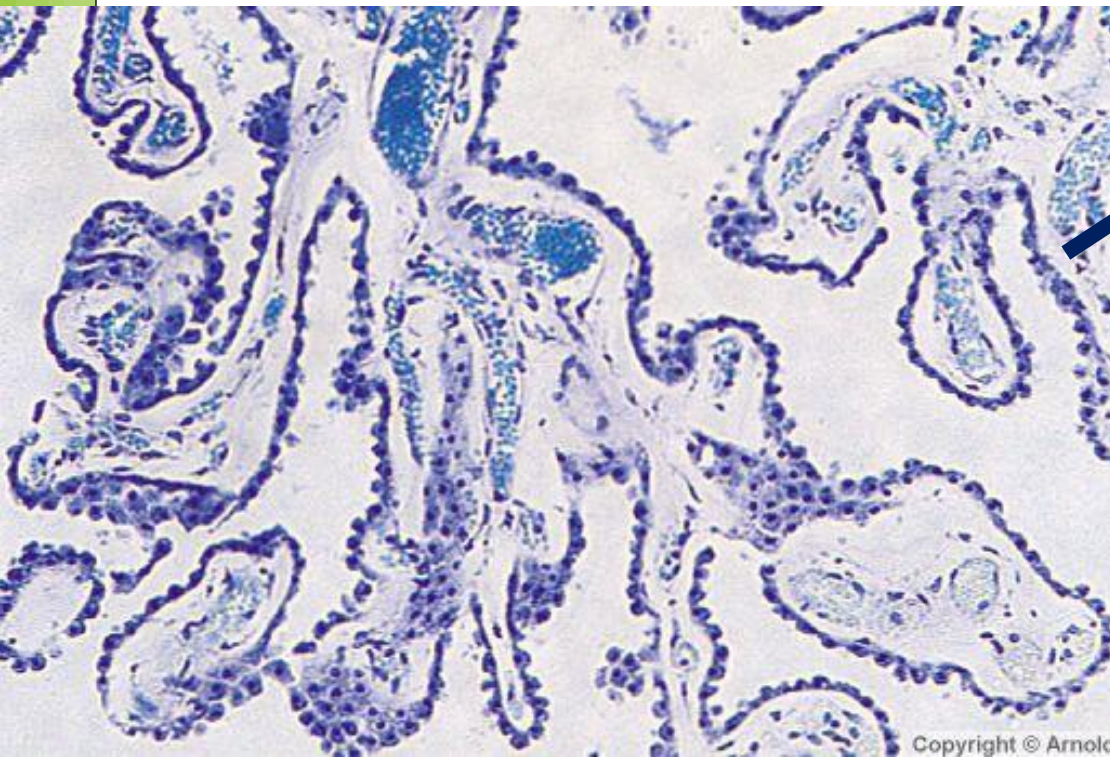
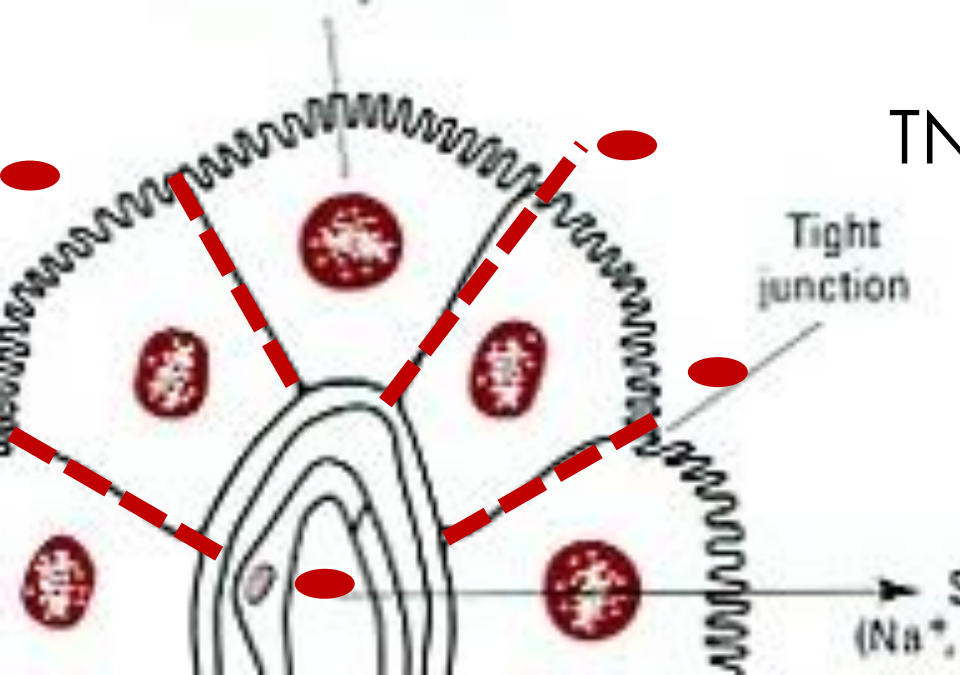
- Results from meningitis associated with septicemia (meningococcemia) with hemorrhagic infarction of adrenal glands & petechial rash

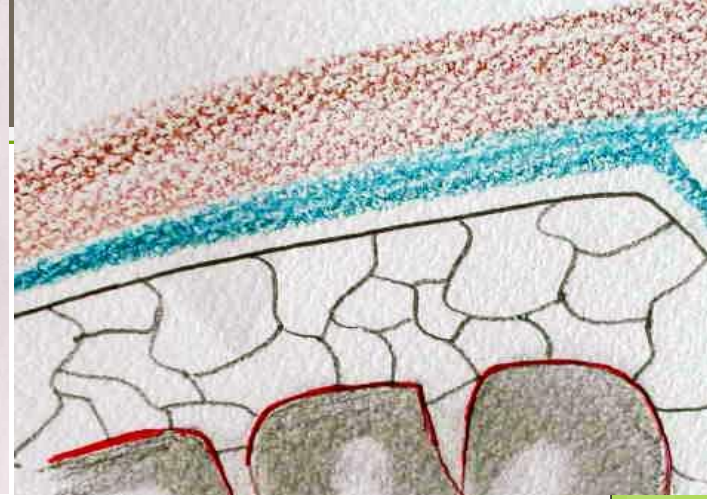
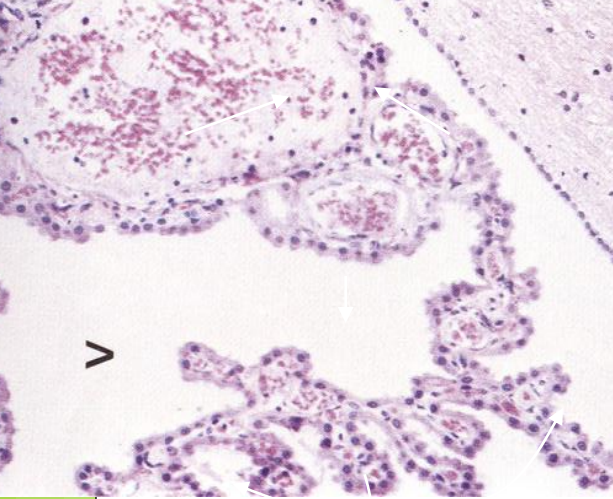


blood-brain barrier (blood-
CSF barrier)

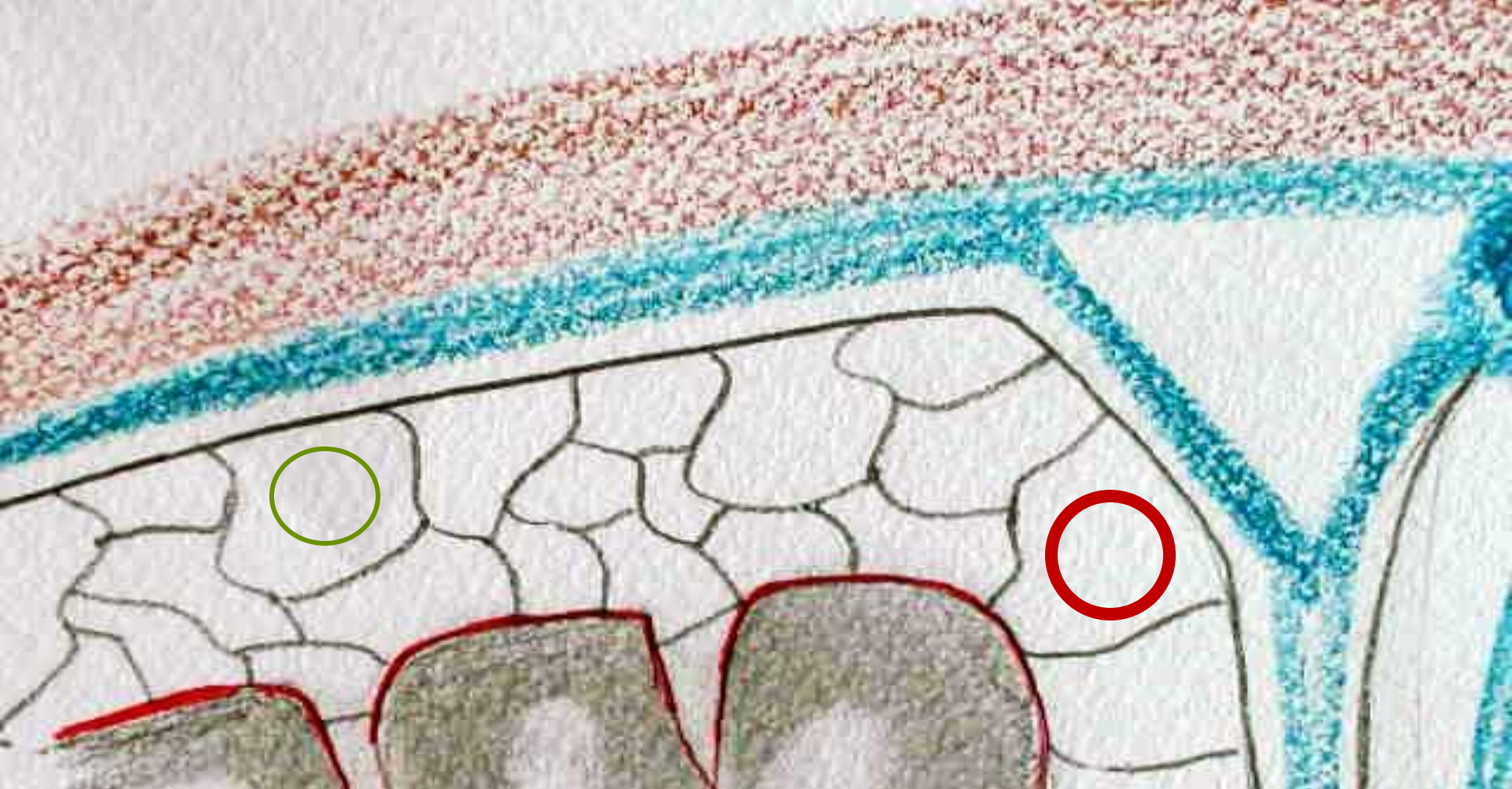


TNF — ~~Blood brain barrier~~ →





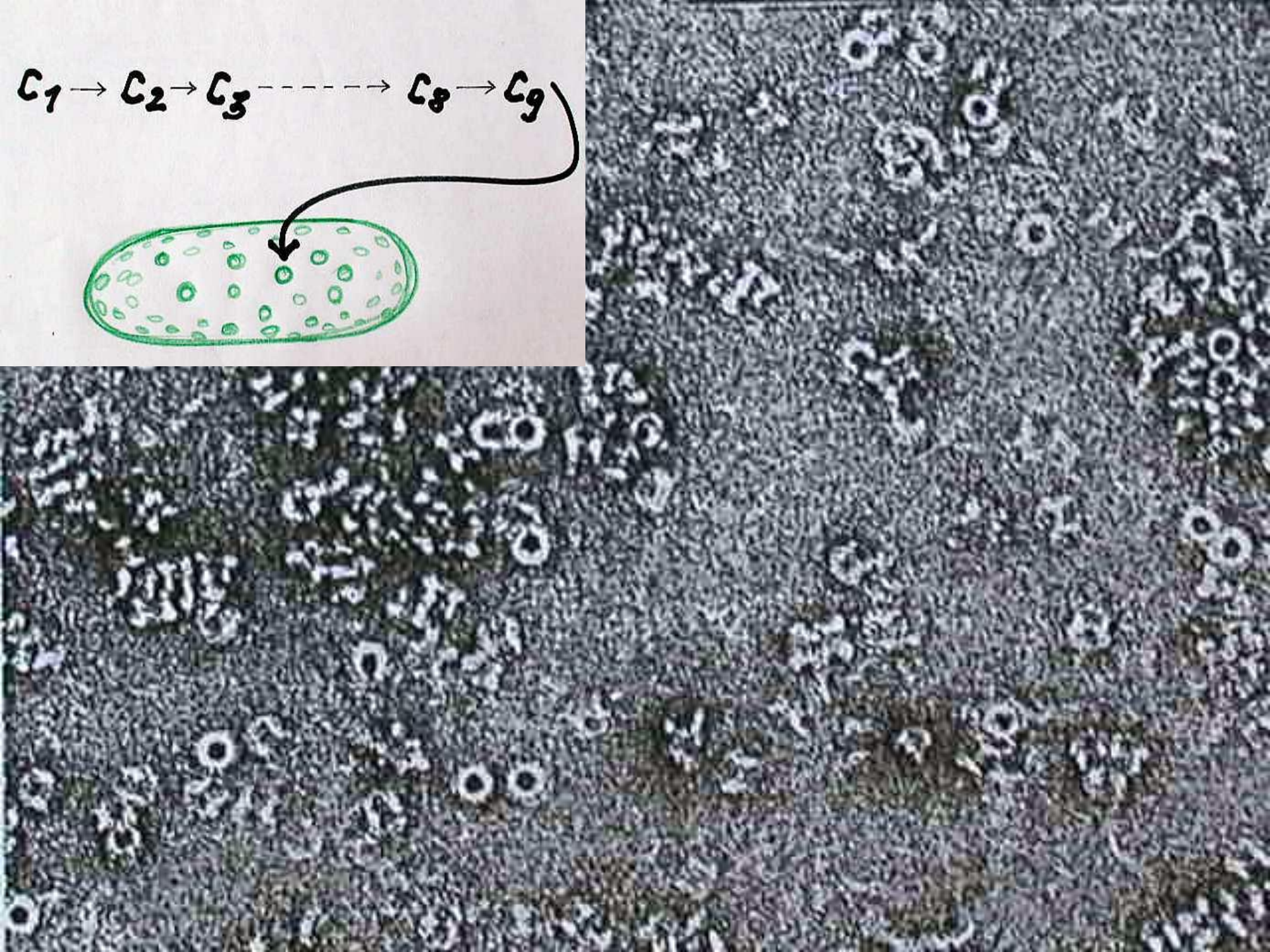
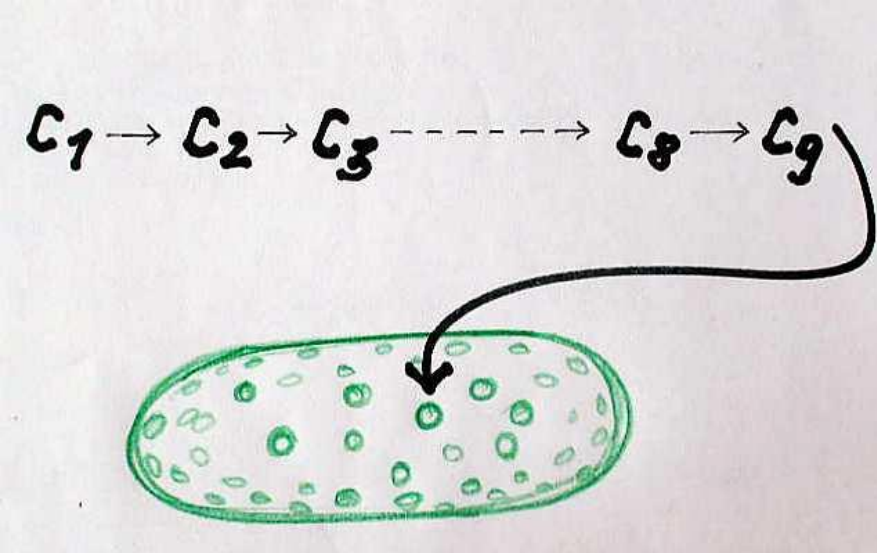
Substance		CSF	Plasma	Ratio	
[Medical Physiology, Ganong, 18 th edit.]				CSF/plasma	
Na ⁺	[meq/kg H ₂ O]	147.0	150.0	0.89	
K ⁺	[meq/kg H ₂ O]	2.9	4.6	0.62	
Mg ⁺⁺	[meq/kg H ₂ O]	2.2	1.6	1.39	
Ca ⁺⁺	[meq/kg H ₂ O]	2.3	4.7	0.49	
Osmolality	[mosm/kg H ₂ O]	289.0	289.0	1.0	
Protein	[mg/dl]	20.0	6000.0	0.003	
Glucose	[mg/dl]	64.0	100.0	0.64	
Cholesterol	[mg/dl]	0.2	175.0	0.001	

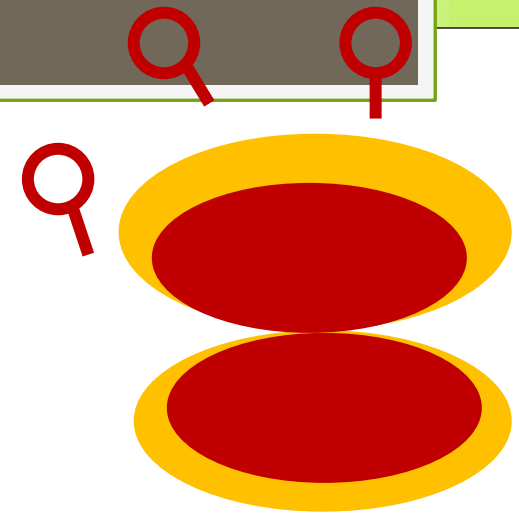
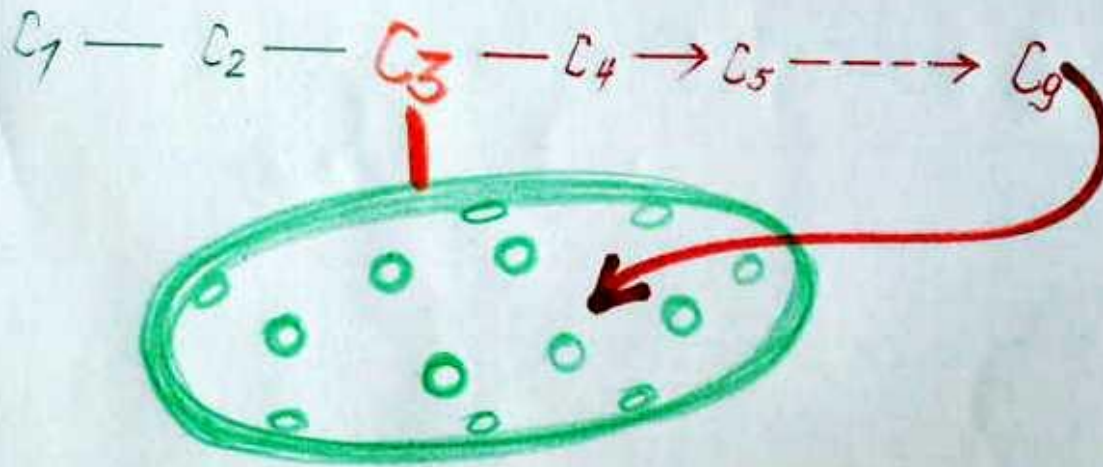


CSF:

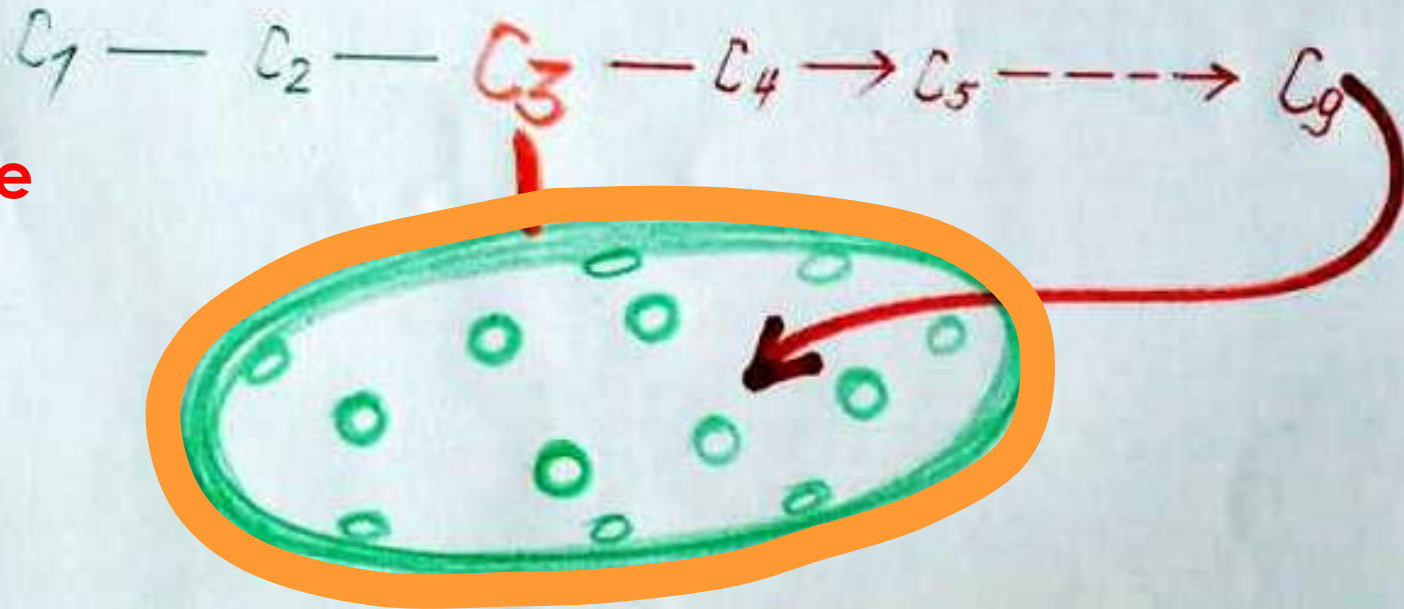
few cells: < 5 lymphocytes/mm³

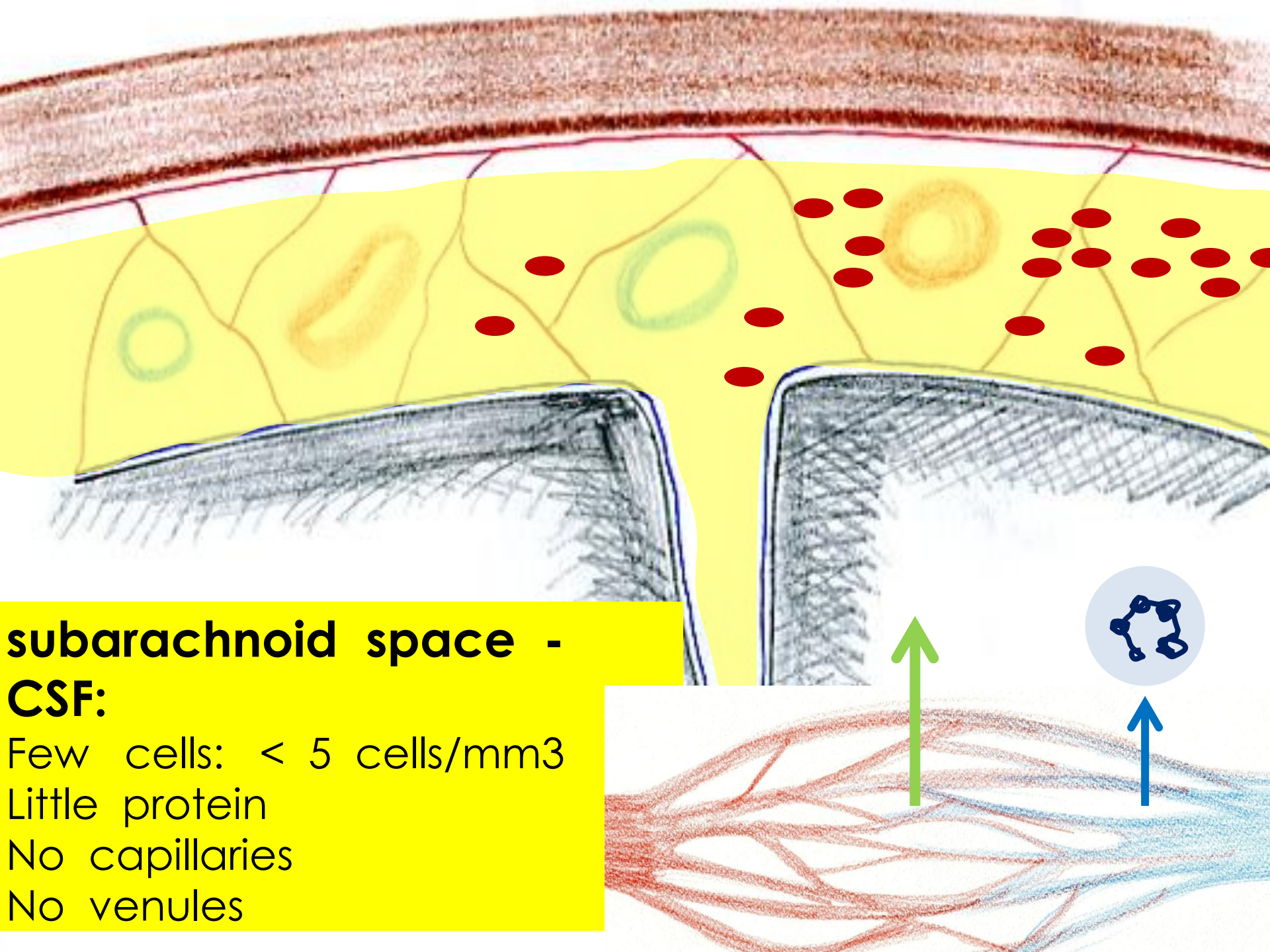
small amount of protein





polysaccharide
capsule

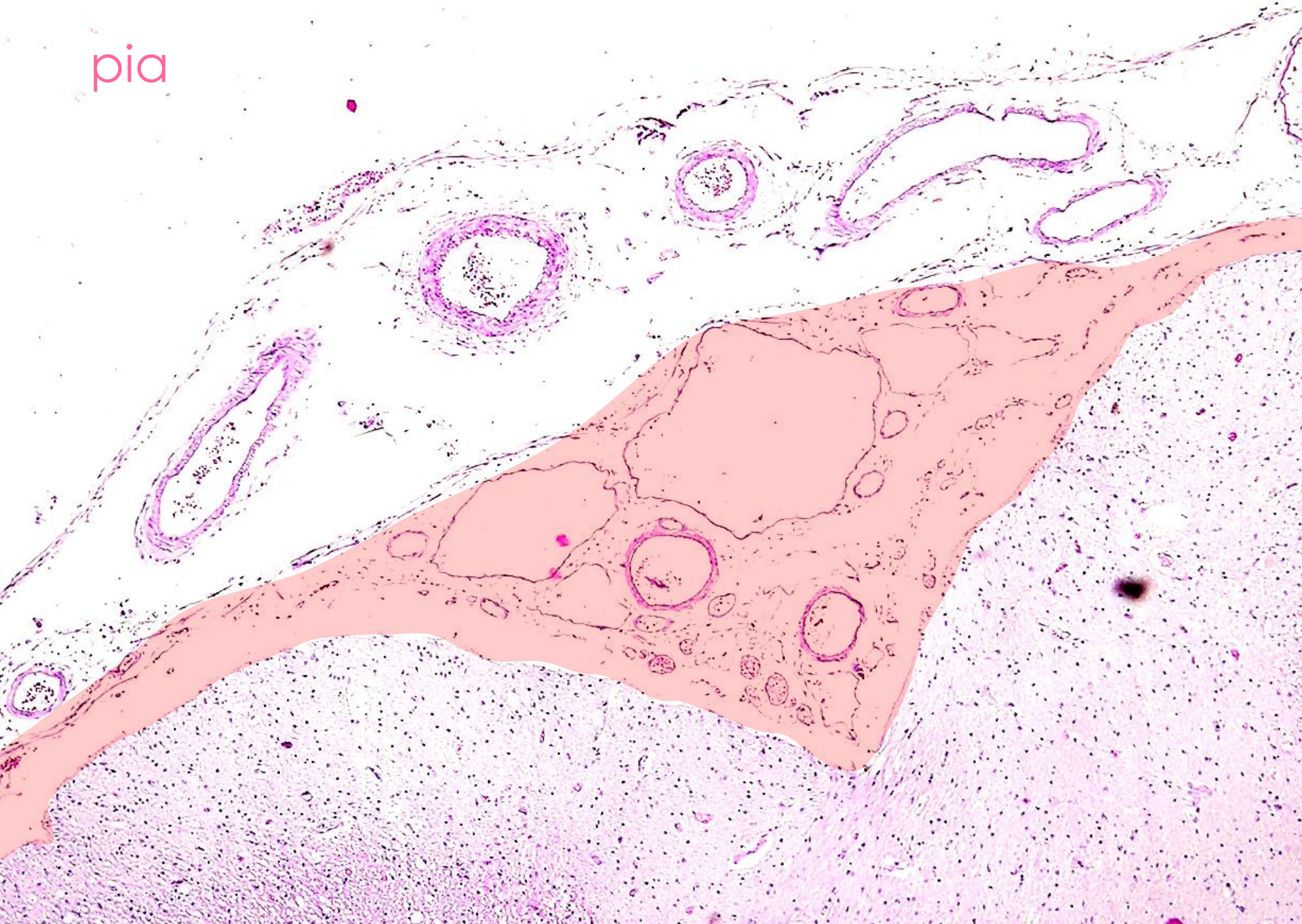


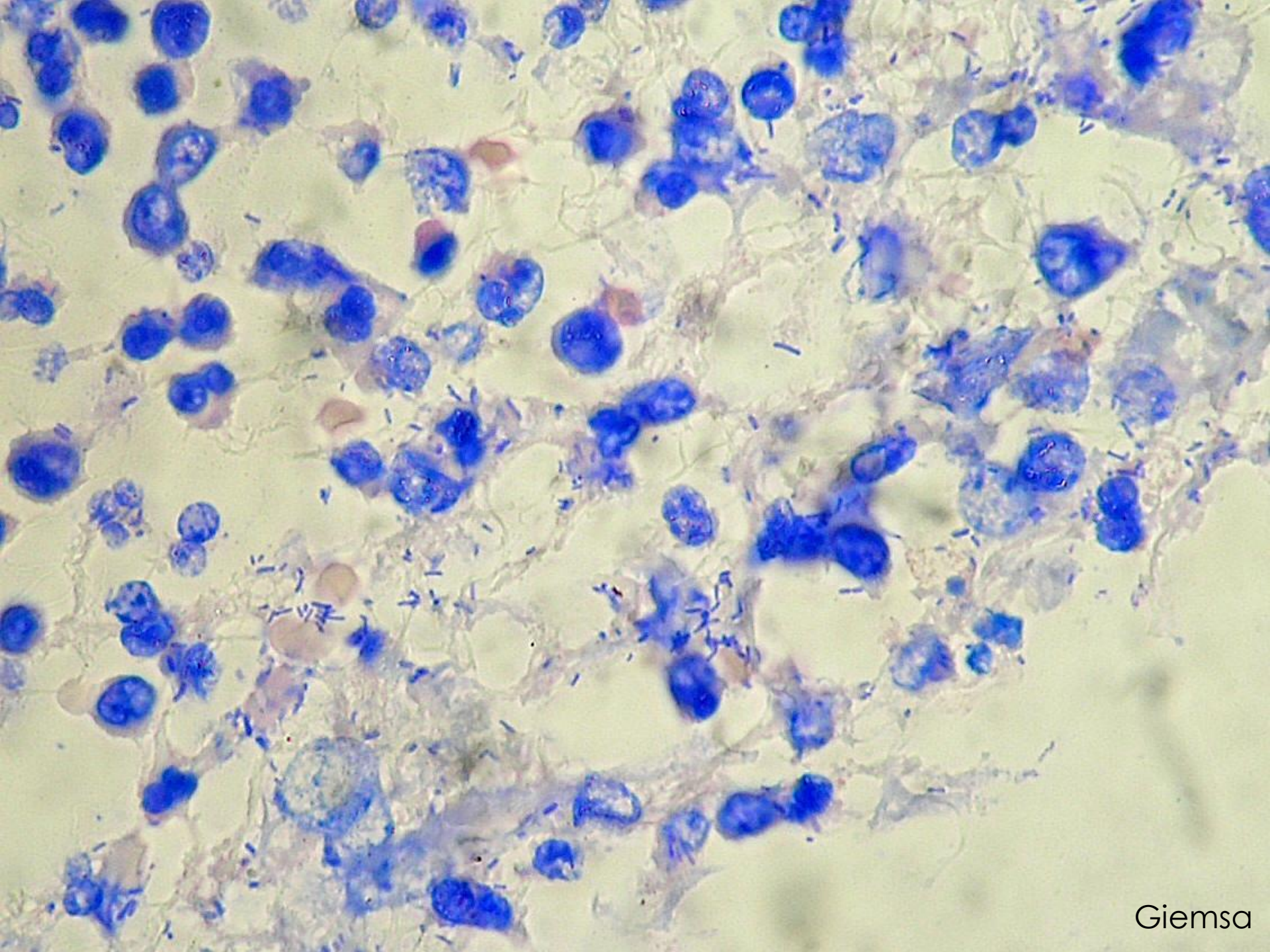


subarachnoid space - CSF:

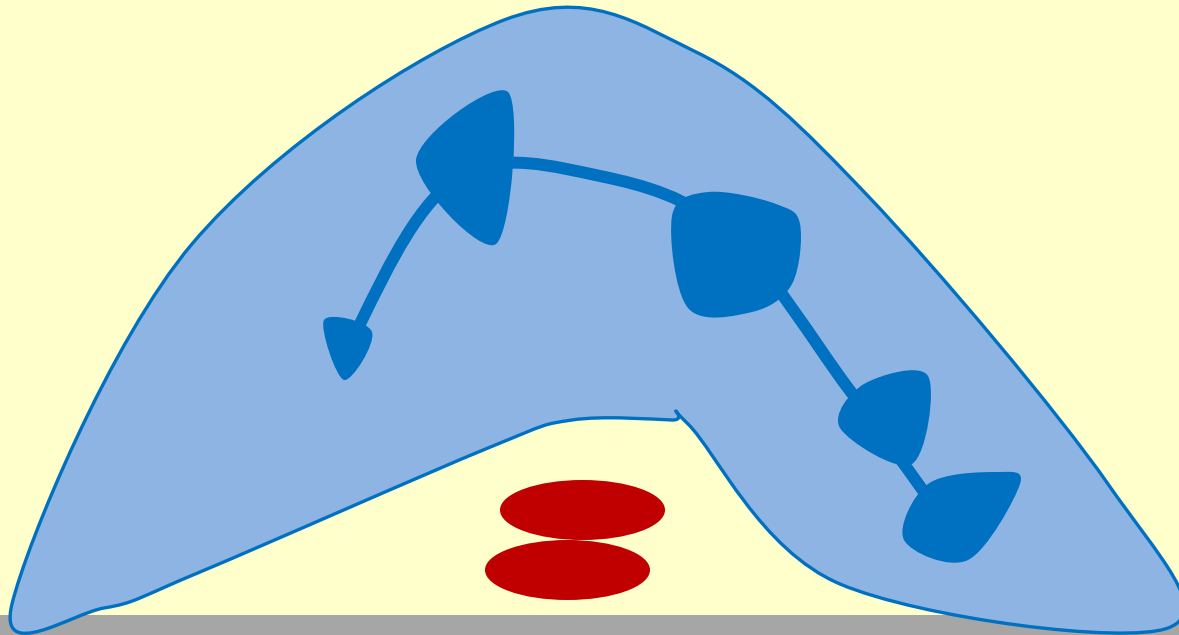
Few cells: < 5 cells/mm³
Little protein
No capillaries
No venules

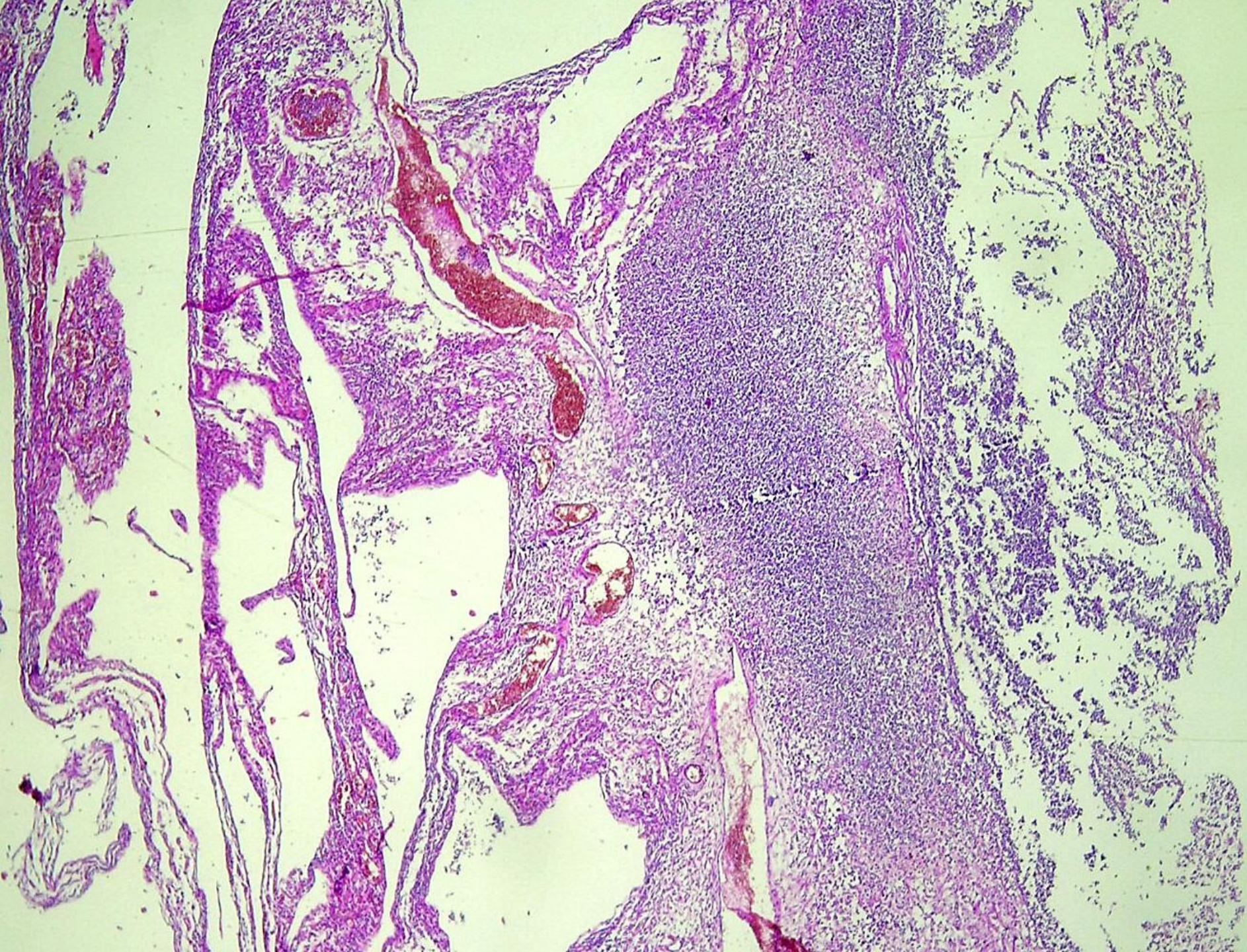
pia

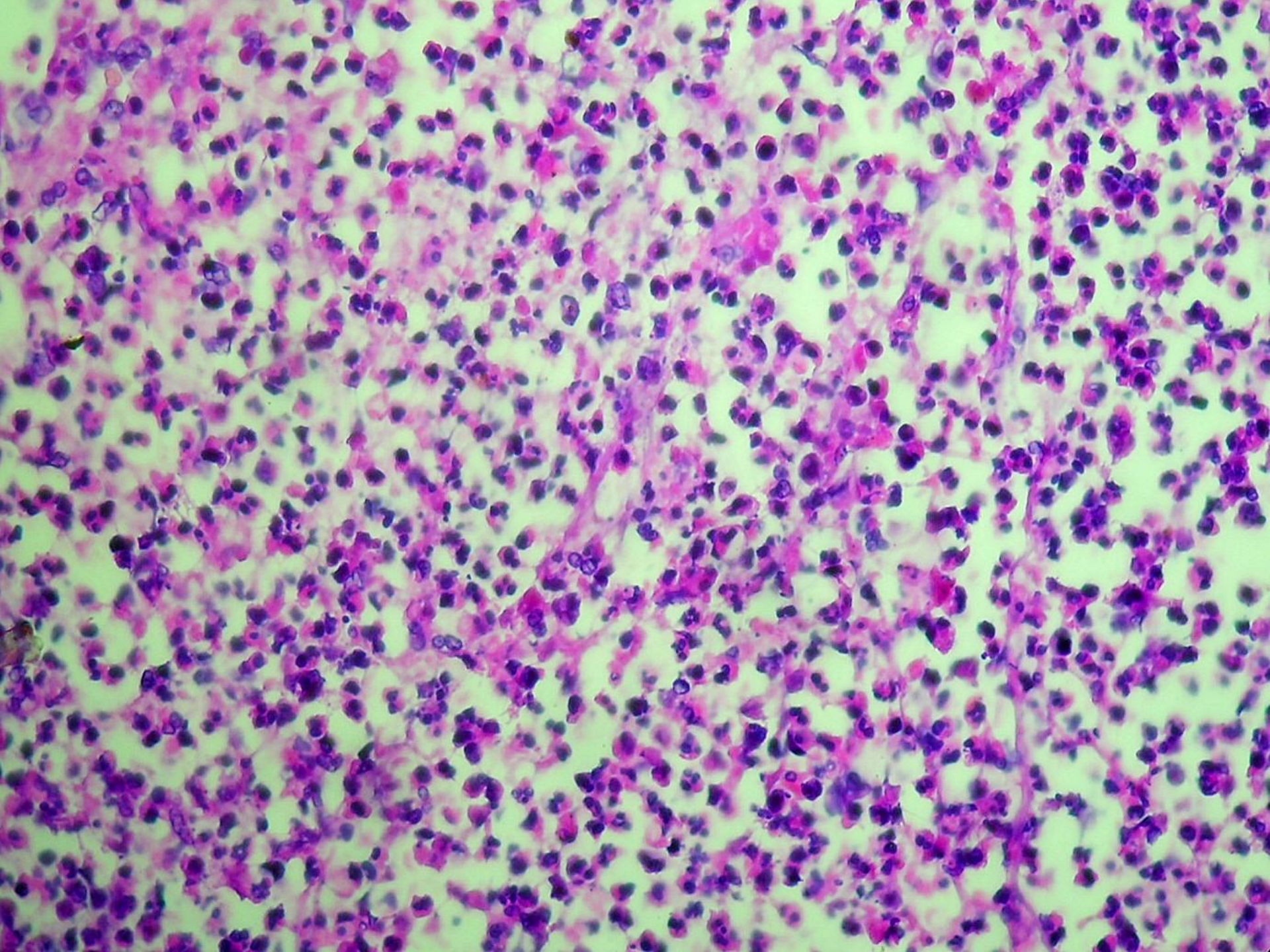




Giemsa



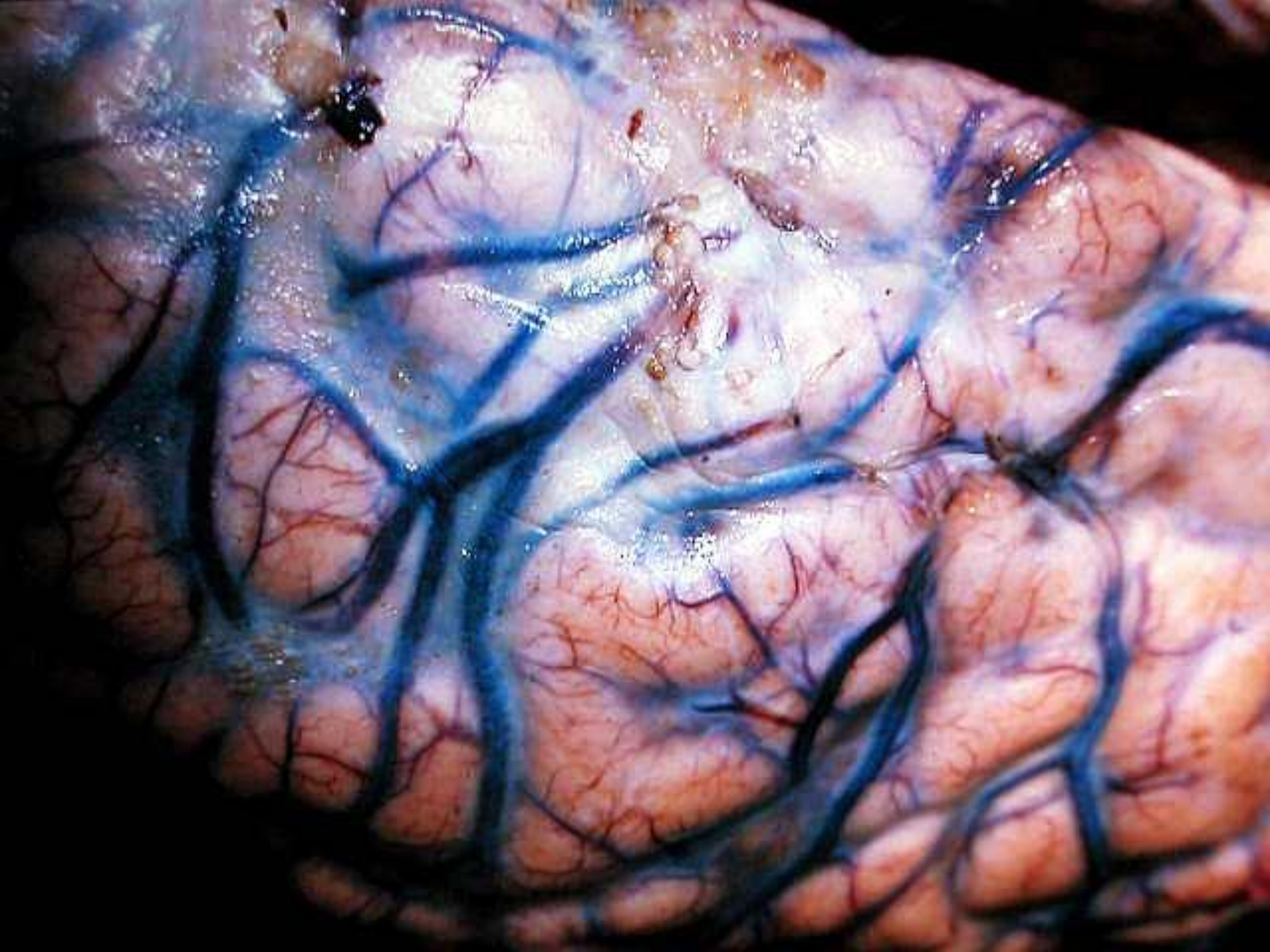




Morphology

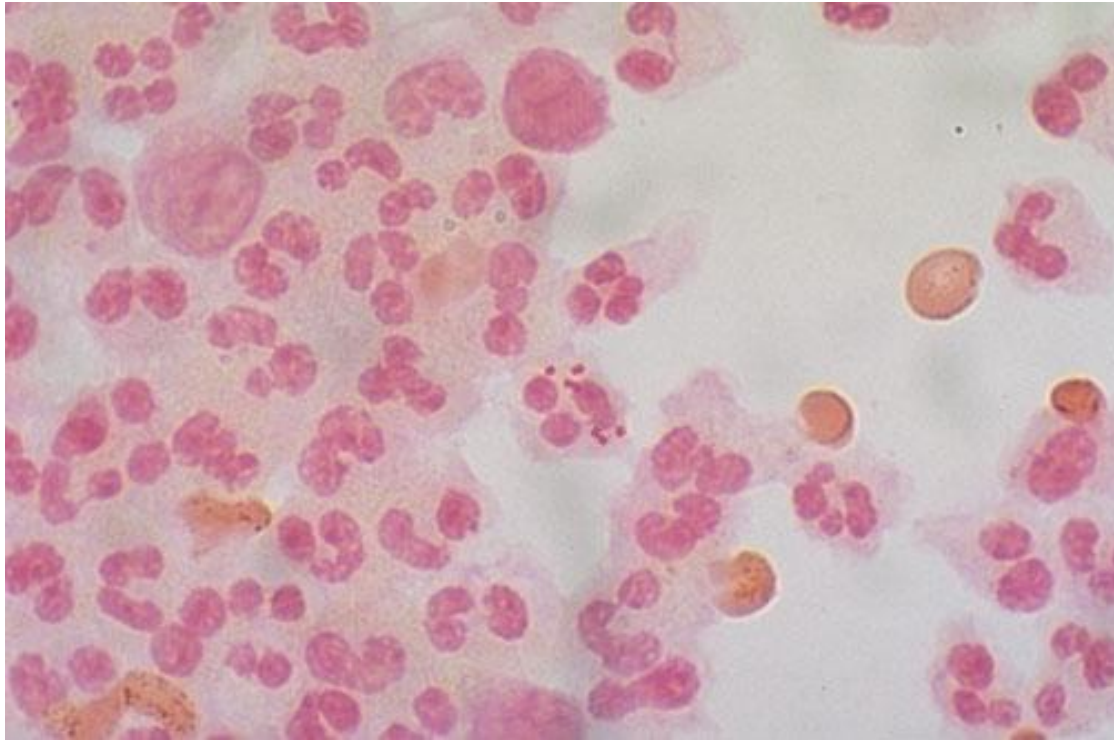
- An exudate is seen on surface within the leptomeninges over the surface of the brain





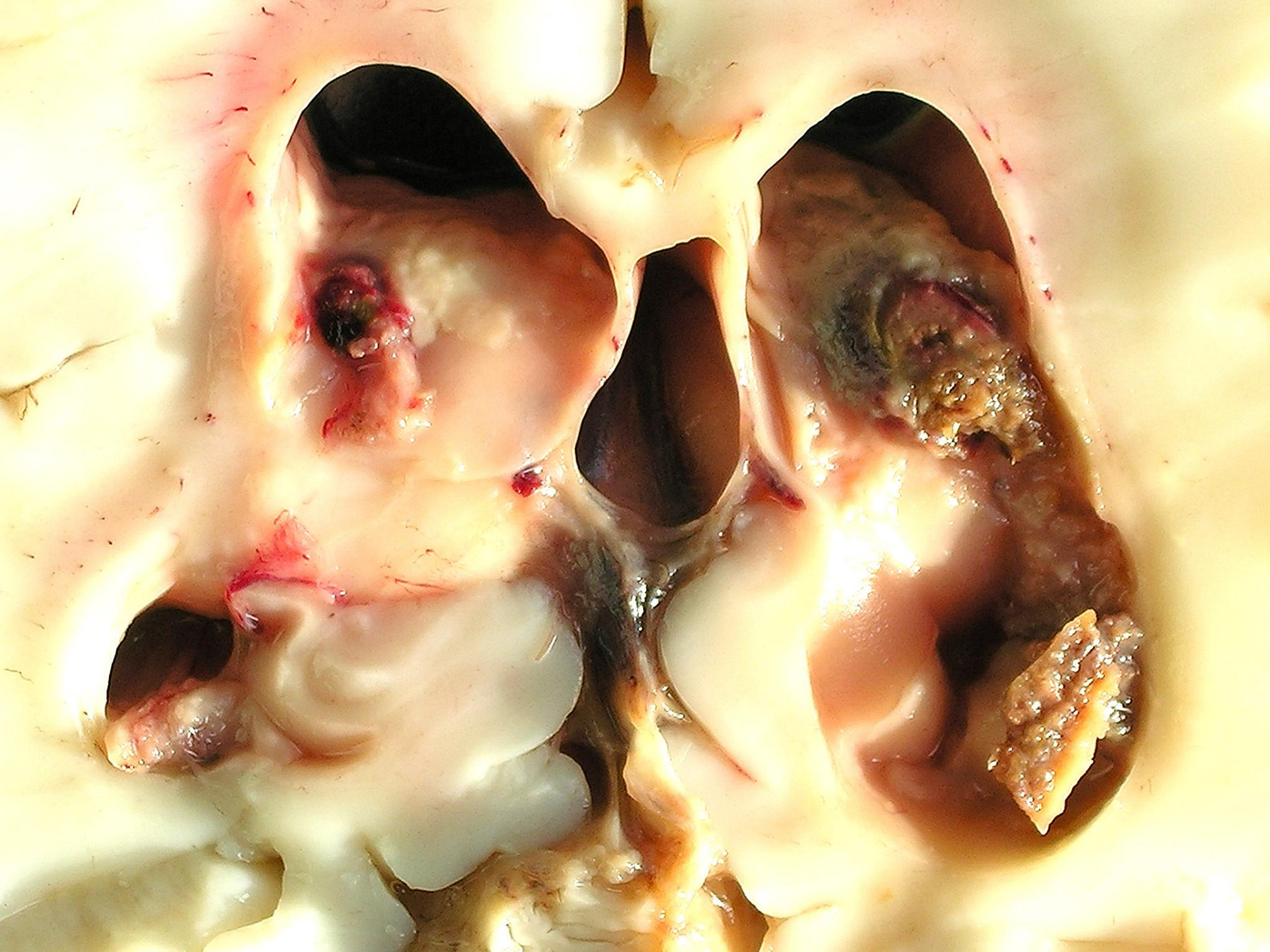
C/F

- Systemic sign of infections with evidence of meningeal irritation including headache, photophobia, clouding of consciousness , neck stiffness
- A spinal tap yields cloudy or frankly purulent CSF under increased pressure with as many as ~ 90,000/mm³ neutrophils , increased protein & reduced glucose content
- Organisms can be seen on smear or can be cultured

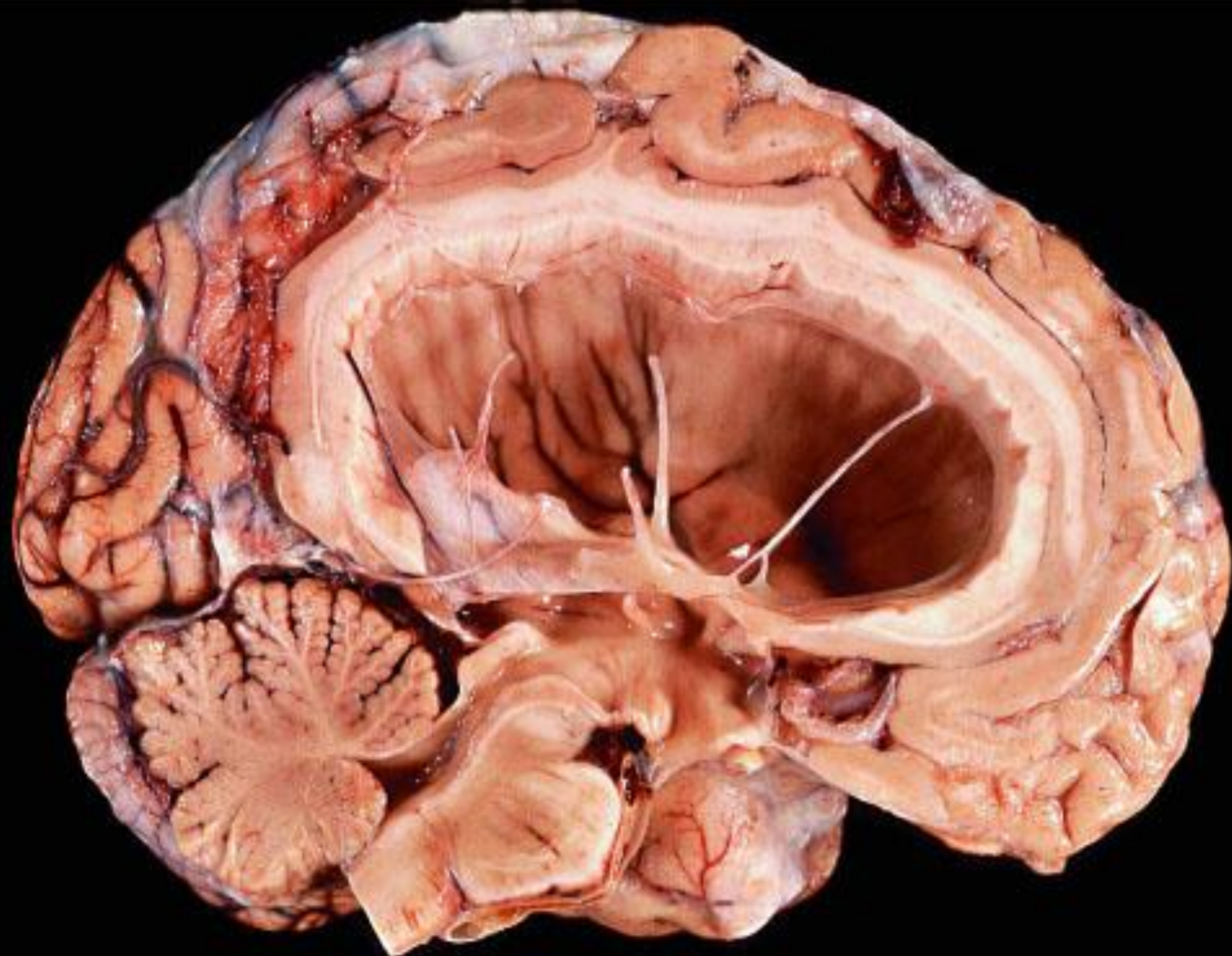


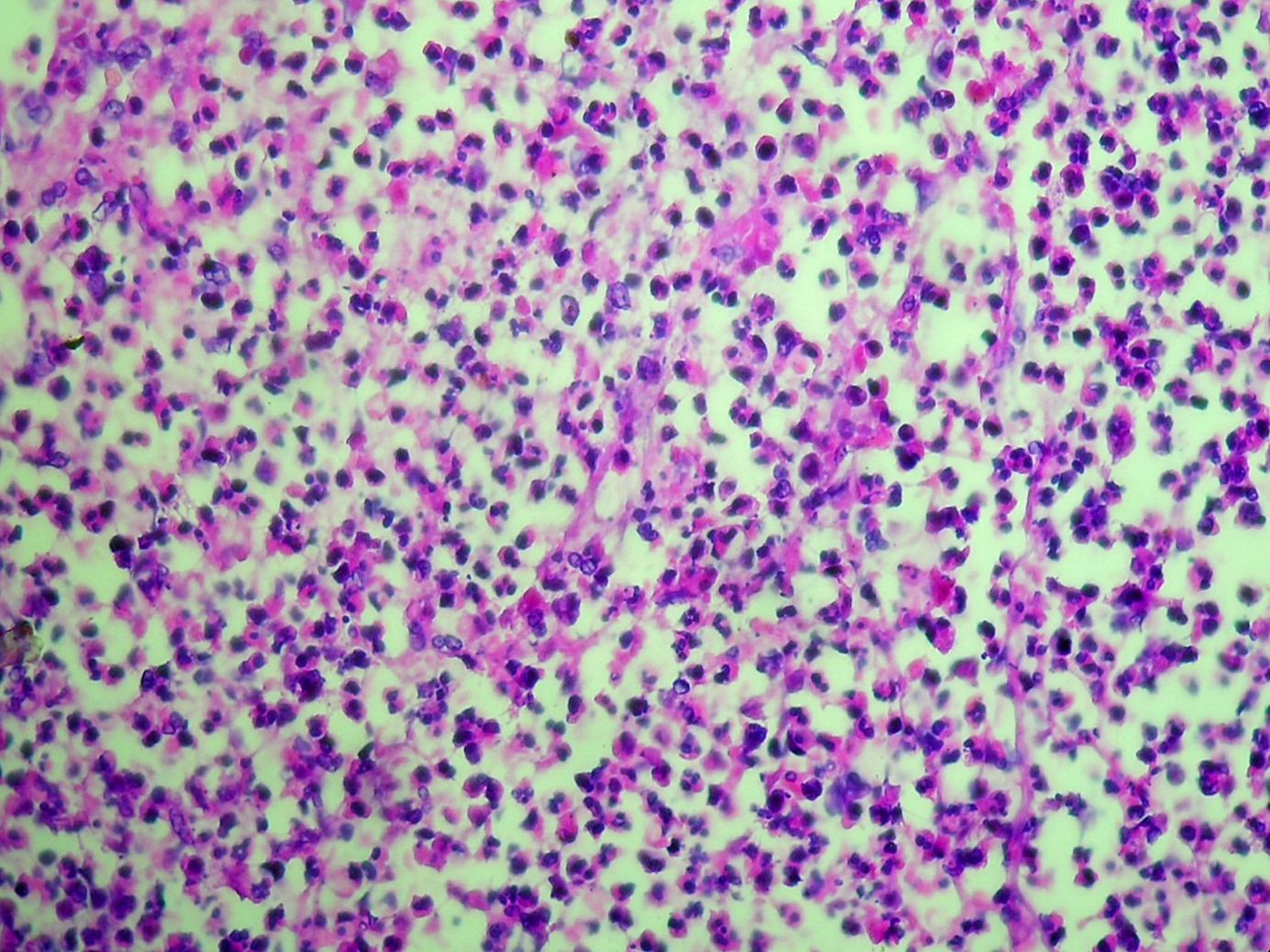
Complication

- Hydrocephalus can be a long-term complication of infection. A basilar meningitis, for example, may lead to scarring that obstructs CSF outflow via the foramina of Luschka or Magendie. (*S.Pneumoniae*).
- Inflammation with scarring of arachnoid granulations at the vertex may diminish CSF absorption.



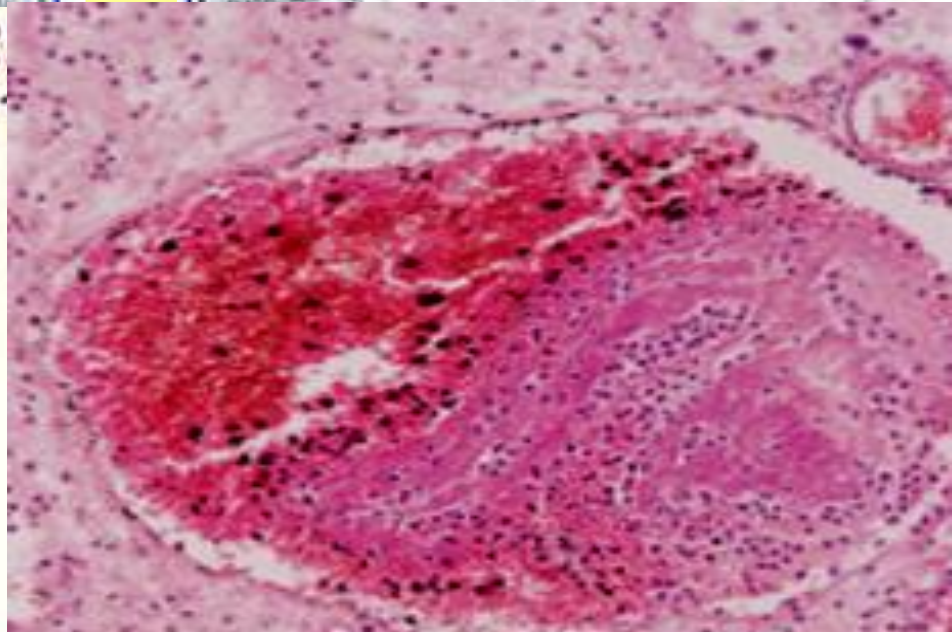
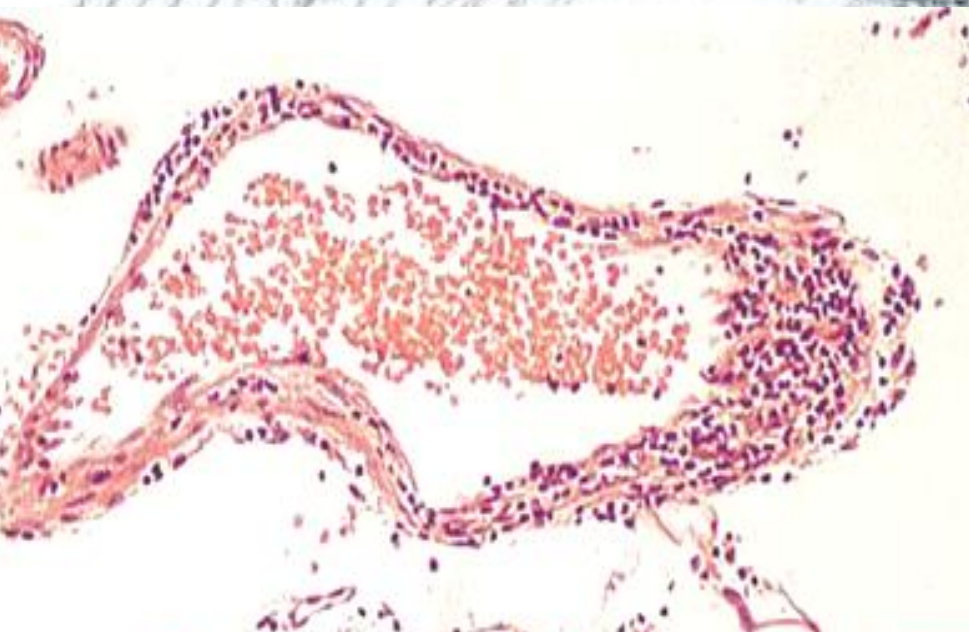
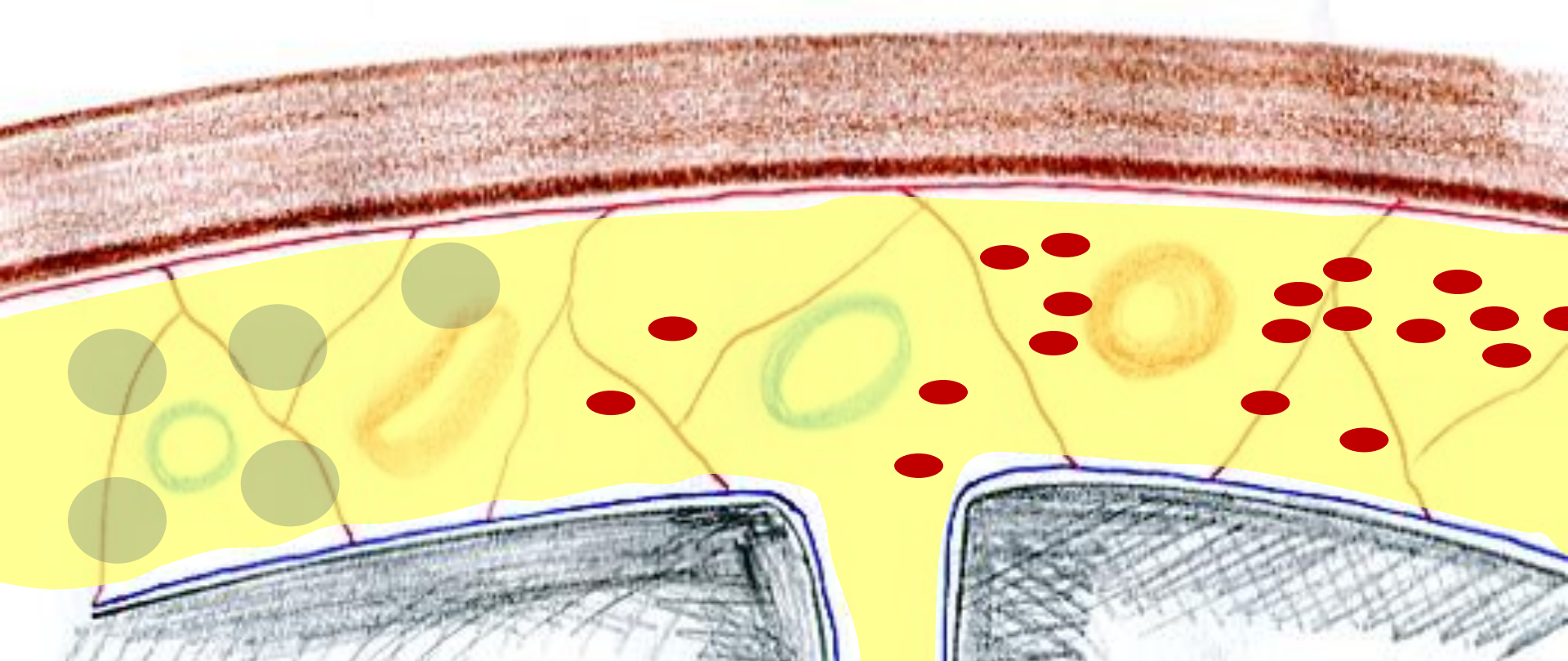


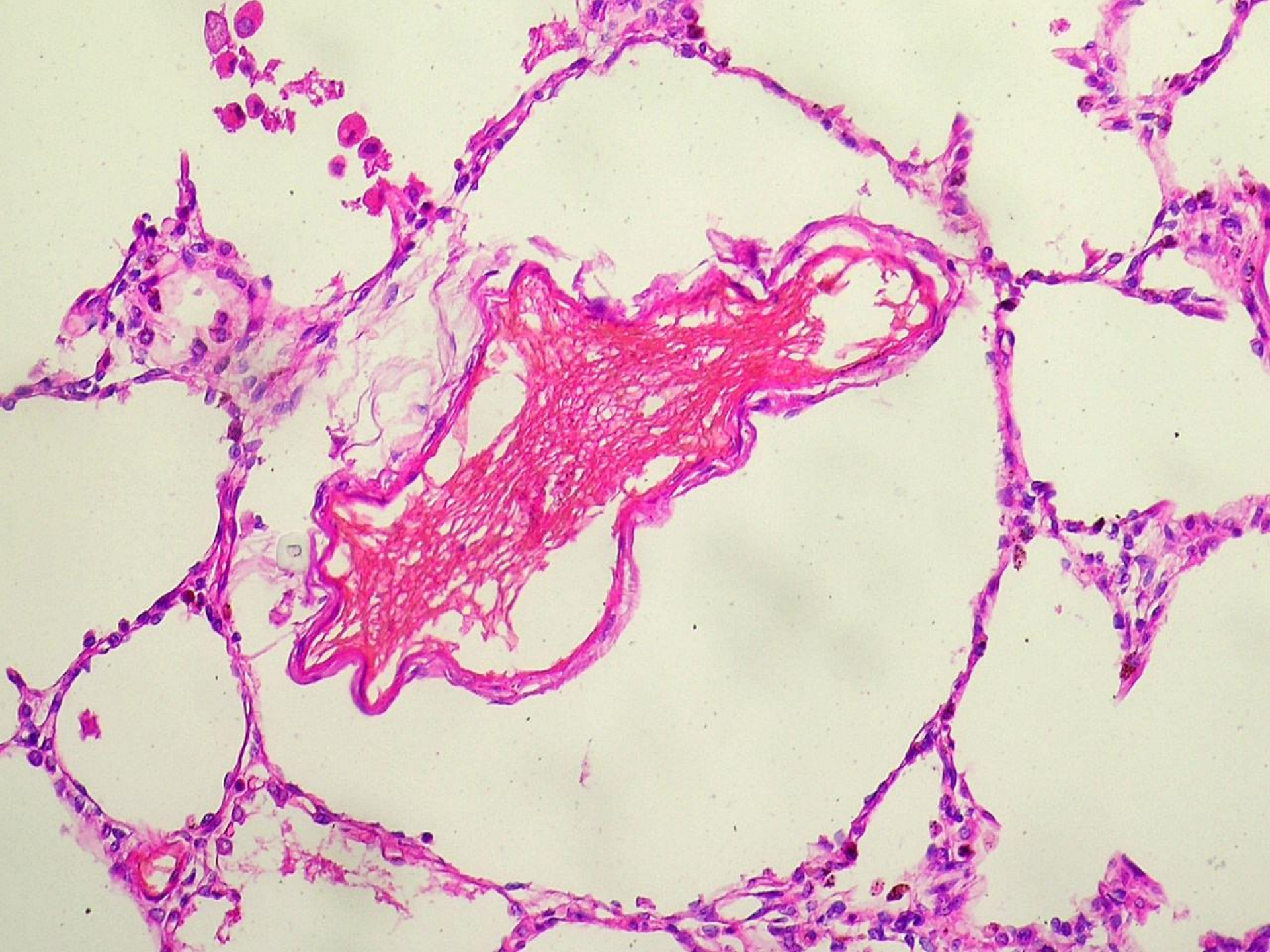




Complication ctd

- ◉ Extension to the brain- encephalitis
- ◉ Phlebitis=> Endothelial Damage=> Thrombosis





Acute aseptic (viral) meningitis

- It is of viral rarely bacterial origin
- Clinical course is less fulminant than pyogenic meningitis & self limiting
- CSF finding – lymphocytic pleocytosis, protein elevation is moderate , glucose level normal
- The pathogens are most commonly are enterovirus- Echovirus, coxasackie virus & non paralytic poliomyelitis

Acute focal suppurative infections

Brain abscess

- Brain abscess may arise by direct implantation of organisms, local extension from adjacent foci (mastoiditis) or hematogenous spread (usually from the primary site in the heart ,lung)
- Predisposing conditions include infective endocarditis, cyanotic heart disease.
- Streptococci & staphylococci are the most common offending organisms

Morphology

- Abscesses are discrete lesions with central liquefactive necrosis , a surrounding fibrous capsule & edema

Clinical features

- Patients present with progressive focal deficit in addition to the general signs of raised intracranial pressure



Chronic bacterial meningitis

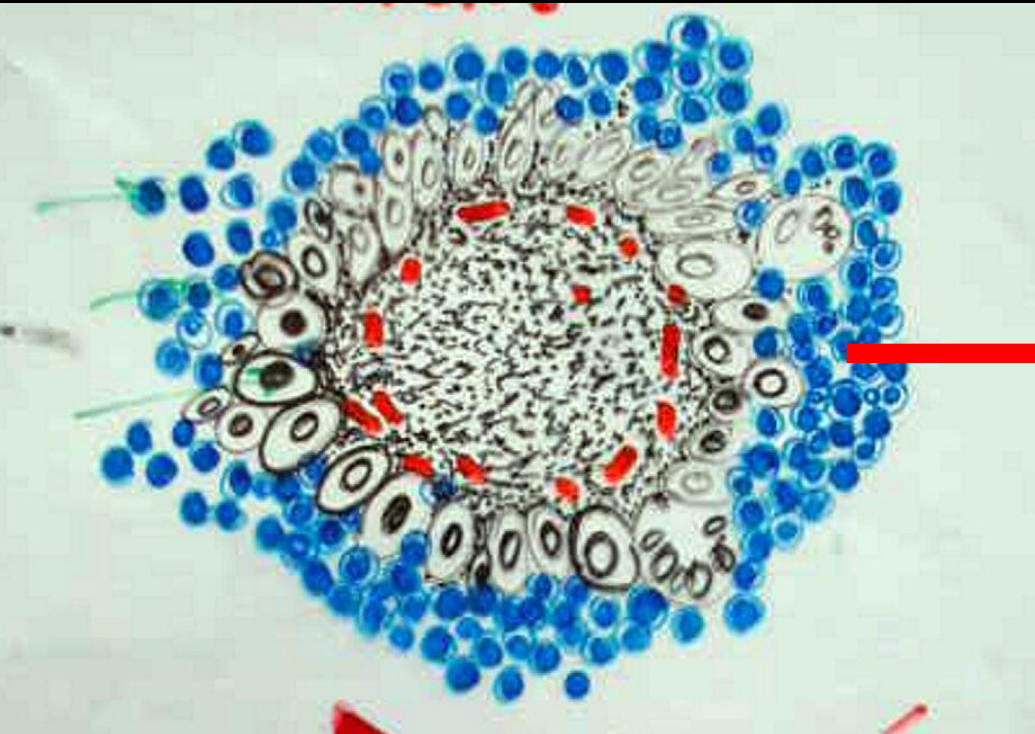
Tuberculous meningitis

- Symptoms are headache, malaise, mental confusion & vomiting
- There is only moderate CSF pleocytosis made up of mononuclear cells or mixture of polymorphonuclear & mononuclear cells, protein level is elevated, glucose content is moderately reduced or normal

Morphology

- The most common pattern of involvement is a diffuse meningitis.

Containment



Progression

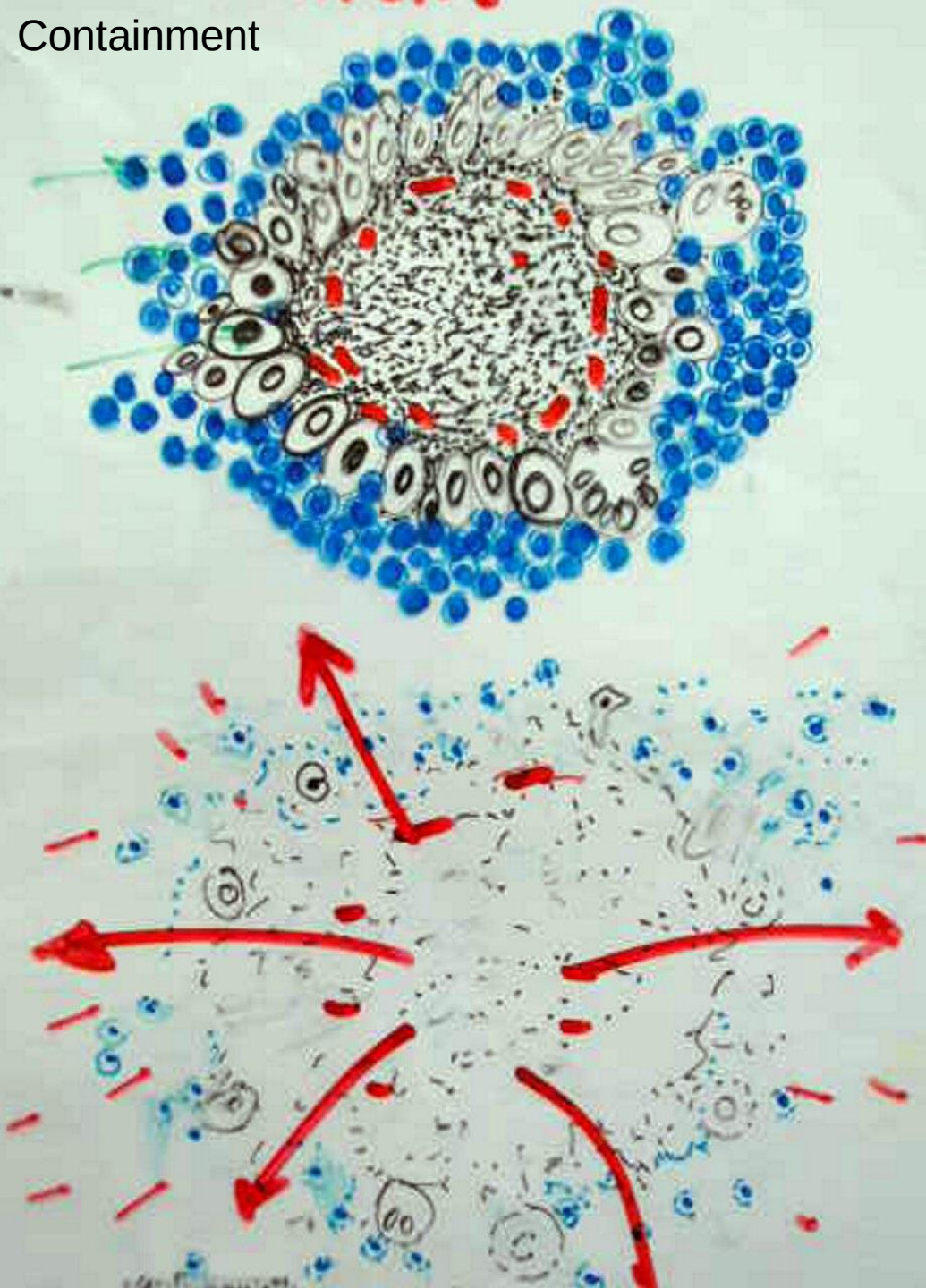
Reactivation

Re-Infection

5%-10%

Not eradicated !

Containment



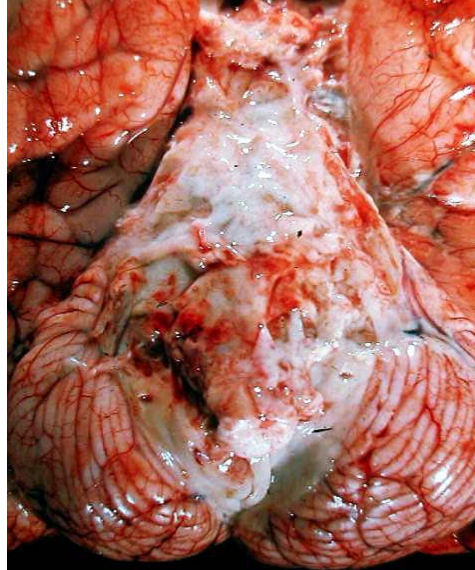
Reactivation: 5%

HIV → 20x

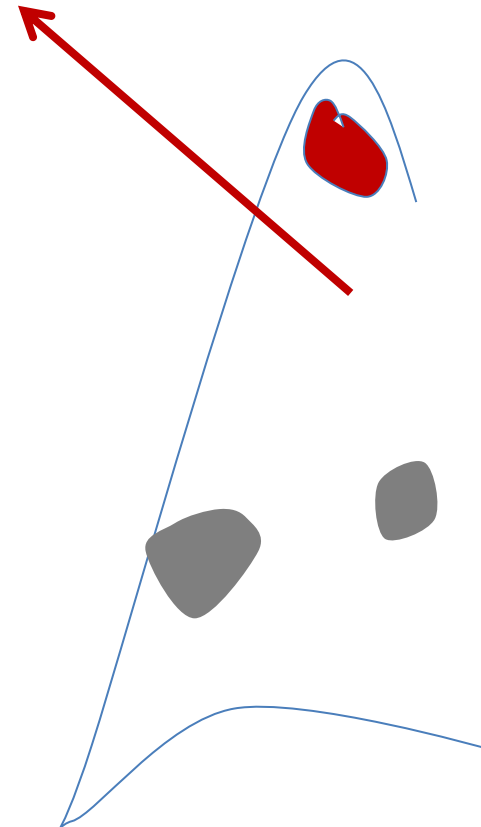
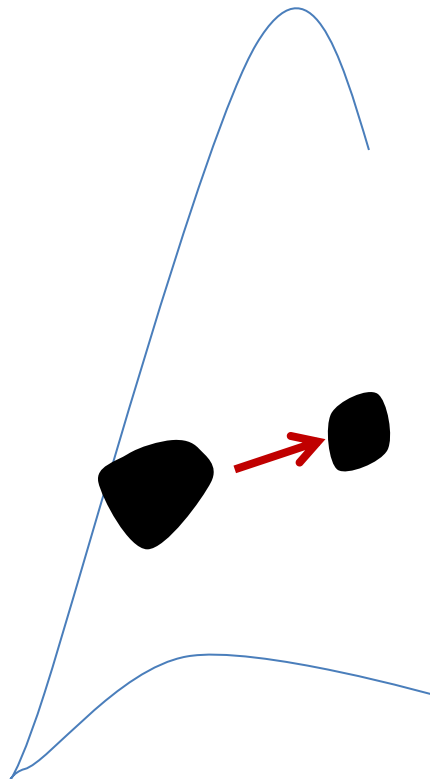
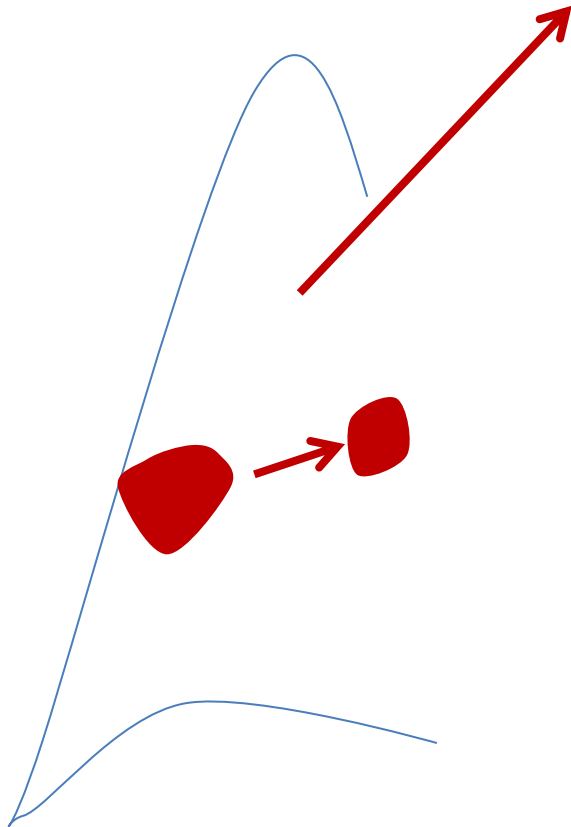
AIDS → 170X

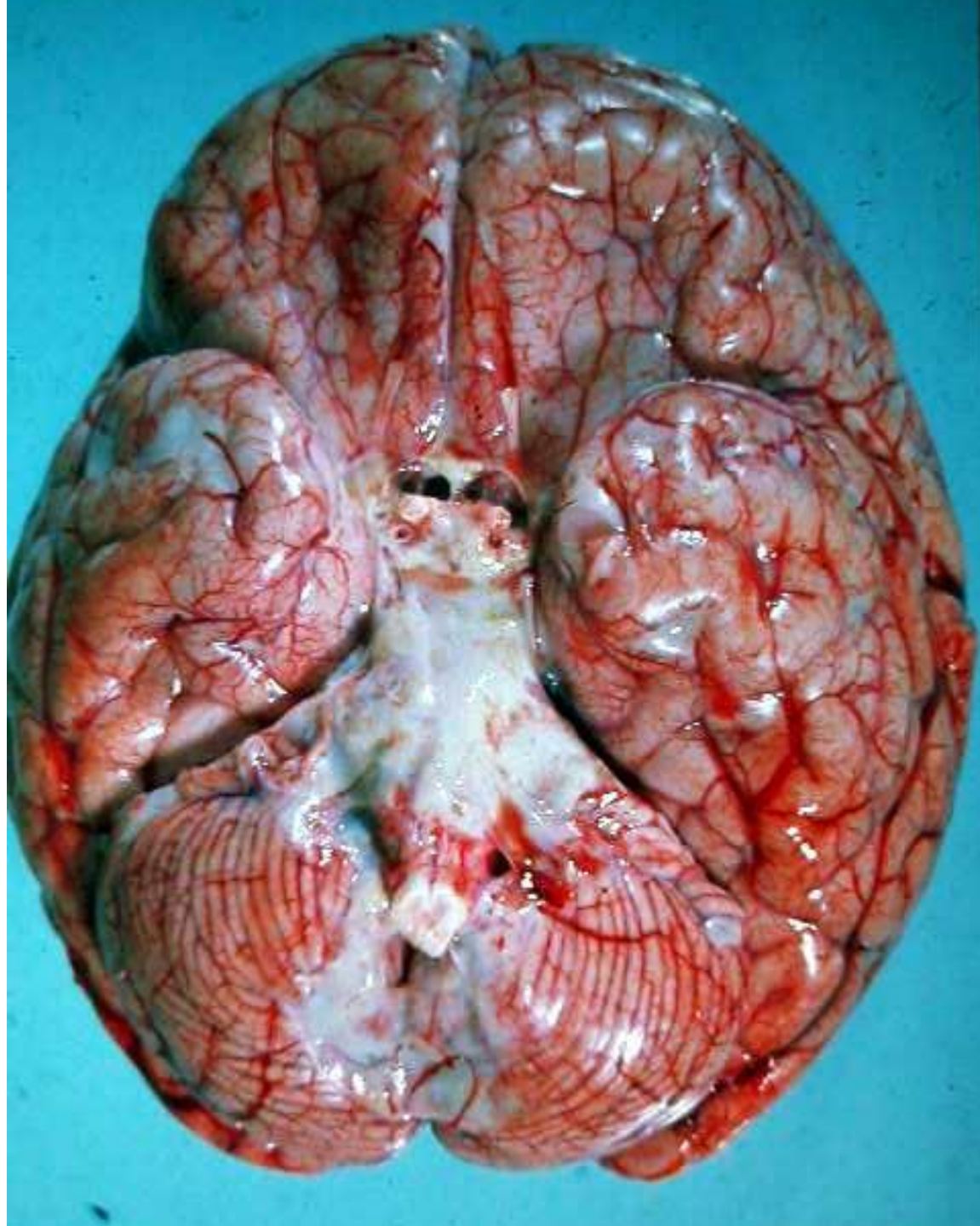


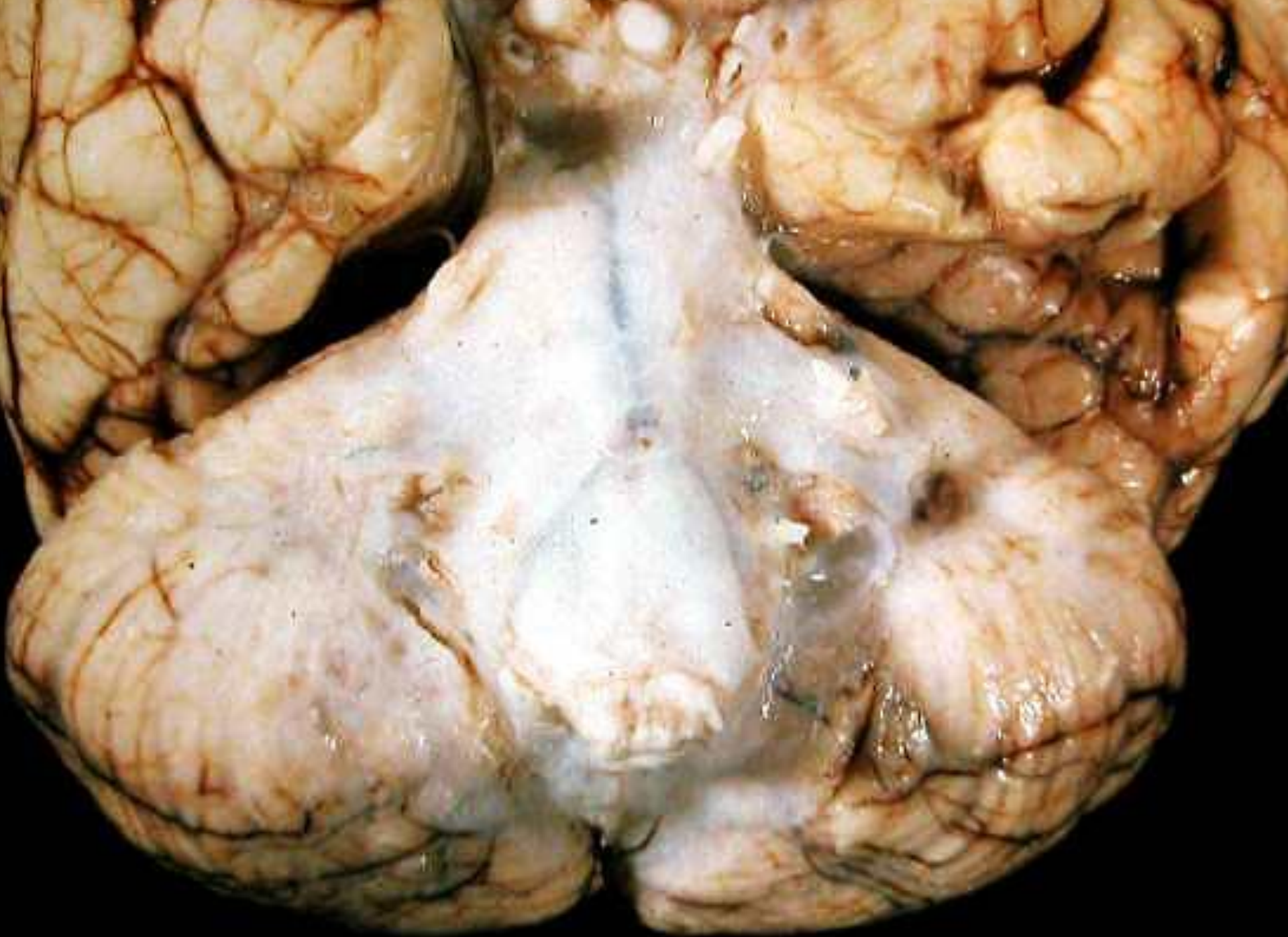
Primary
progression



reactivation
miliary
re-infection



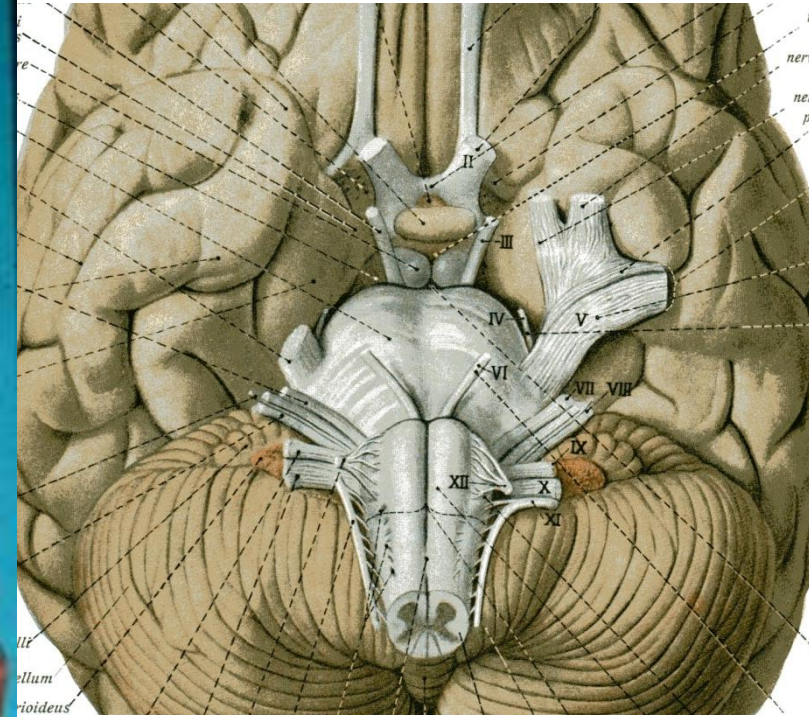
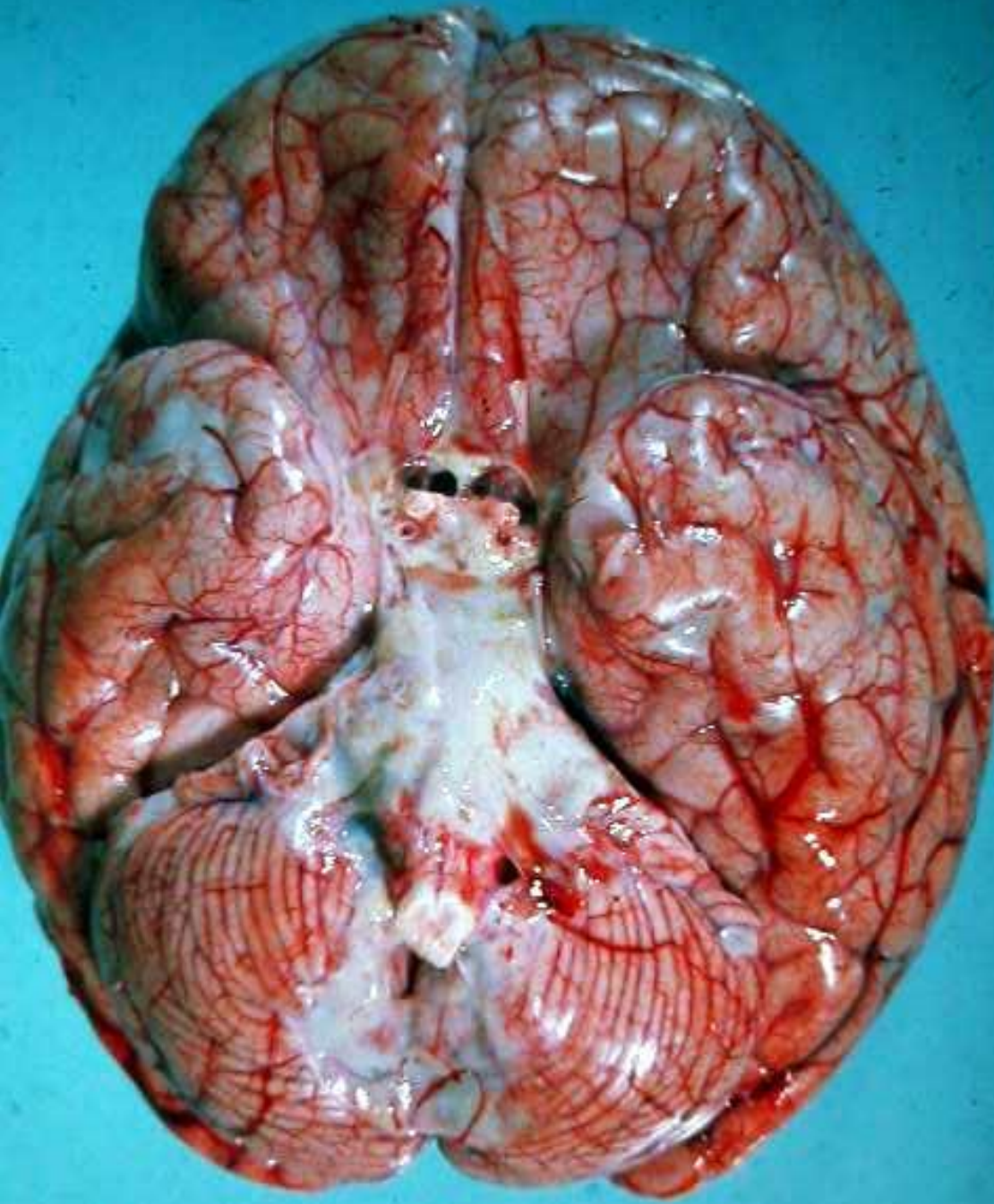




TB : basal meningitis



cranial nerves involved



visual loss optic
hearing loss VIII
double vision III, IV, VI

Normal

water clear

pressure 0 – 10 – 15 mmHg

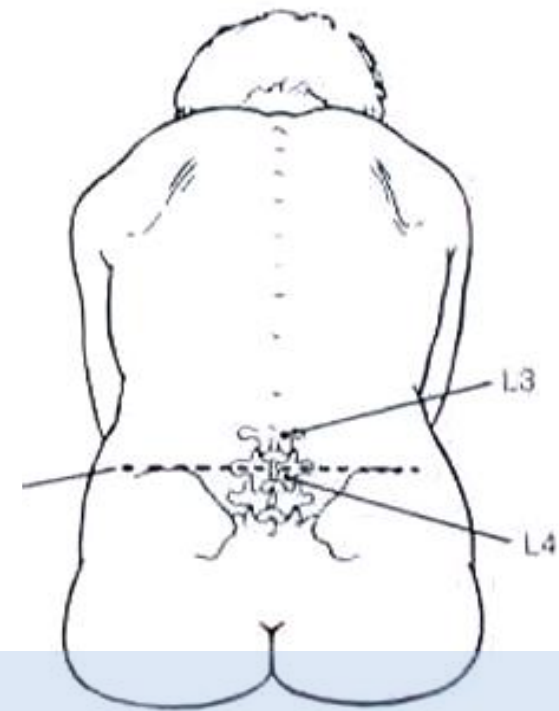
cells: few < 5 l

lymphocytes/mm³

protein small amount

glucose small amount

Spinal tap CSF



Meningitis, purulent

Cloudy, purulent

Pressure: ↑

Cells: up to 90'000
neutrophils

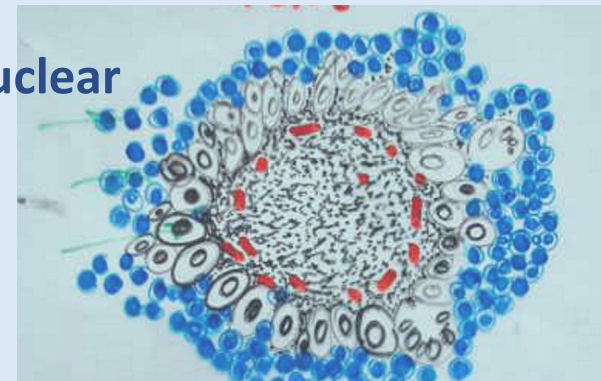
Protein ↑

Glucose ↓

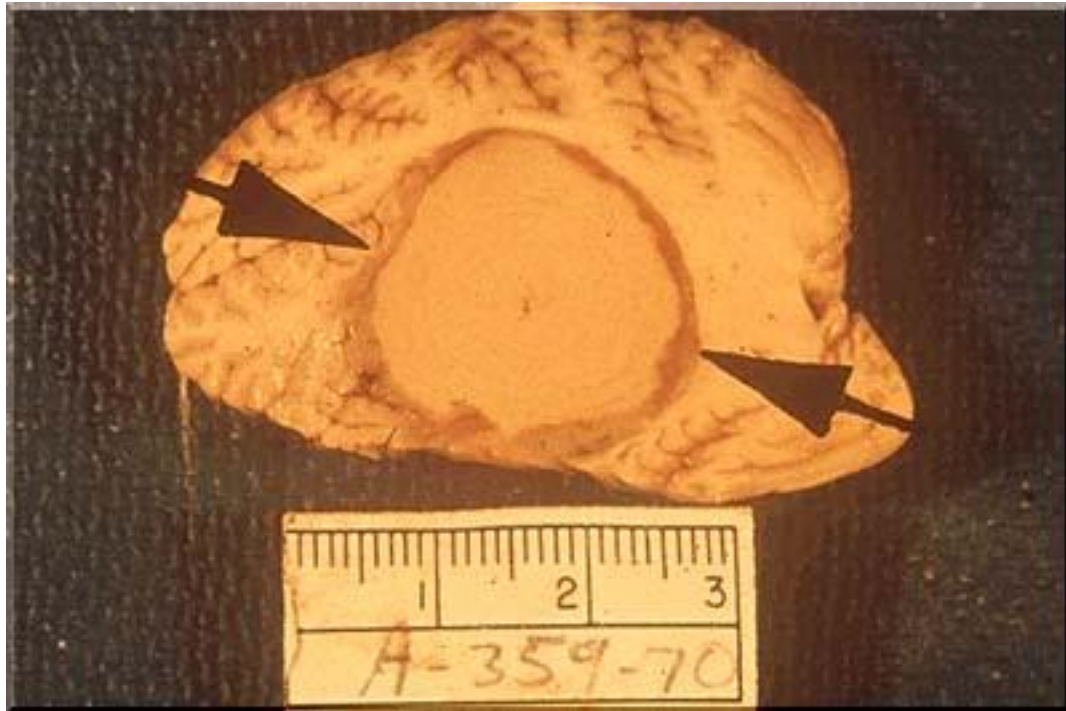
Tuberculosis

clear or slightly opaque

mainly mononuclear

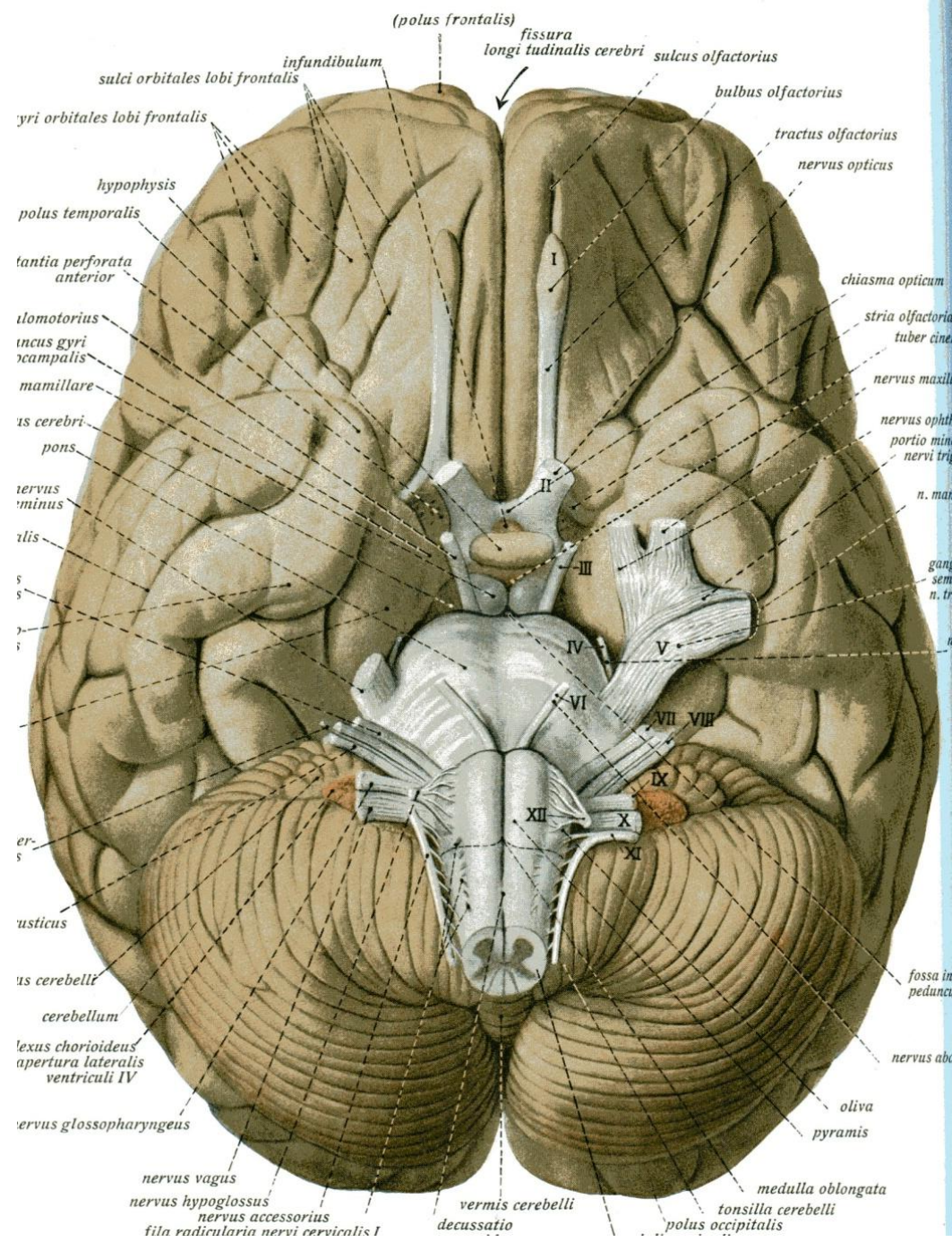
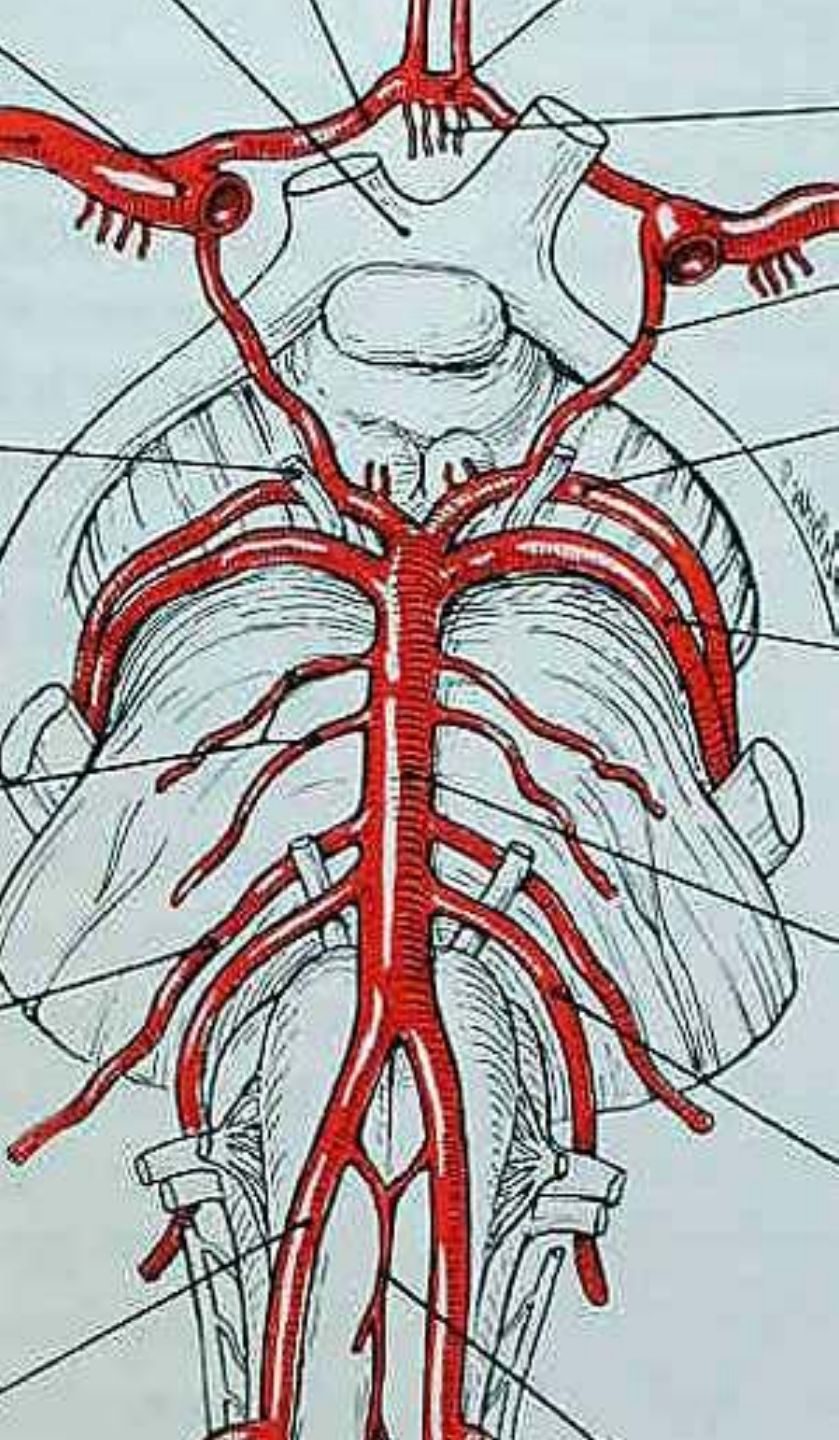


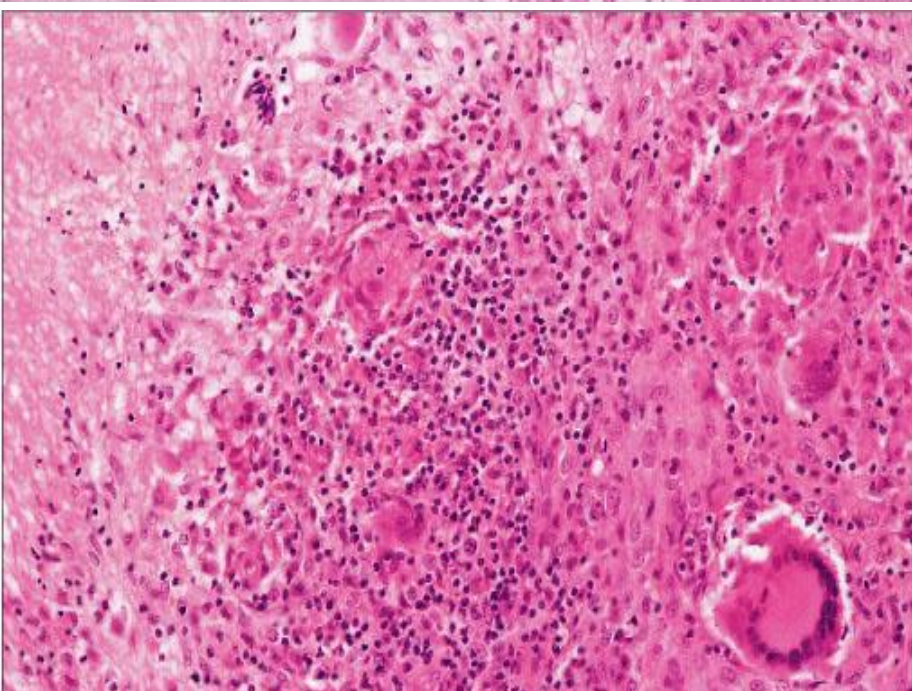
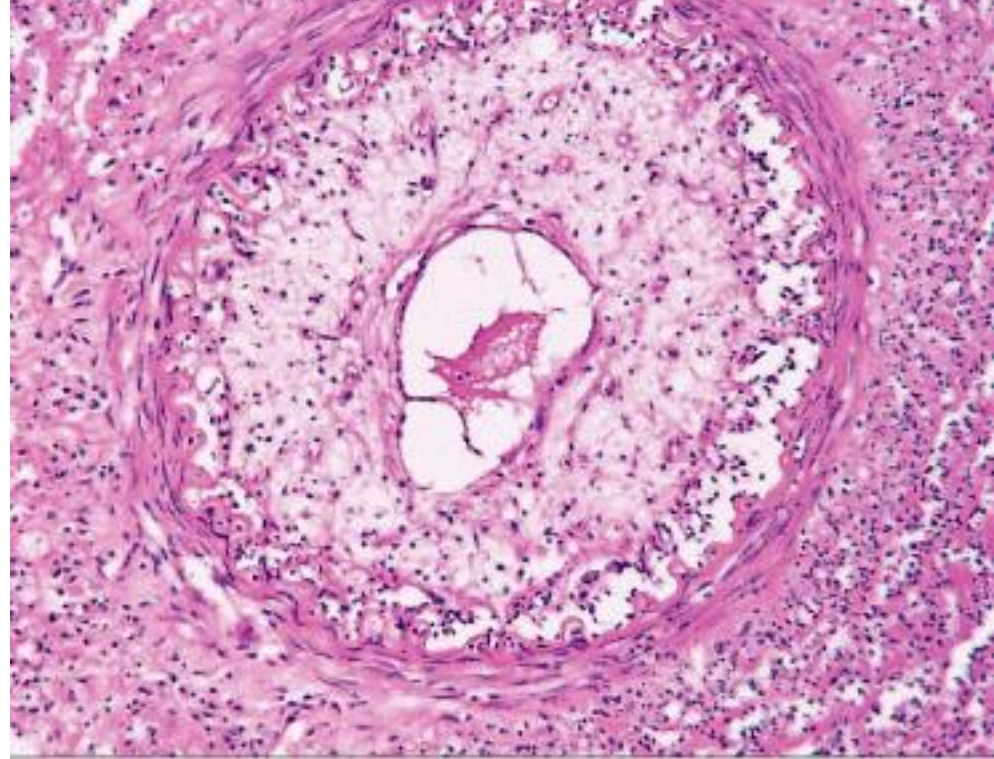
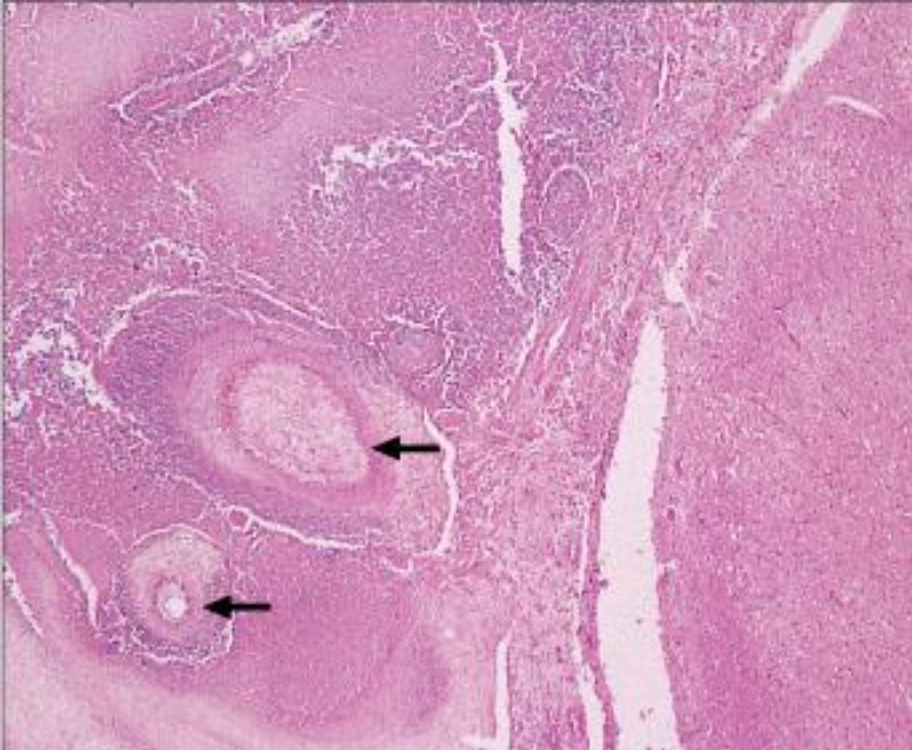
- Another manifestation of the disease is the development of a single (or often multiple) well circumscribed intraparenchymal mass (tuberculoma) associated with meningitis . Tuberculoma may be up to several cms in diameter causing mass effect



Clinical features

- The most serious complication of chronic tuberculous meningitis are arachnoid fibrosis, which may produce hydrocephalus, & obliterative endarteritis , which may produce infarction of underlying brain





Neurosyphilis

Tertiary syphilis

- It occurs in 10% of untreated patients

1. **Meningovascular neurosyphilis**

- Chronic meningitis involving the base of the brain & variably cerebrum, spinal leptomeninges.
- It may be associated with obliterative endarteritis.

2. **Paretic neurosyphilis**

- is caused by invasion of the brain by *Treponema pallidum* & characterized by insidious but progressive loss of mental & motor functions with mood alteration finally resulting in severe dementia (general paresis of insane).

3. **Tabes dorsalis**

- Damage by spirochetes to sensory nerves in the dorsal roots resulting in impaired joint position & sense resulting in ataxia, loss of pain sensation leading to skin & joint damage (charcot joint)



- Patients with HIV infection are at increased risk for neurosyphilis.

Viral encephalitis

- Parenchymal infection of the brain almost invariably associated with meningeal infection (meningoencephalitis)

Varicella Zoster virus

- Primary varicella infection presents as chickenpox in children without any evidence of neurological involvement but reactivation in adults (shingles) usually manifest as painful skin eruptions in single or limited dermatomal distribution.
- Herpes zoster reactivation is usually self limited process , but there may be persistent postherpetic neuralgia syndrome in 10% of patients



Cytomegalovirus

- Infection of the nervous system occurs in fetuses & immunosuppressed individuals.
- CMV is the most common opportunistic viral pathogen in patients with AIDS affecting the CNS in 15% to 20% of cases.

- CMV can affect any cell. Greater tendency to localize at subependymal, paraventricular area=> Ventriculoencephalitis=> Calcification
- Lower spinal Cord roots=> Painful Radiculoneuritis

Poliomyelitis

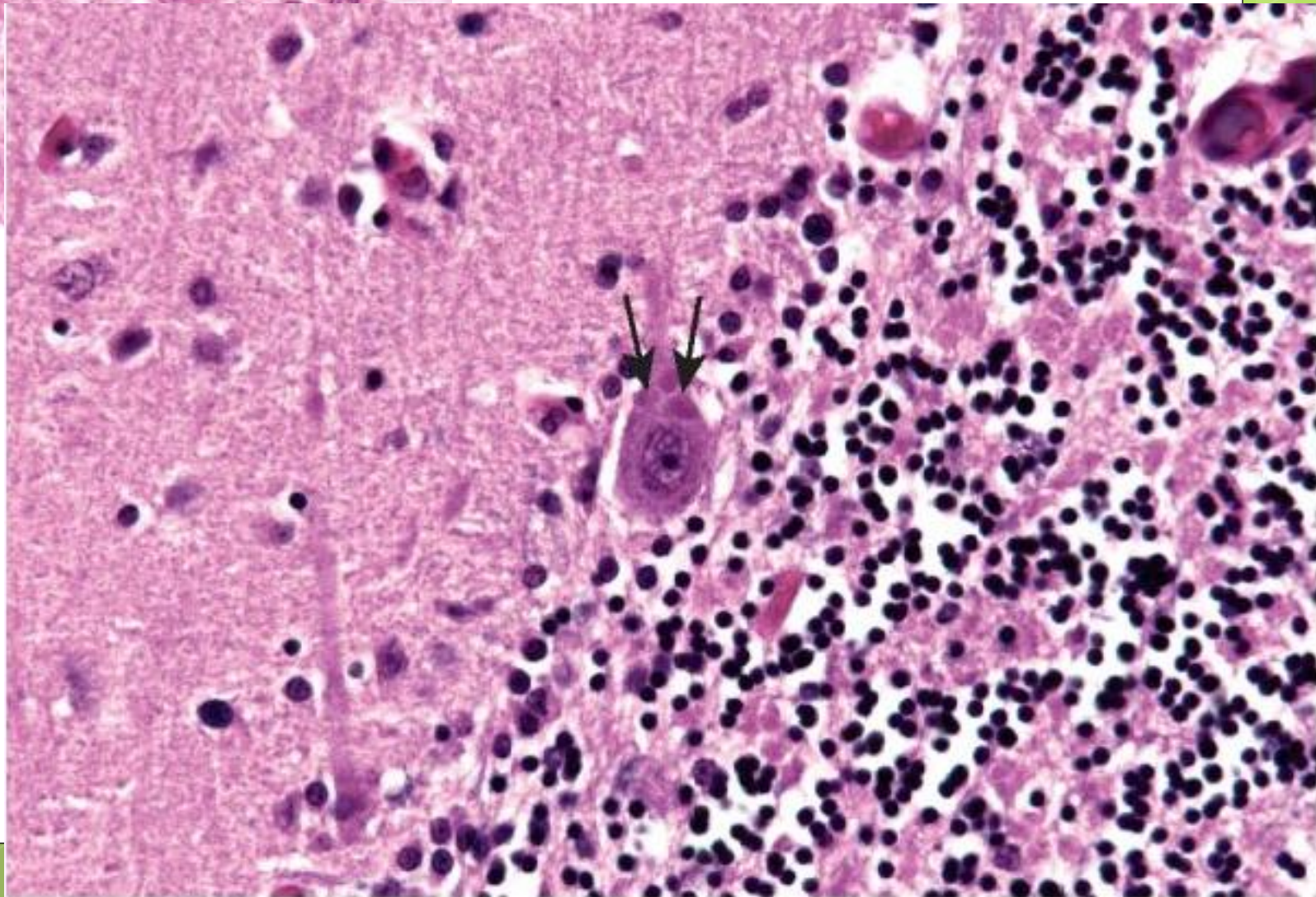
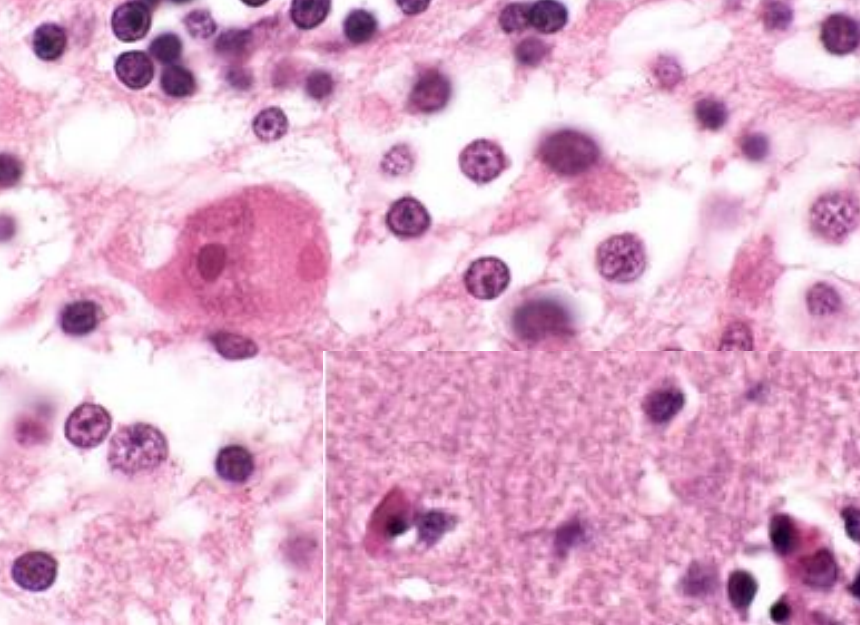
- Poliovirus is member of enteroviruses
- In non immunized individuals infections cause mild gastroenteritis & in small fraction of those , it invades the nervous system.
- CNS infection results in meningeal irritation & in some patients it progresses & affects the spinal cord with loss of **anterior horn motor neuron** , it produces flaccid paralysis with **muscle wasting & hyporeflexia** in the corresponding region of body—permanent sequel of poliomyelitis

Rabies

- Severe encephalitis transmitted to humans by bite of rabid animal (dogs, bats)
- The virus enters CNS by ascending along the peripheral nerves from wound site.
- Brain Stem, Basal ganglia, spinal cord, dorsal root ganglia.

Morphology

- Negri bodies, the pathognomonic microscopic finding, are cytoplasmic, round to oval, eosinophilic inclusions that can be found in pyramidal neurons of the hippocampus & Purkinje cells of the cerebellum



C/F

- The incubation period is usually b/n 1 & 3 months but it depends on the distance b/n the wound & the brain
- It manifests initially with non specific symptoms
- then **extraordinary CNS excitability occurs** .
- Contraction of the pharyngeal muscles on swallowing produces foaming at mouth & may create aversion to swallowing even to water (hydrophobia)
- There is meningismus, flaccid paralysis and also periods of mania & stupor progress to coma & death from respiratory center failure.

Human Immunodeficiency virus

- As many as 60% of patients with AIDS develop neurologic dysfunction during the course of their illness.

CNS and HIV

```
graph TD; A[CNS and HIV] --> B[HIV]; A --> C[HIV-associated [opportunistic]]; A --> D[CNS Lymphoma]
```

A flowchart with a central box at the top labeled 'CNS and HIV'. A vertical line descends from this box and connects to a horizontal line. From this horizontal line, three vertical lines descend to three separate boxes below: 'HIV', 'HIV-associated [opportunistic]', and 'CNS Lymphoma'.

HIV

HIV-associated
[opportunistic]

CNS Lymphoma

CNS and HIV

```
graph TD; A[CNS and HIV] --> B[HIV]; A --> C["HIV-associated [opportunistic]"]; A --> D[CNS Lymphoma]
```

A flowchart with a central box labeled 'CNS and HIV'. A horizontal line extends from the bottom of this box, with three vertical lines descending from it to three separate boxes below. The boxes are labeled 'HIV', 'HIV-associated [opportunistic]', and 'CNS Lymphoma' from left to right.

HIV

HIV-associated
[opportunistic]

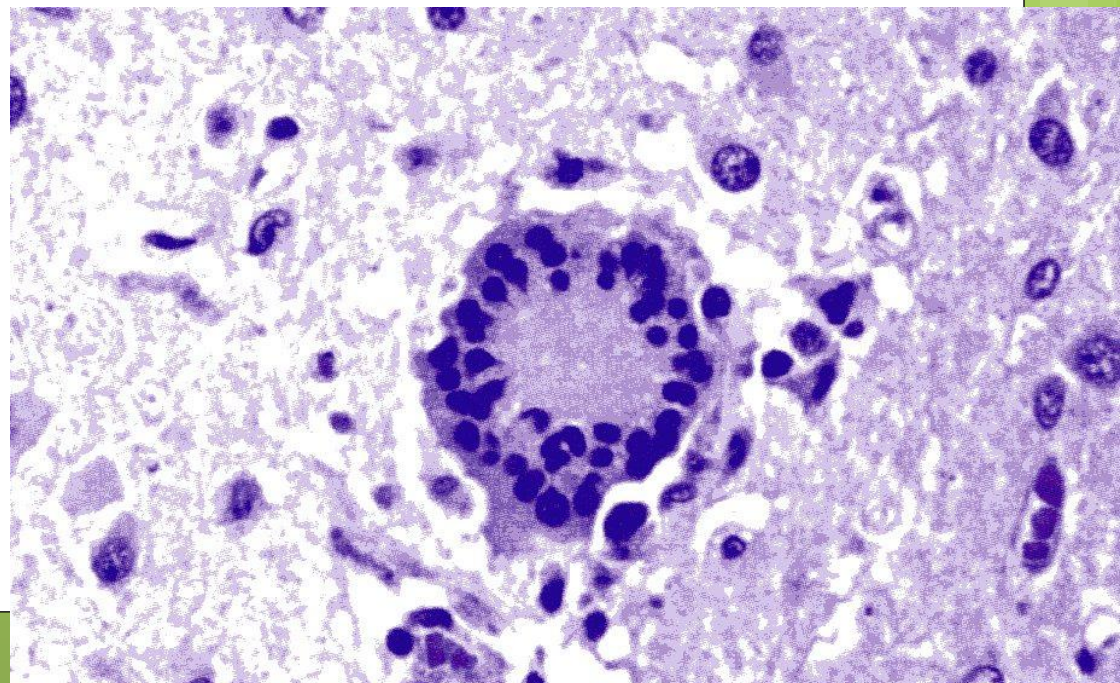
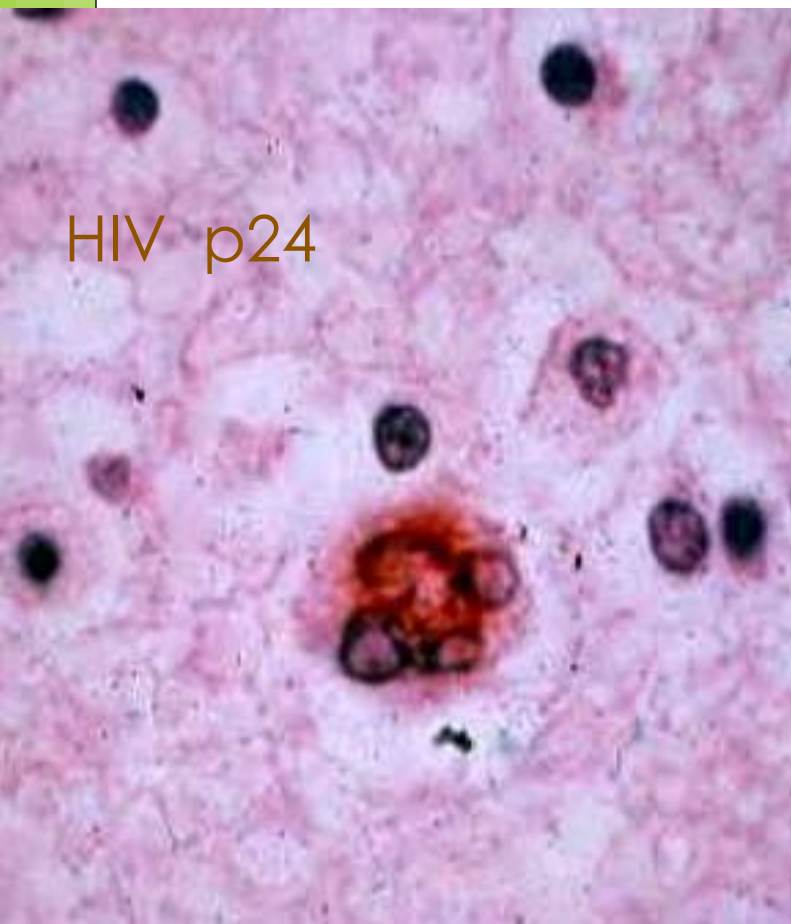
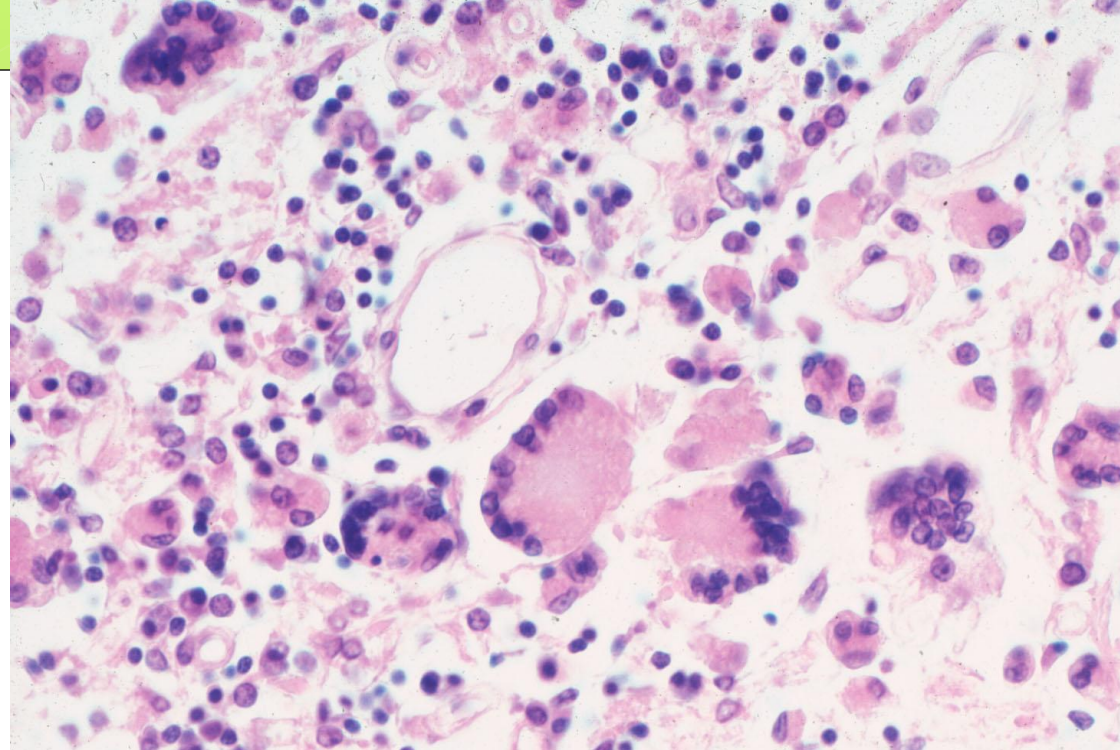
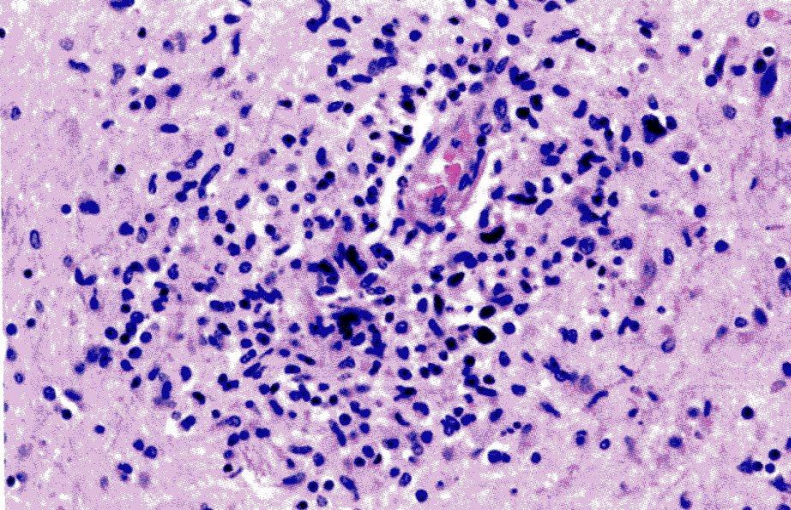
CNS Lymphoma

**HIV
Meningoencephalitis**

- ✓ HIV aseptic meningitis occurs in 1 to 2 wks of seroconversion in about 10% of patients

- ✓ HIV meningoencephalitis (subacute encephalitis)
- Patients with this disorder can manifest clinically with dementia referred to as AIDS dementia complex
- The dementia begins insidiously with mental slowing , memory loss, & mood disturbances such as apathy & depression .Motor abnormalities, ataxia, bladder & bowel incontinence & seizure can also be present

- It is likely that neurons & oligodendrocytes are not directly infected by HIV & damage to these occurs indirectly through the release of cytokines & alterations of the blood brain barrier
- Morphology- widespread chronic inflammation with microglial nodules.



- ✓ AIDS in children
 - These include microcephaly with mental retardation & motor developmental delay with spasticity of limbs

Progressive multifocal leukoencephalopathy

- PML is a viral encephalitis caused by the JC polyomavirus preferentially infects oligodendrocytes . Demyelination is the principal pathologic effect
- It occurs invariably in immunocompromised individuals such as AIDS

Poliomavirus (JC virus)

acquired: oral or respiratory (>75% of adults)

latent or low-grade persistent infection in:

- B-lymphocytes
- kidney
- probably CNS

usually entirely benign

Poliomavirus (JC virus)

immuno - deficiency



hematogenous dissemination → CNS



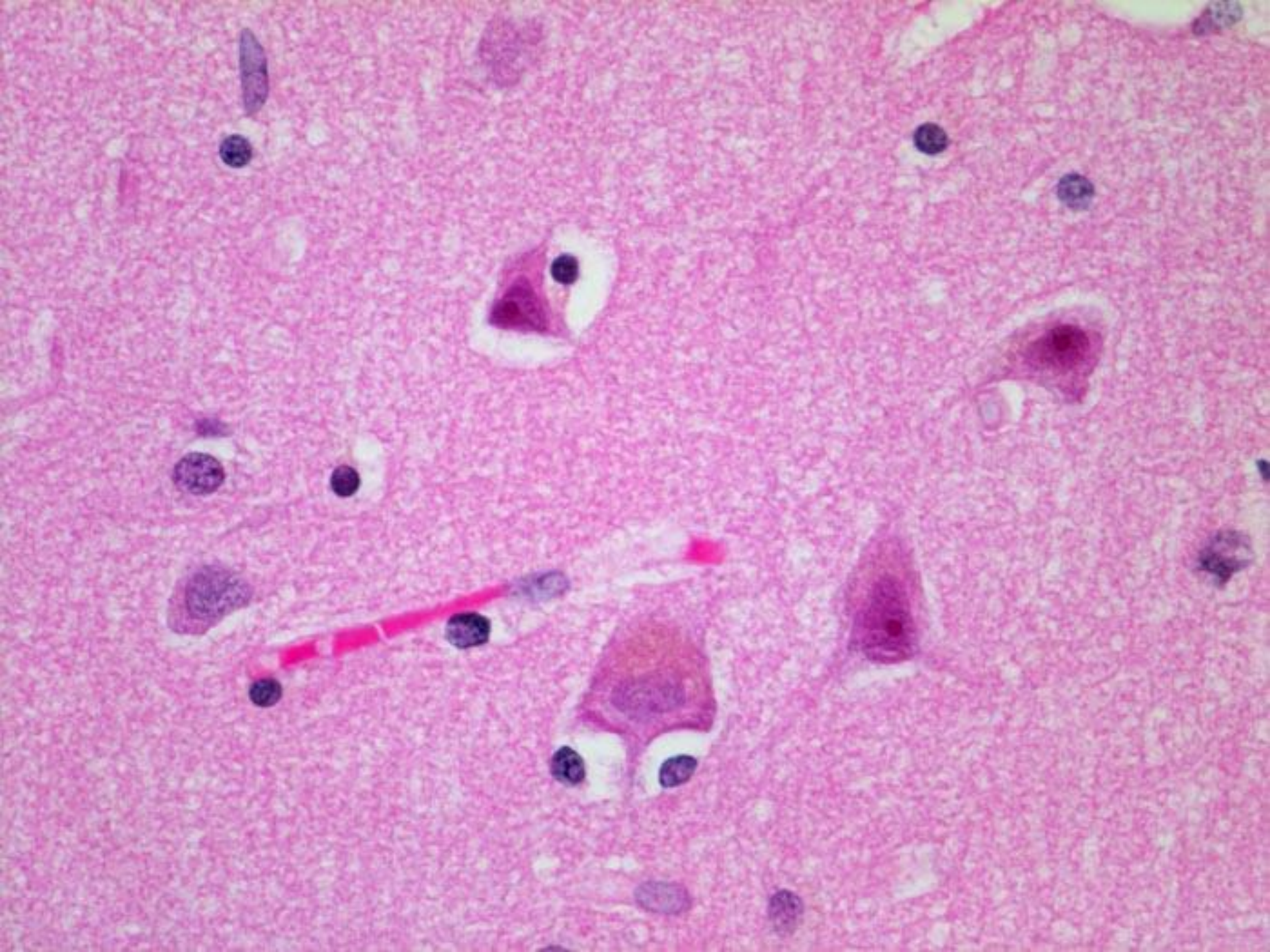
oligodendrocyte: productive infection with JC virus

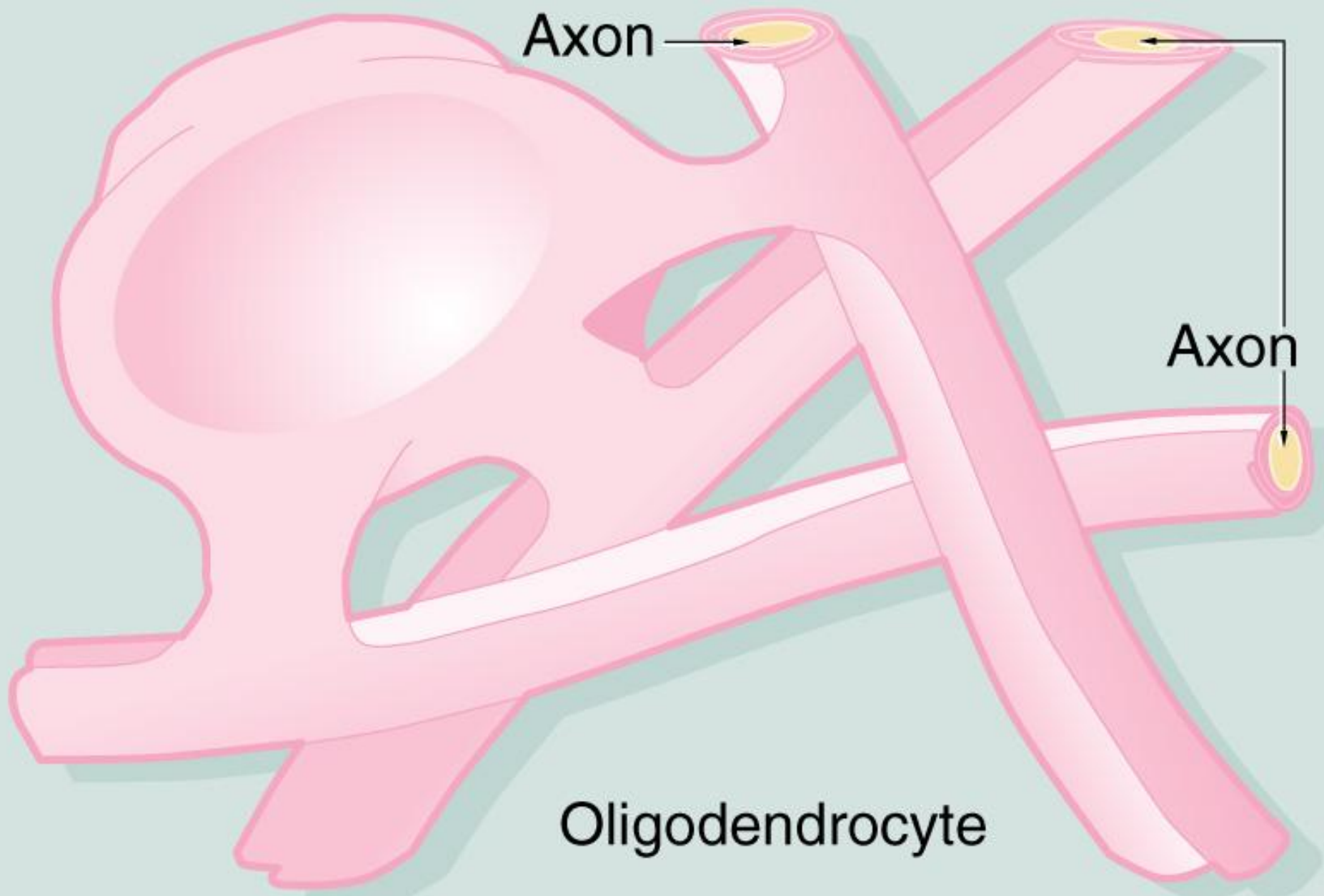


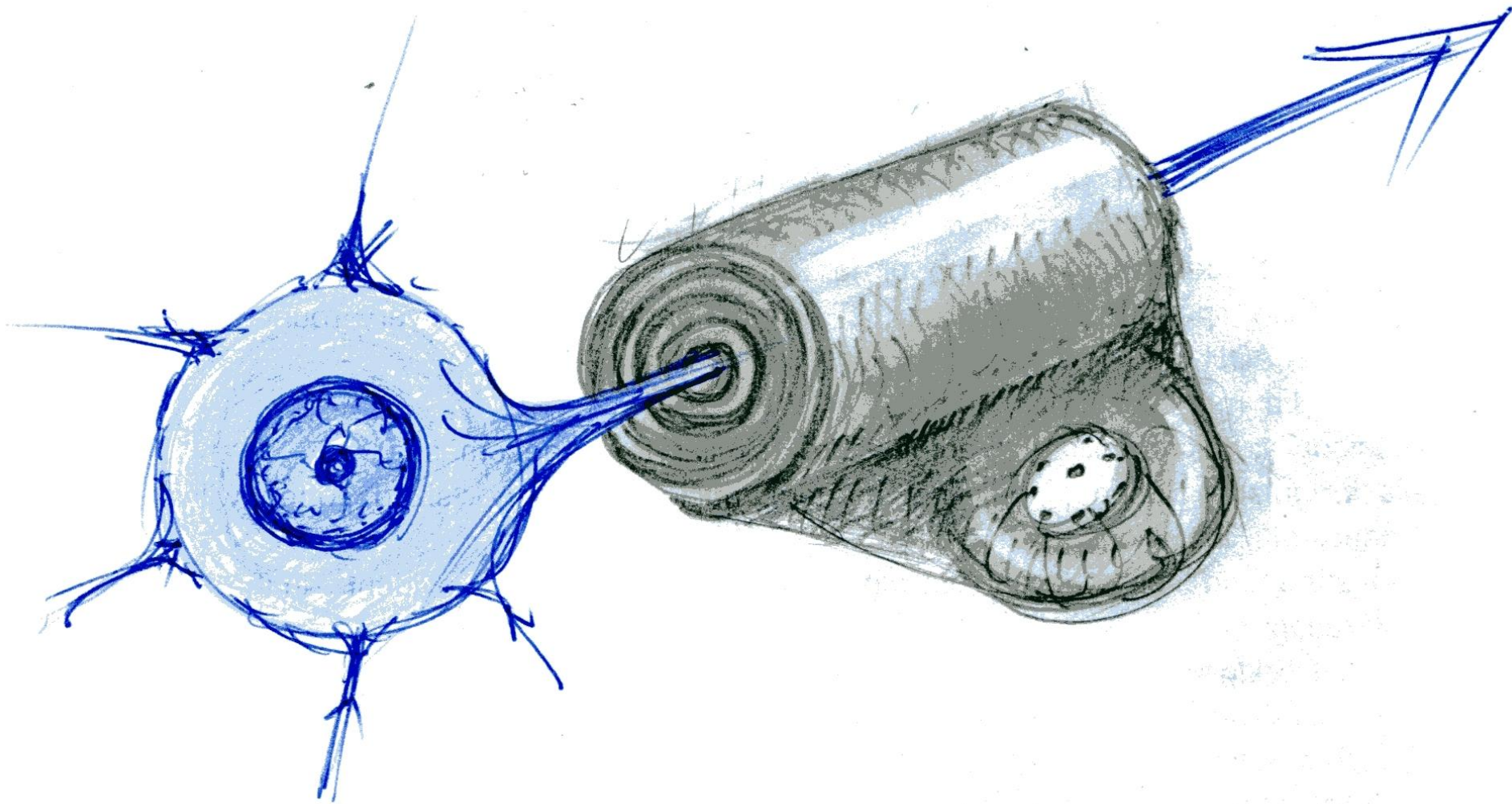
interaction with specific glial transcription factor

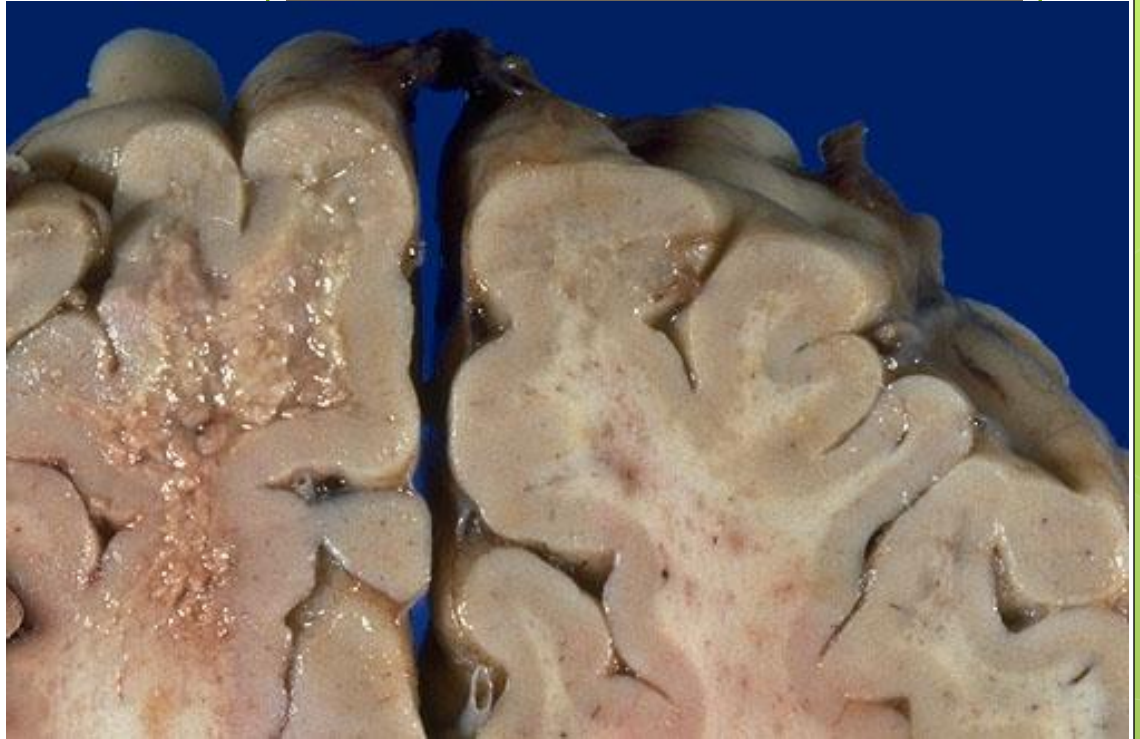
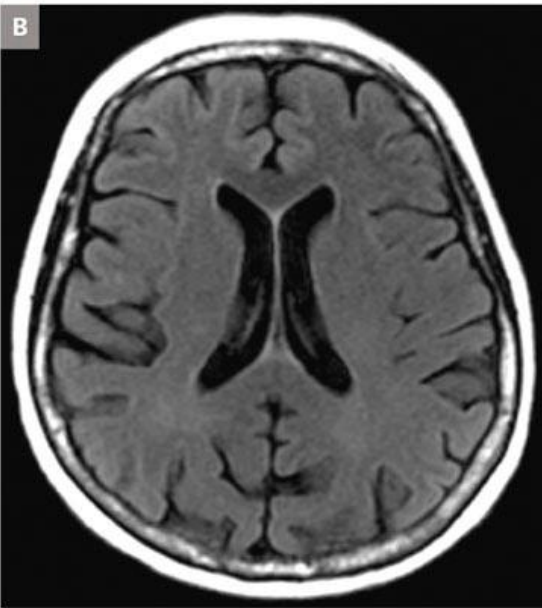
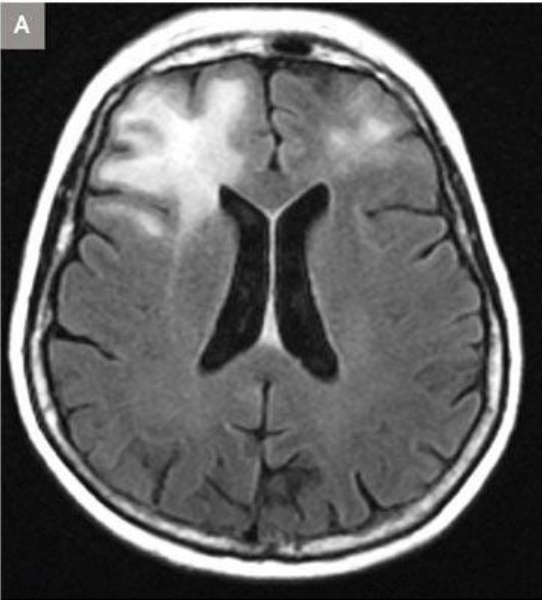


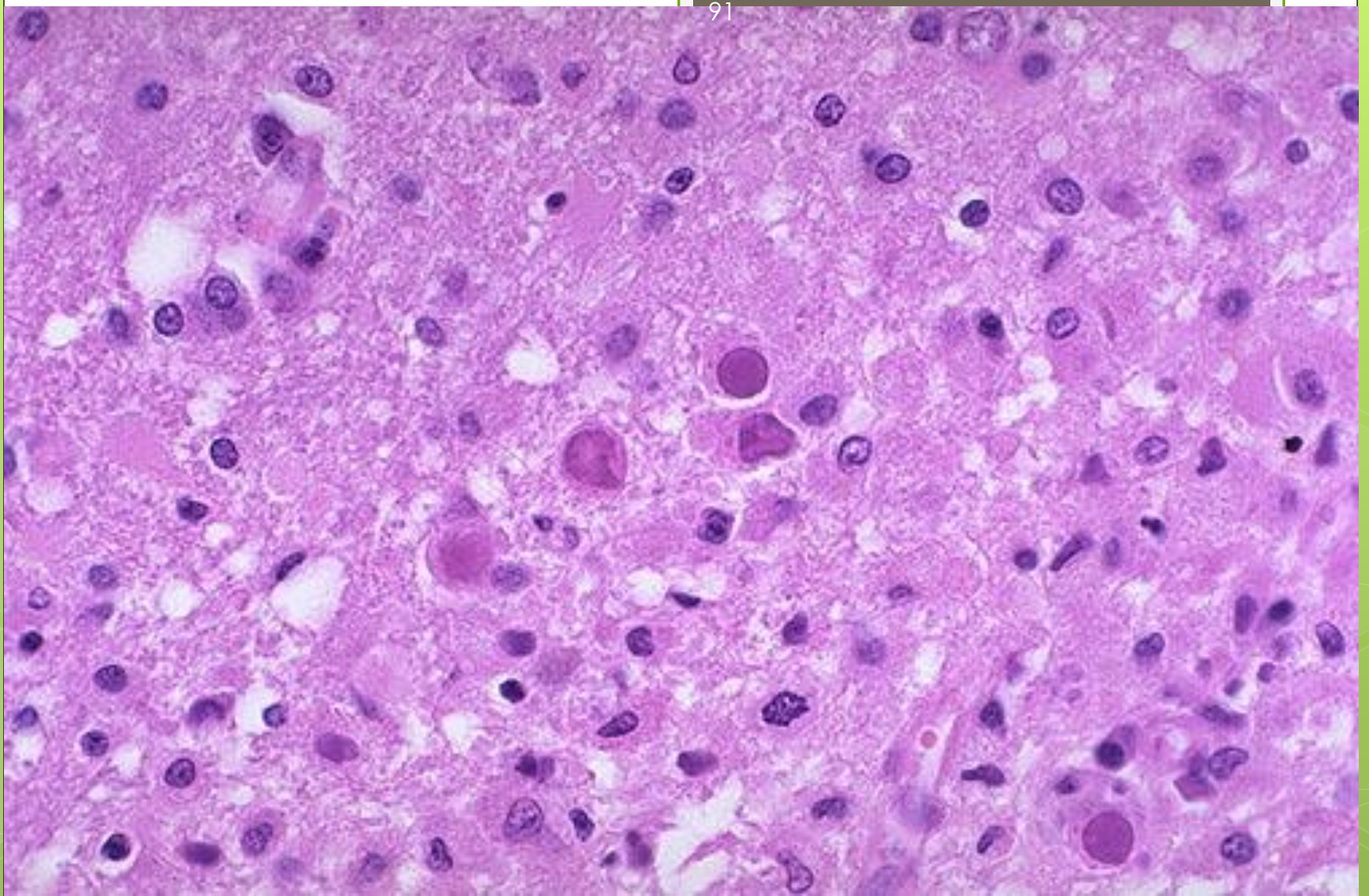
demyelination → neurological deficits (insidious)

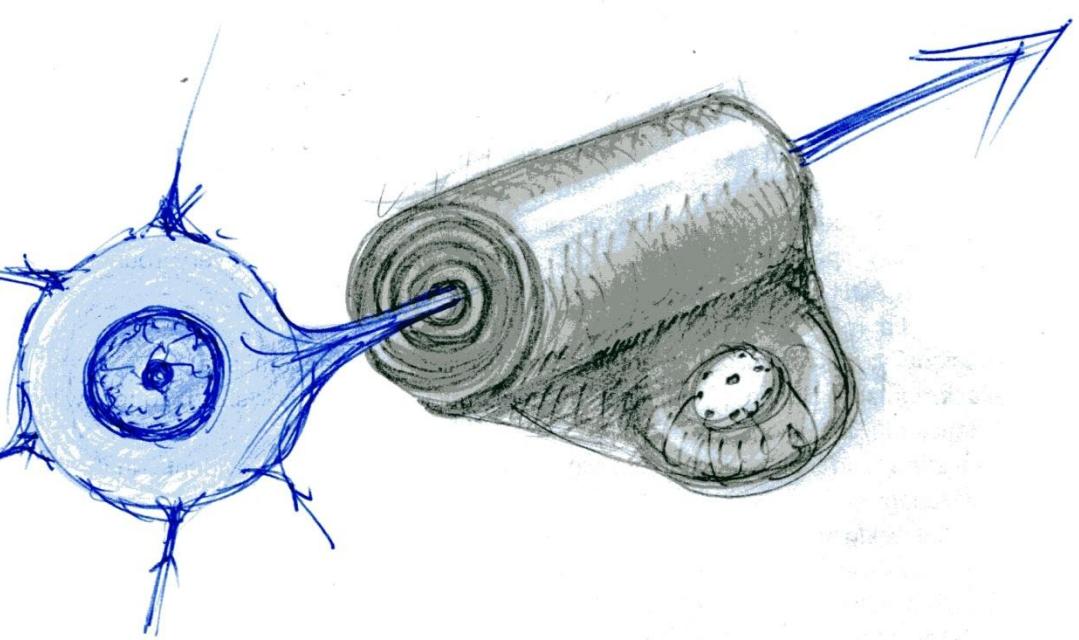




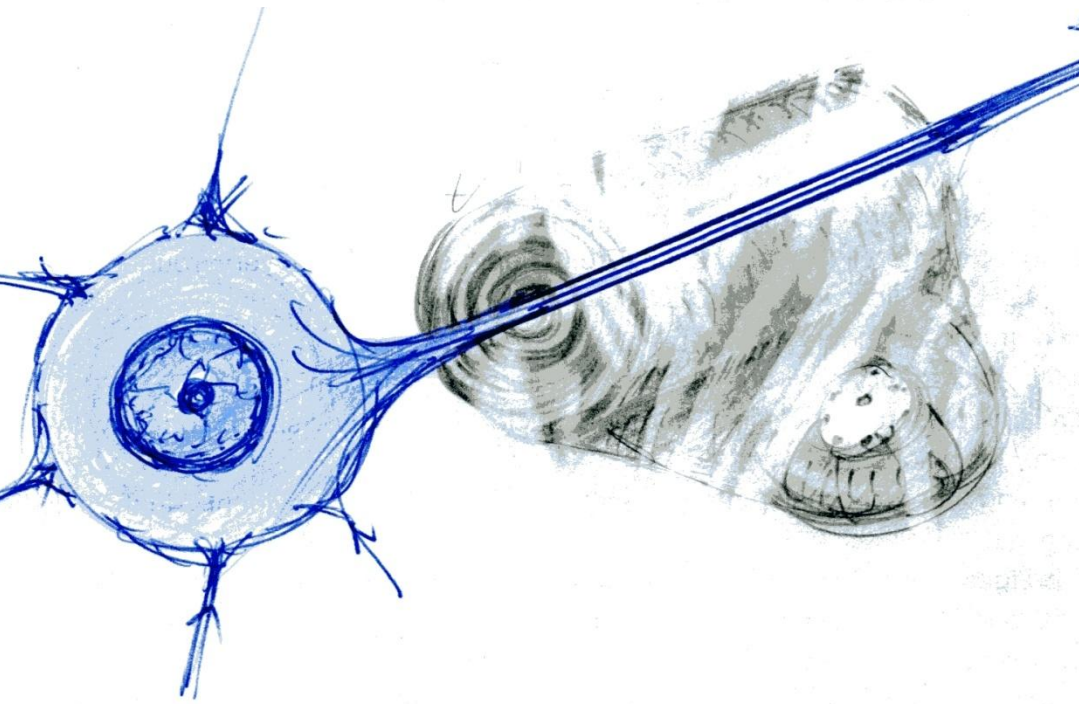
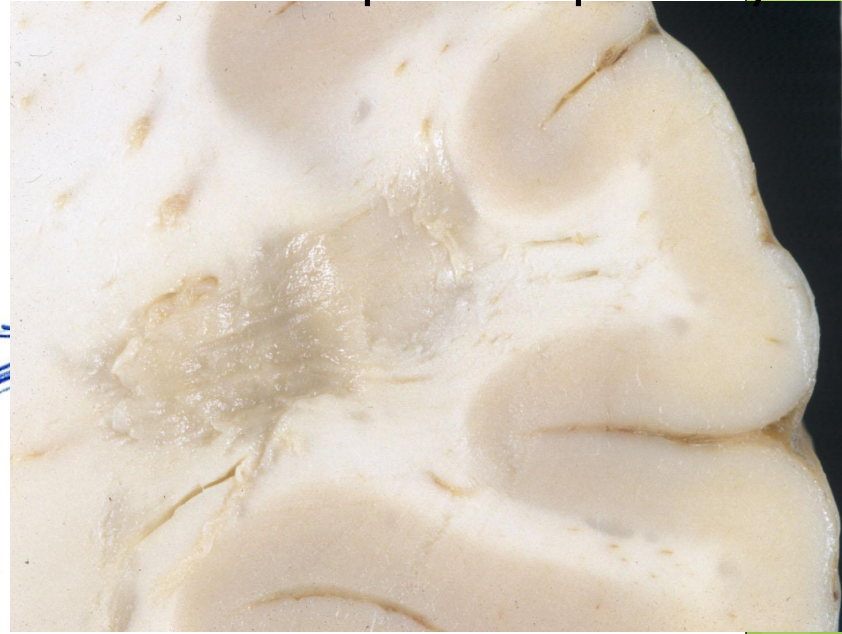








progressive
multifocal
leukencephalopathy



poliomavirus JC virus

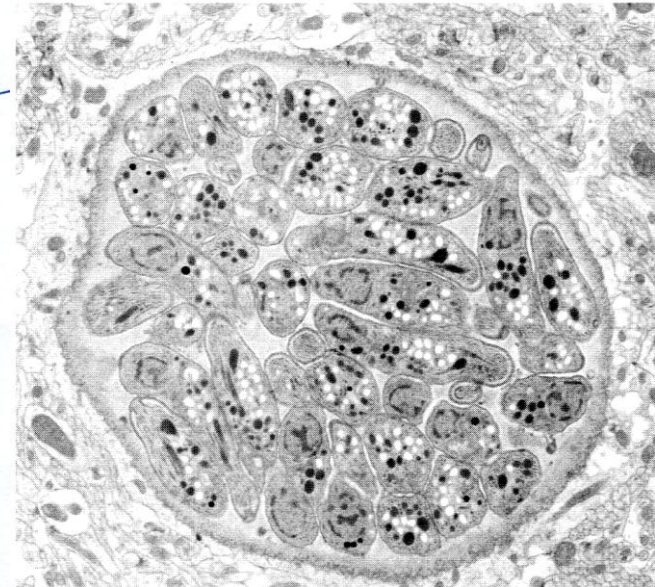
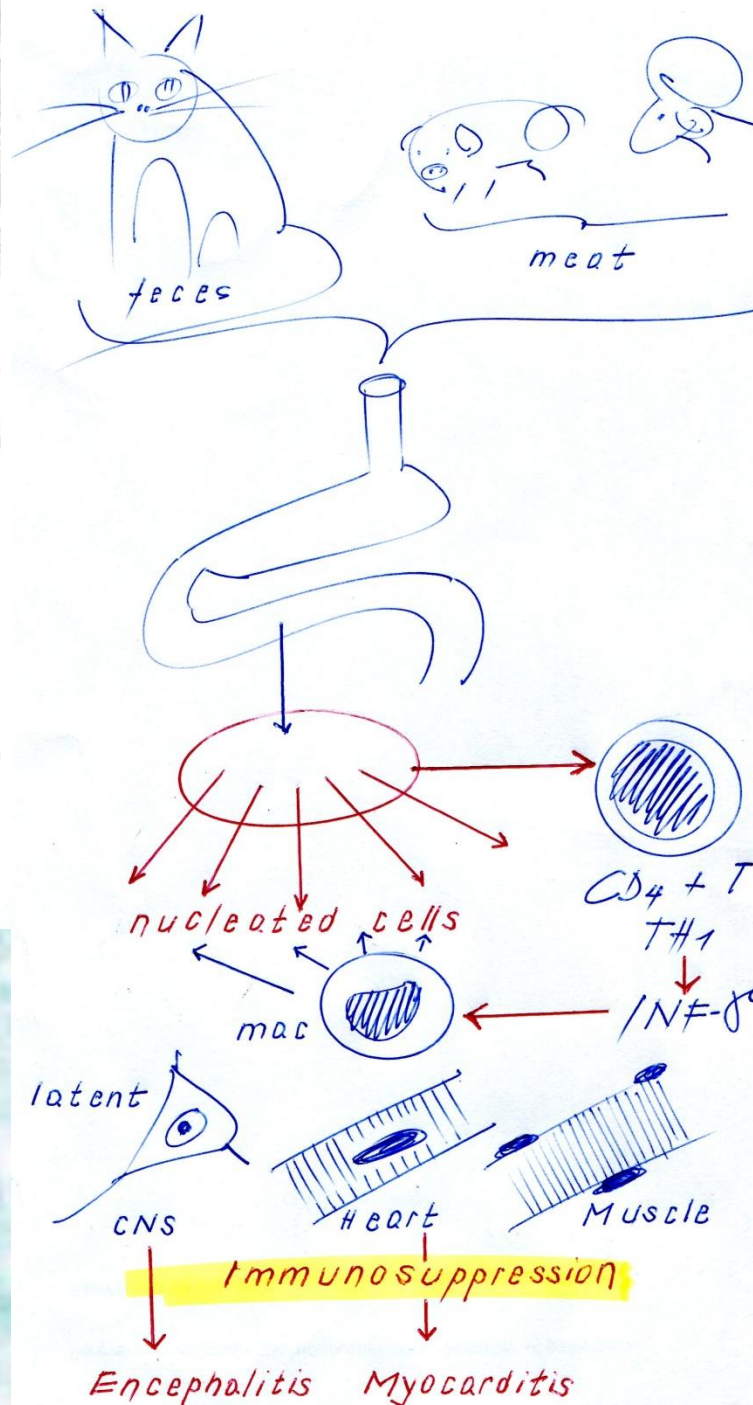
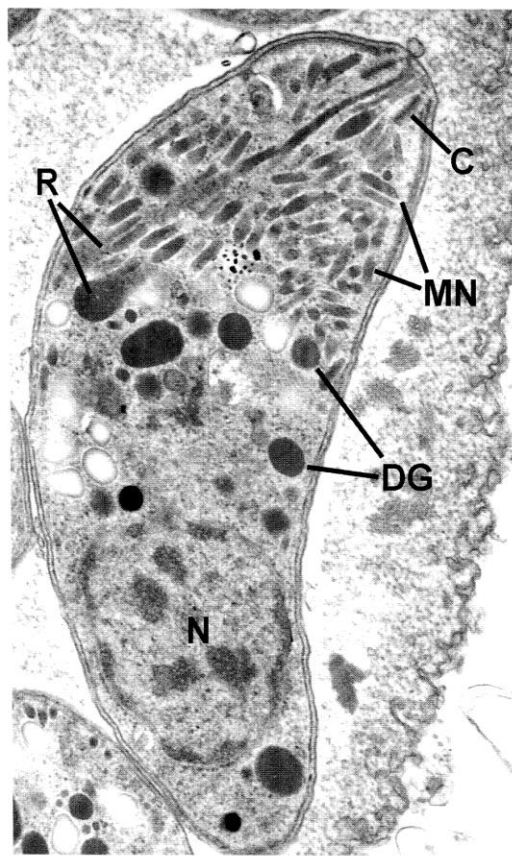
demyelination
relative preservation of
axon

Protozoal

Cerebral toxoplasmosis

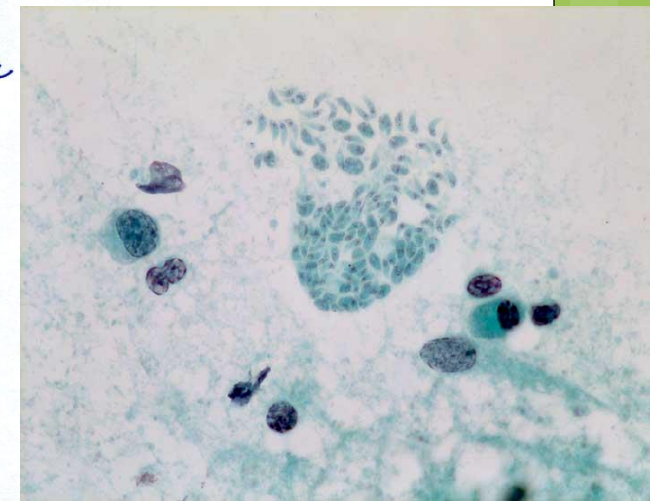
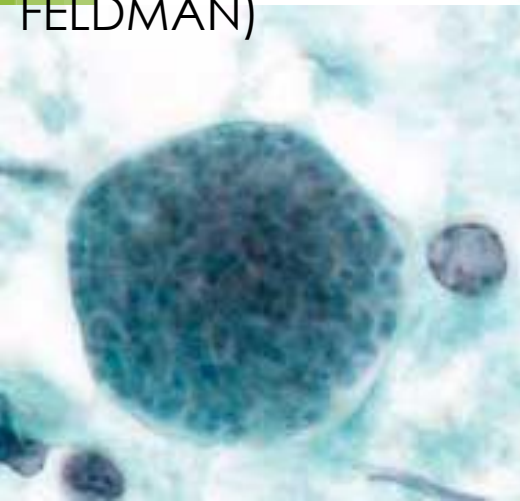
- Sub acute(1-2weeks)
- Focal or diffuse
- Site- Cerebral Cortex(near grey-white junction.
 - Cerebellum and brain stem- seldom

Toxoplasma Gondii



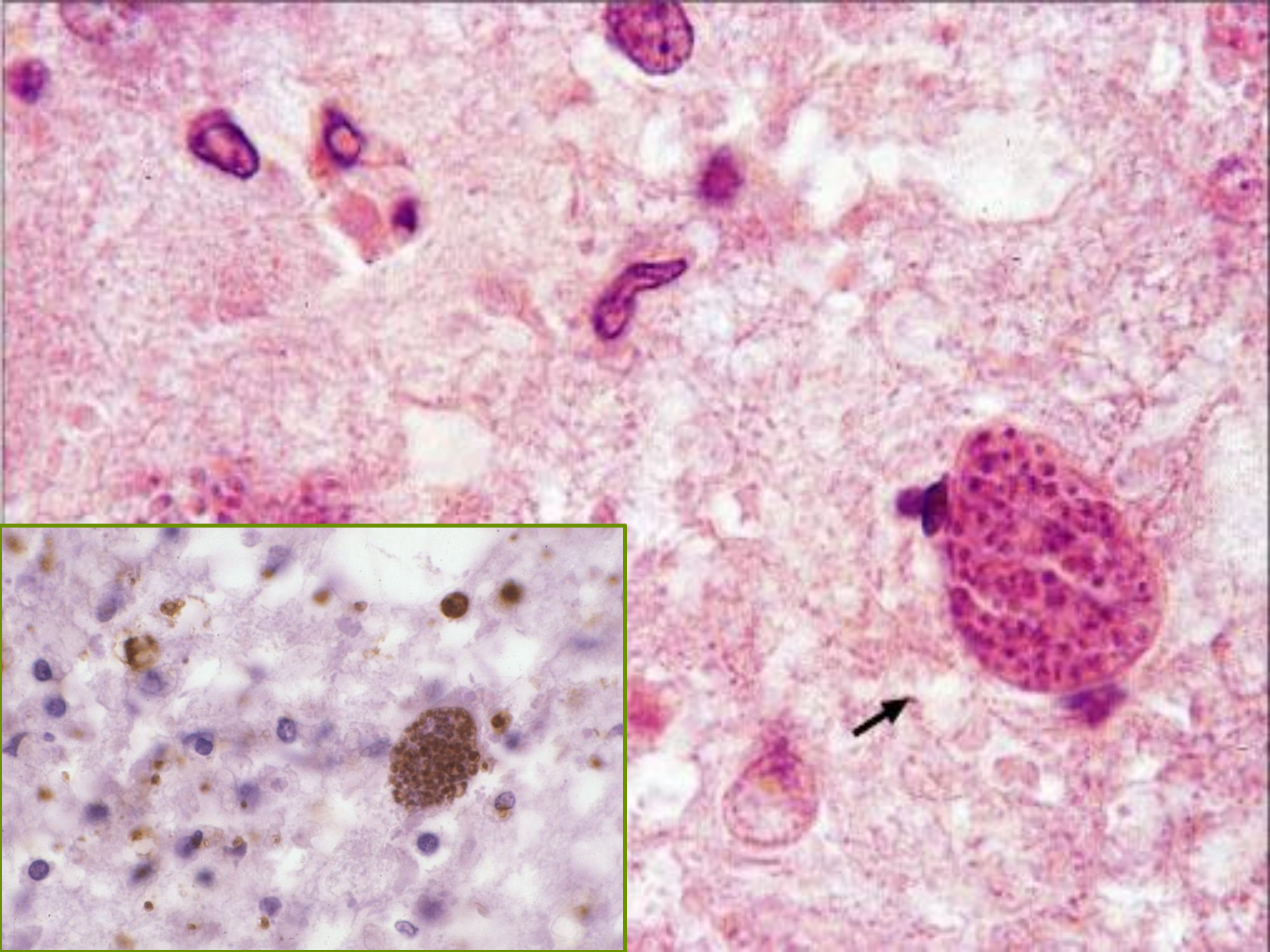
humoral (SABIN-FELDMAN)

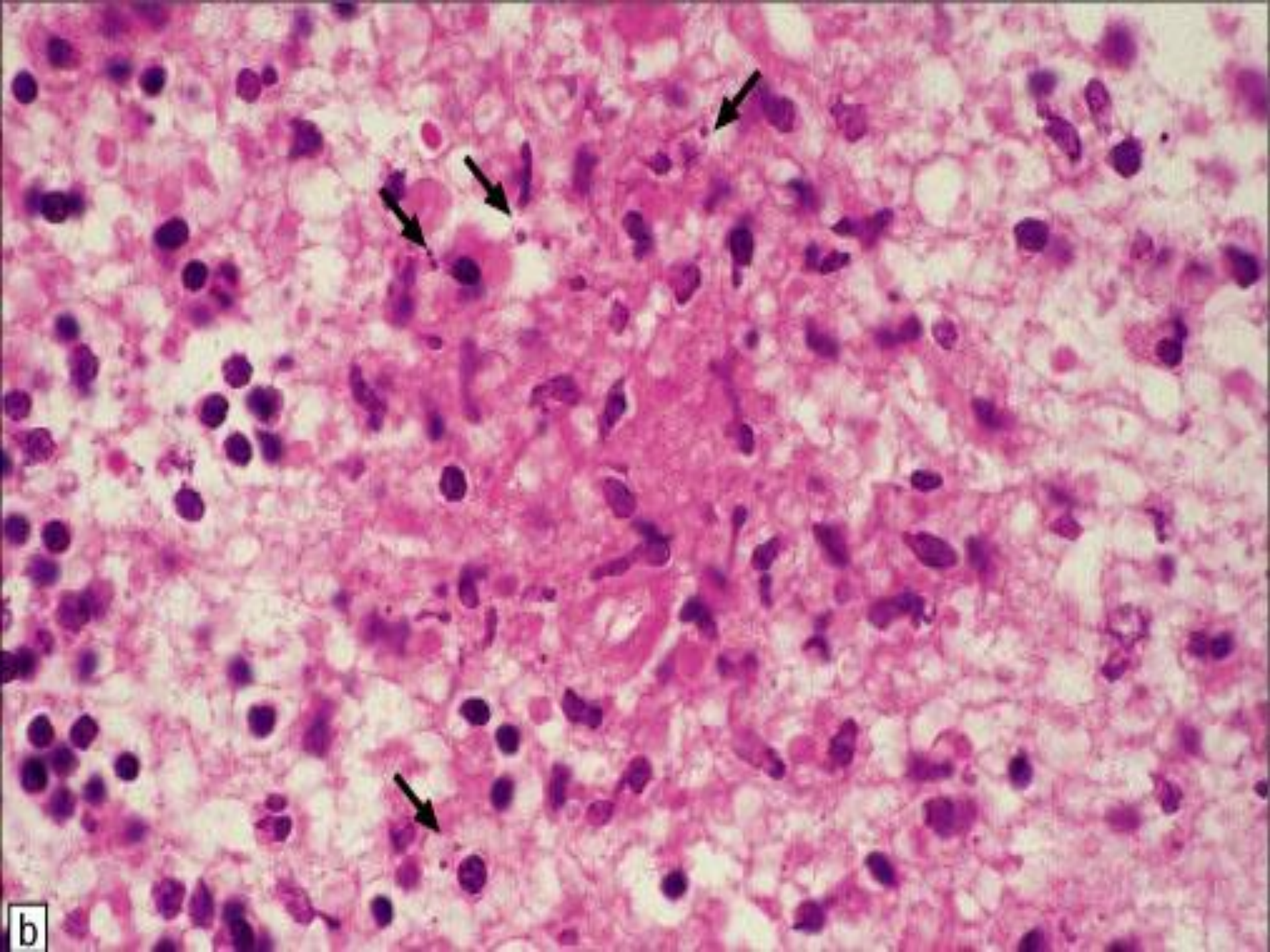
cellular



Morphology

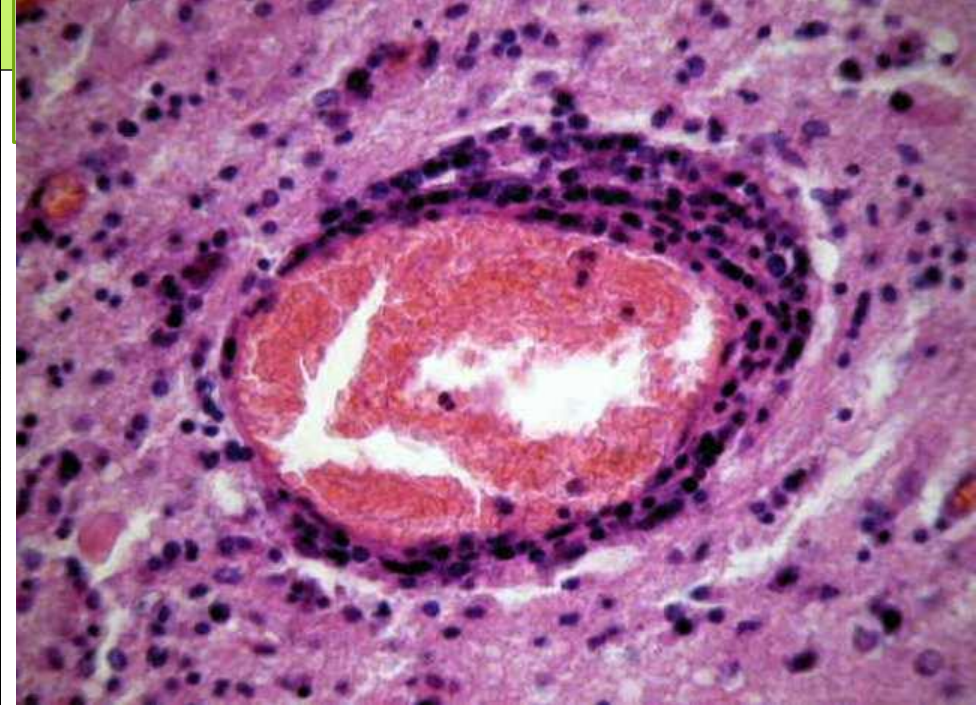
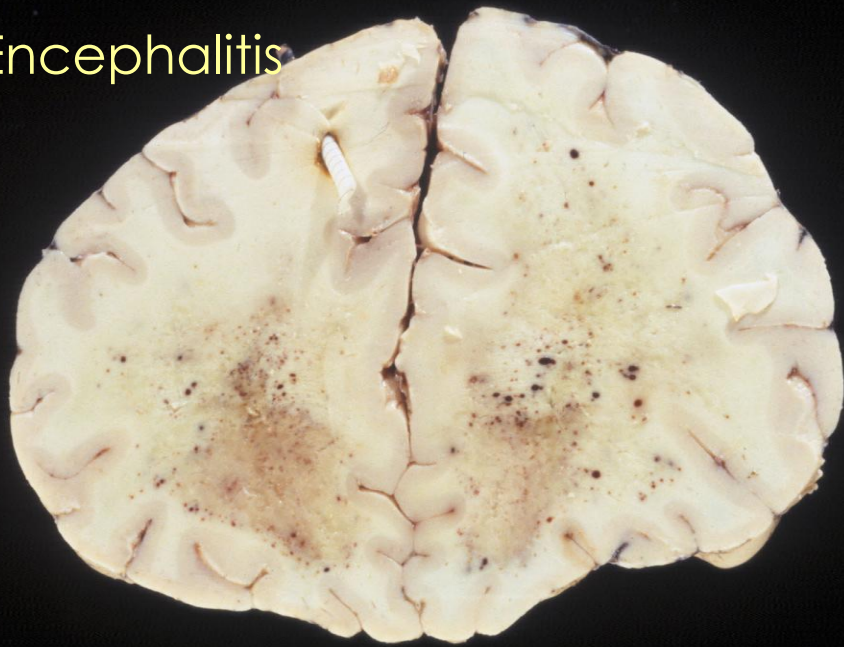
- ◉ Foci of necrosis
 - ◉ With free tachyzoites and encysted bradyzoites at the periphery
- ◉ Acute and chronic inflammation
- ◉ Vascular proliferation
 - ◉ Intimal proliferation
 - ◉ Vasculitis with sometimes fibrinoid necrosis and thrombosis



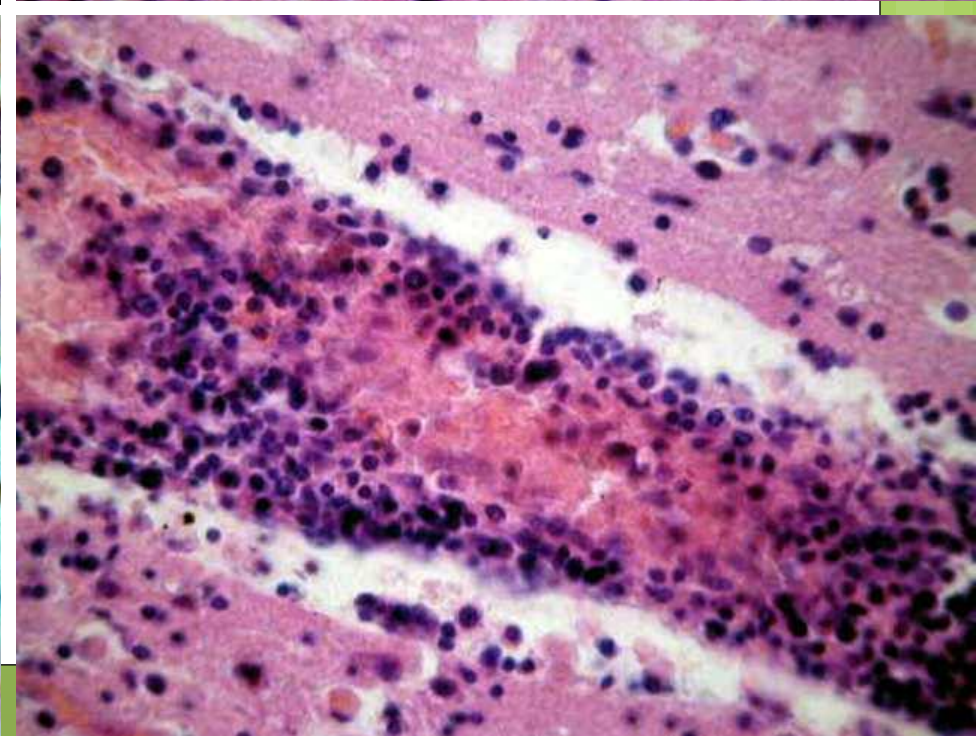
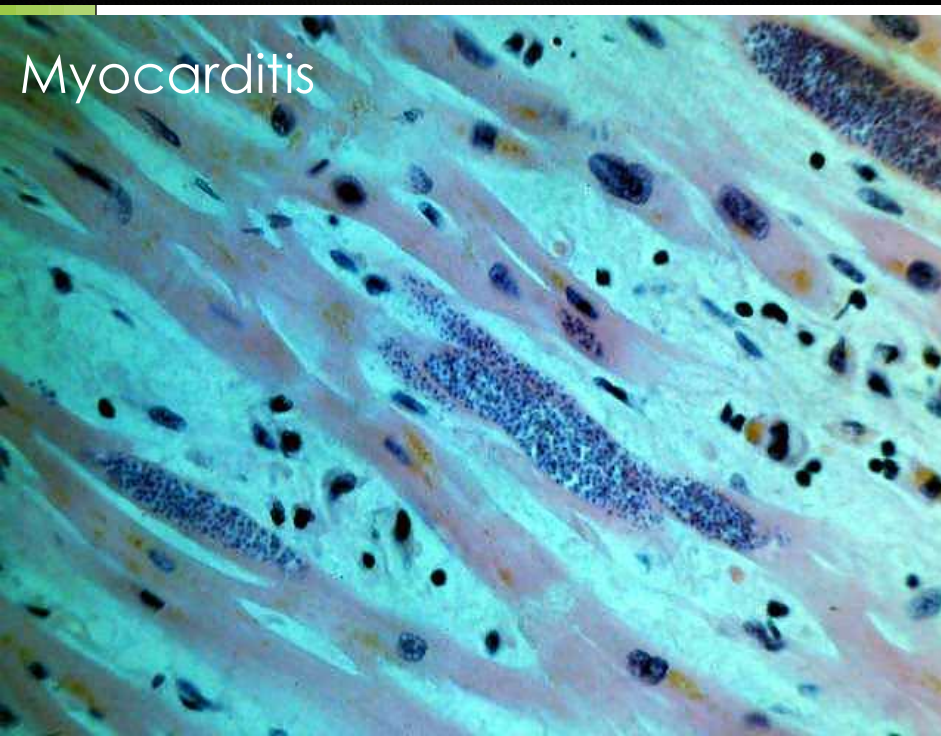


b

Encephalitis



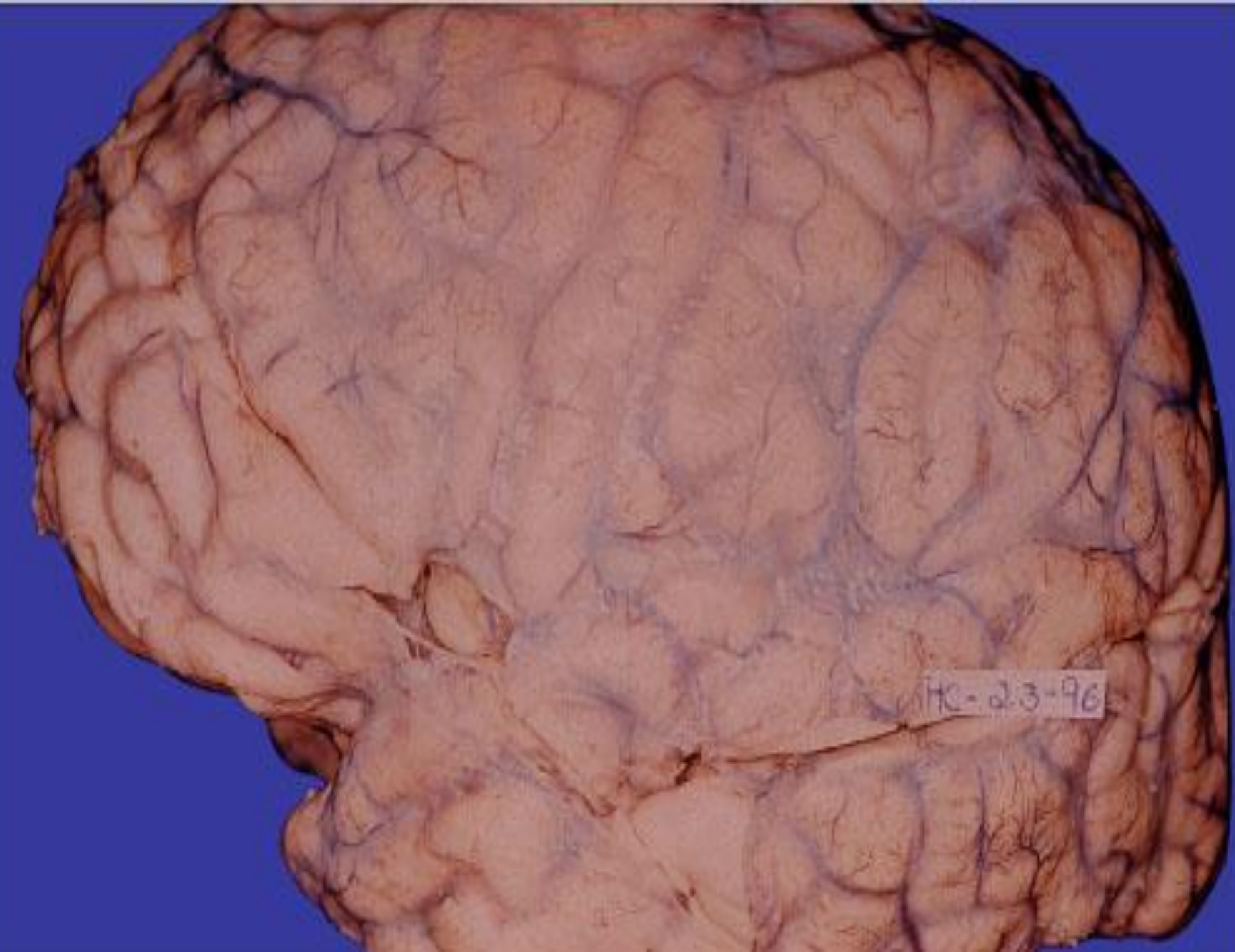
Myocarditis

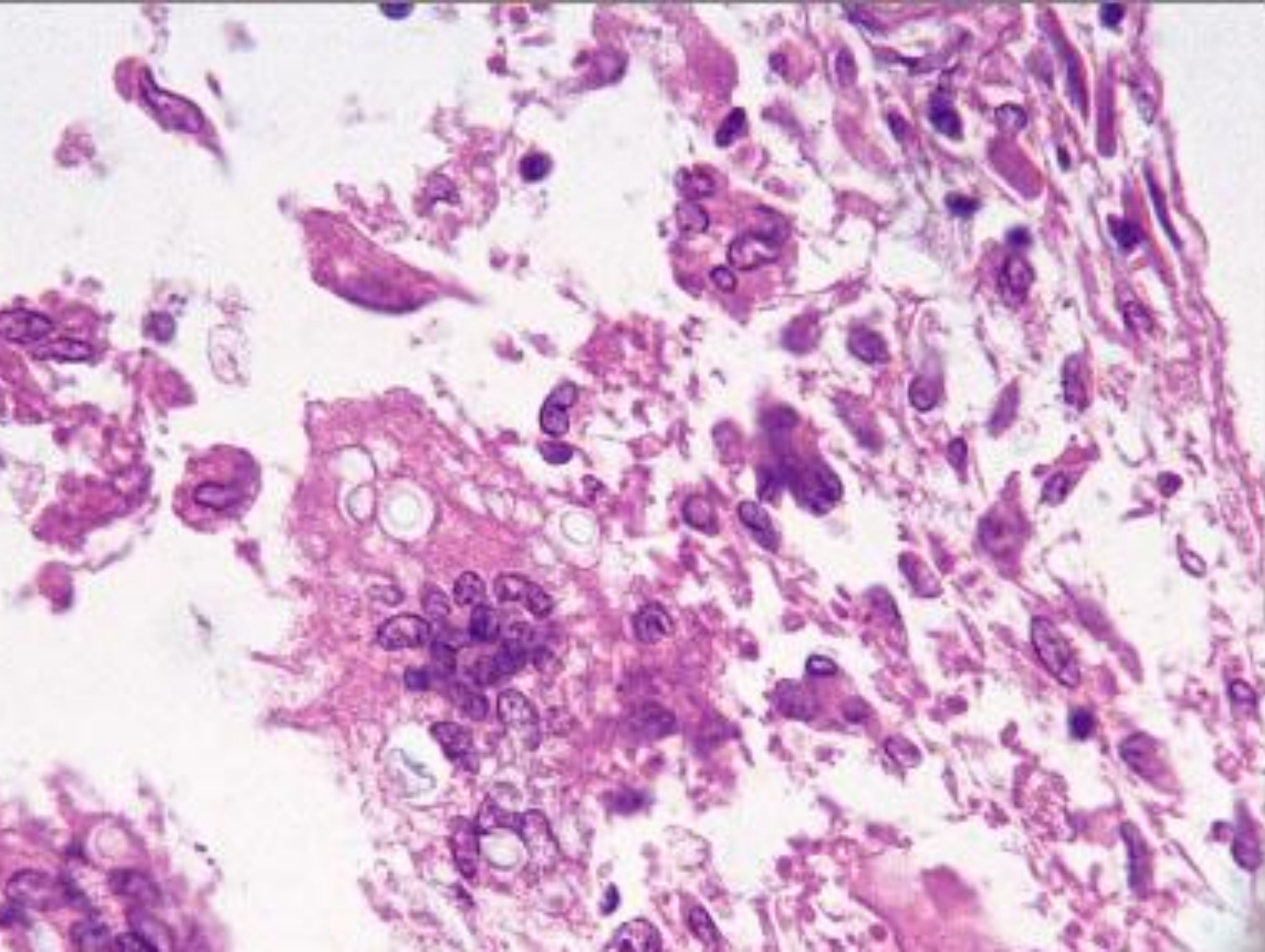


Fungal Meningoencephalitis

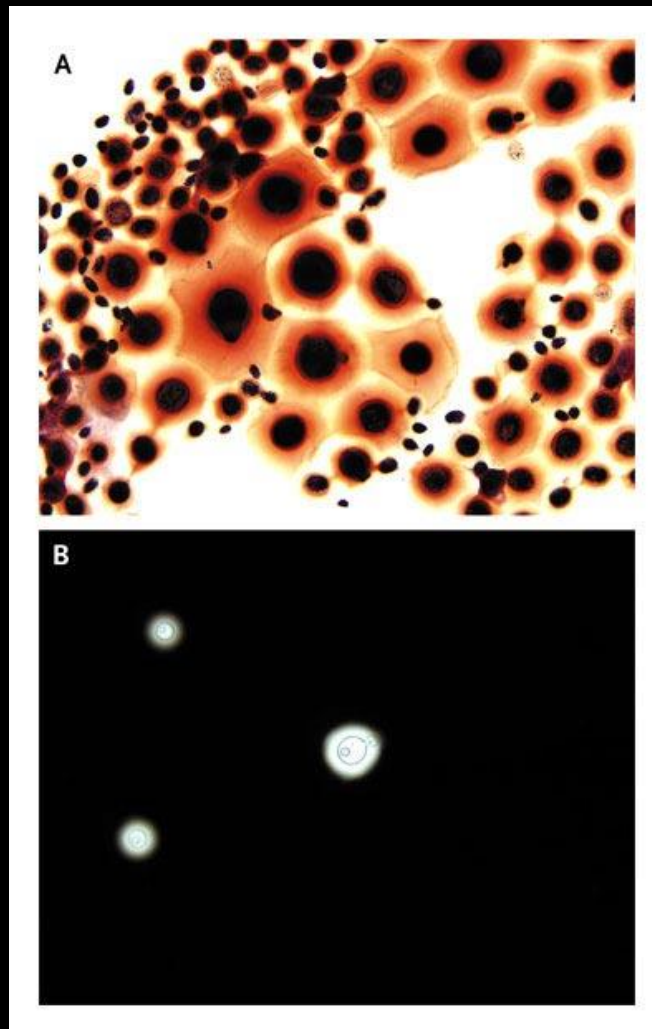
- Vasculitis-Mucormycosis, Aspergillosis
 - Thrombosis=>Infarction(Hemorrhagic, septic)
- Parenchymal Invasion-Cryptococcosis,Candida(Multiple microabscesses)

- Meningitis- Cryptococci
 - Fuminant to chronic and insidious
 - CSF- few cells but high protein
 - With yeasts of cryptococci seen in India INk





Cytocentrifuged Gram's Stain of Cerebrospinal Fluid from Another Patient



Fishman J et al. N Engl J Med 2008;358:1604-1613



The NEW ENGLAND
JOURNAL of MEDICINE

HIV and Cryptococcus in Gondar

Dr. Teshale Seboxa EMJ (2010) 48: 247

Gondar 1994-1997 (preHAART era)

- 375 HIV-infected pats

→ **31 cryptococcal meningitis = 8.3%**

= most common life-threatening invasive fungal infection in AIDS

CNS and HIV

```
graph TD; A[CNS and HIV] --> B[HIV]; A --> C[HIV-associated [opportunistic]]; A --> D[CNS Lymphoma]; B --> E[HIV meningoencephalitis (subacute, giant cells)]; C --> F[Papovavirus (J C Virus)]; C --> G[Progressive multifocal leukencephalopathy]; C --> H[Toxoplasma Gondii]; C --> I[Cryptococcus neoformans];
```

HIV

HIV meningoencephalitis
(subacute, giant cells)

HIV-associated [opportunistic]

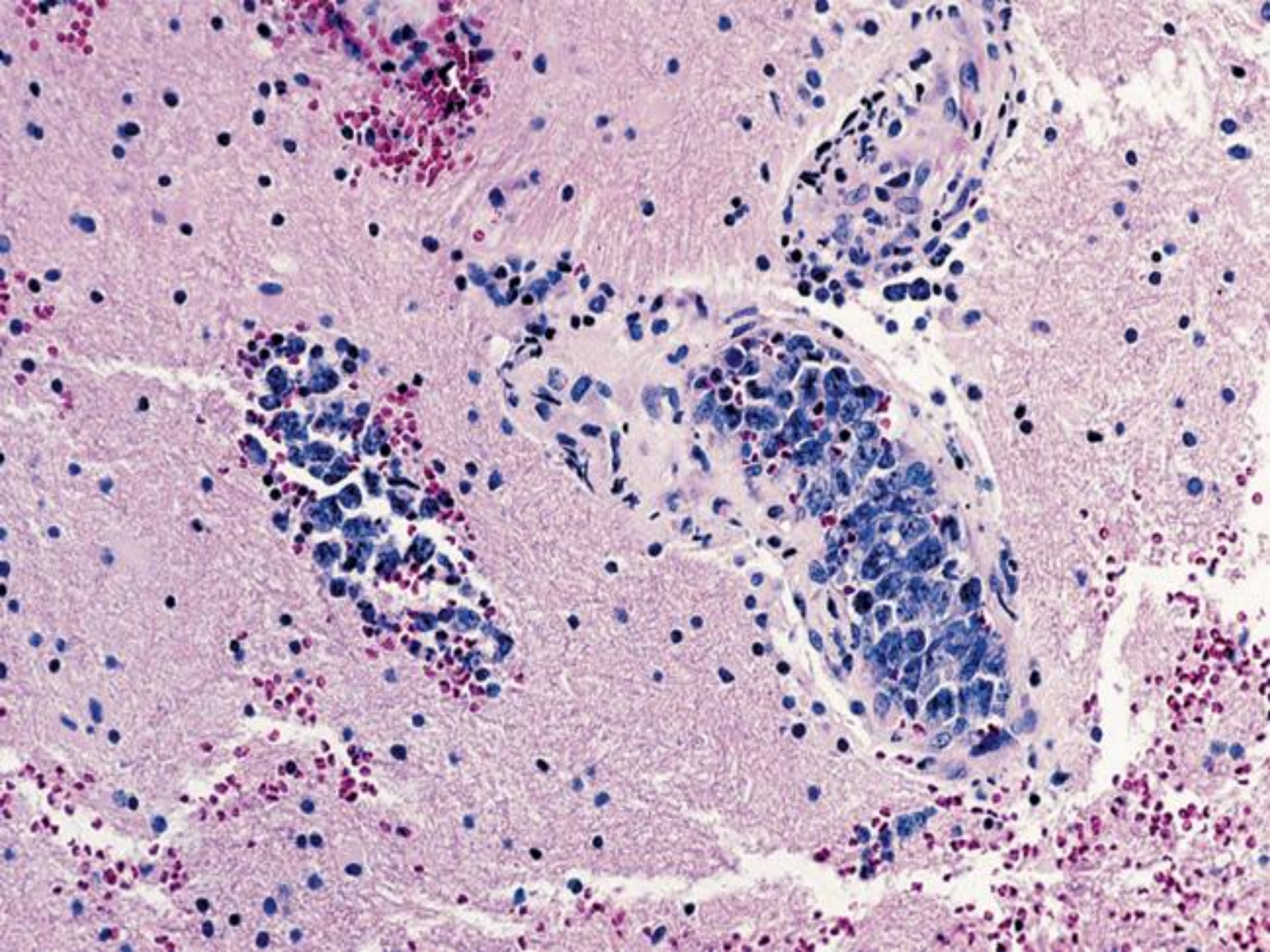
Papovavirus (J C Virus)

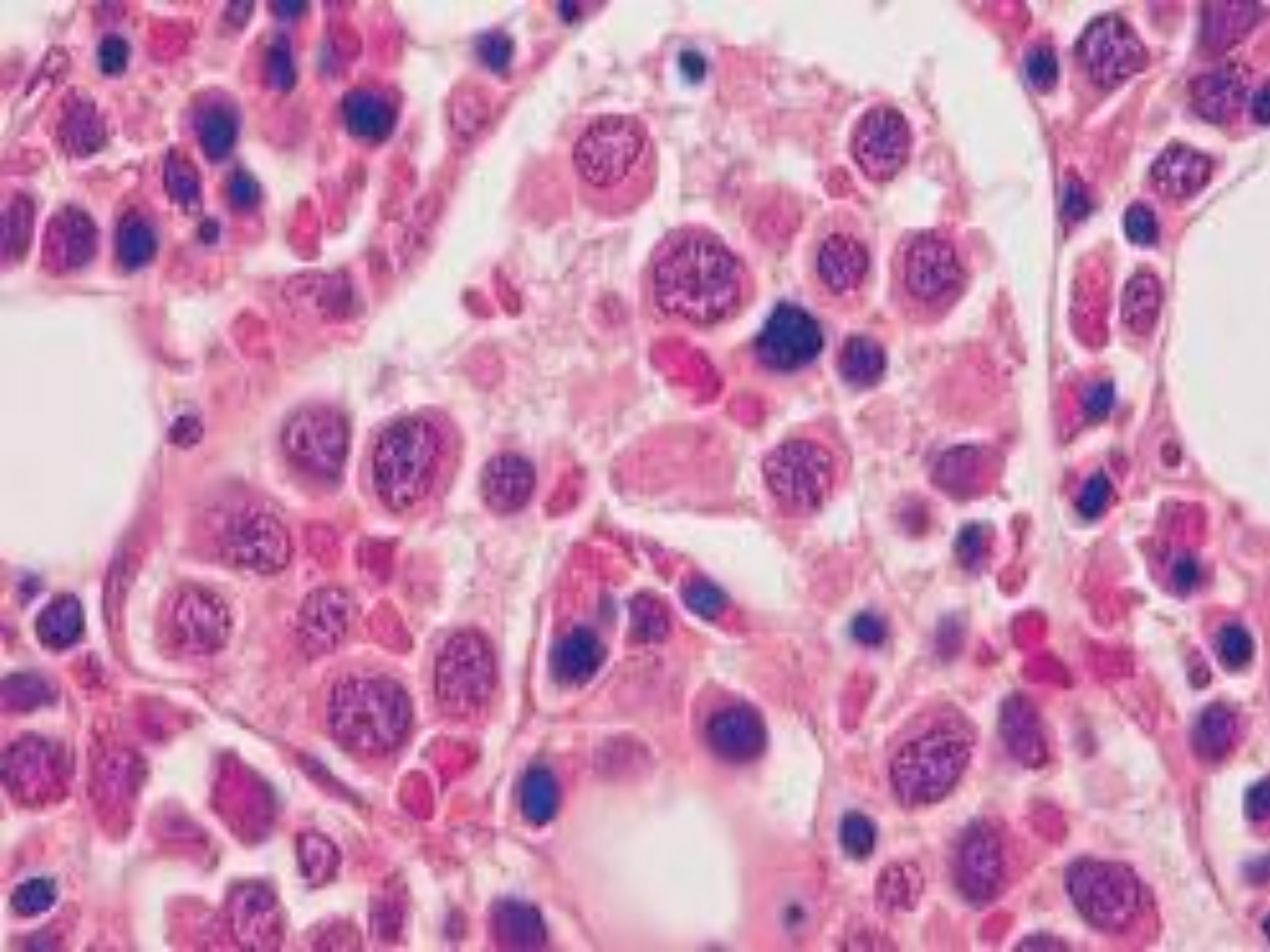
Progressive multifocal
leukencephalopathy

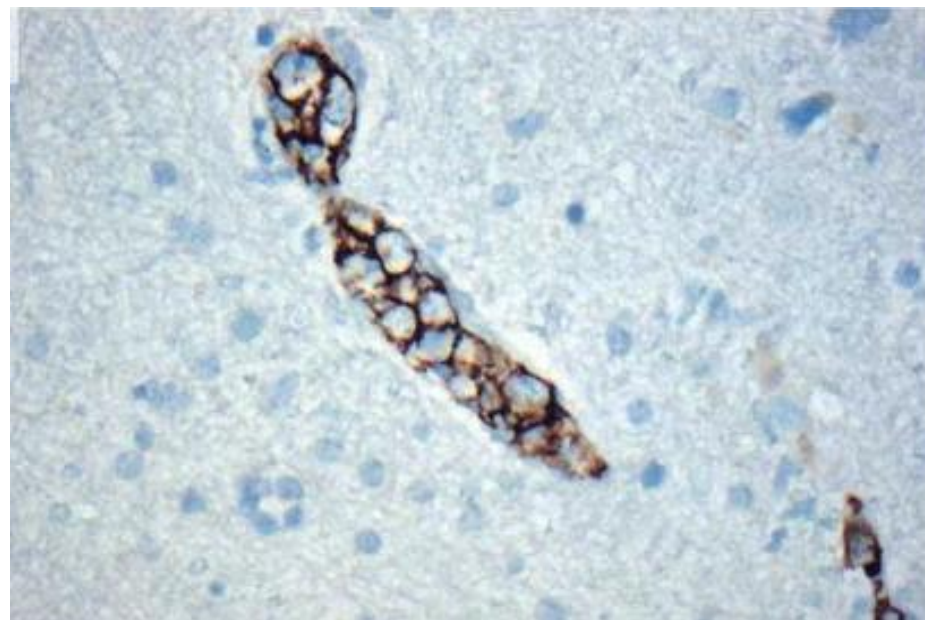
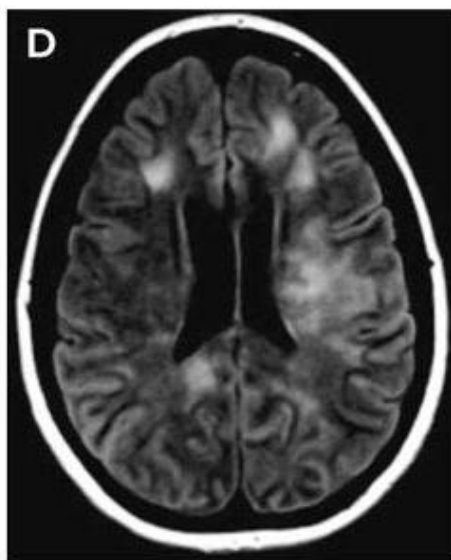
Toxoplasma Gondii

Cryptococcus neoformans

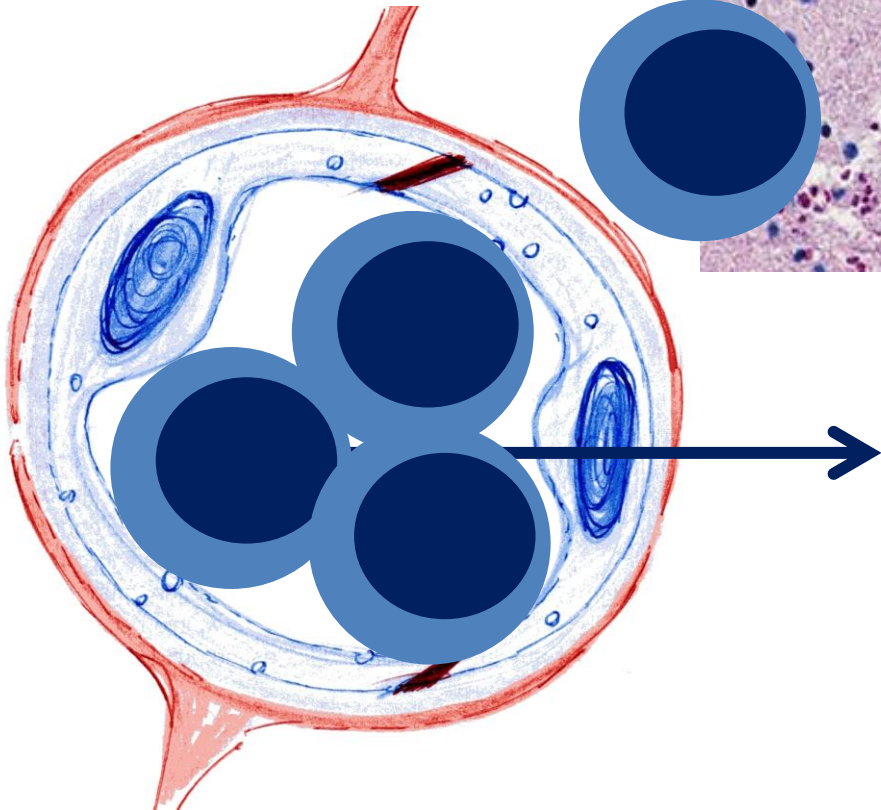
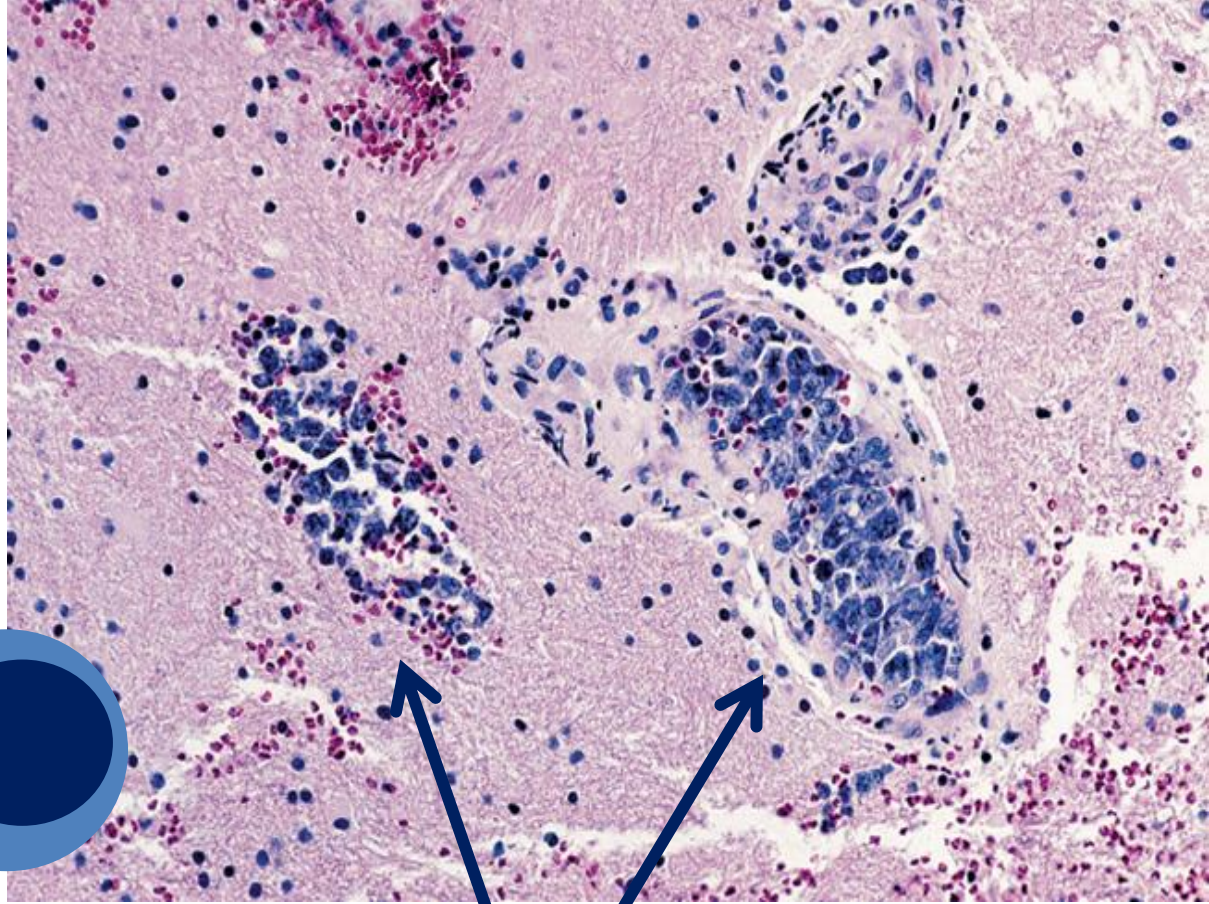
CNS Lymphoma



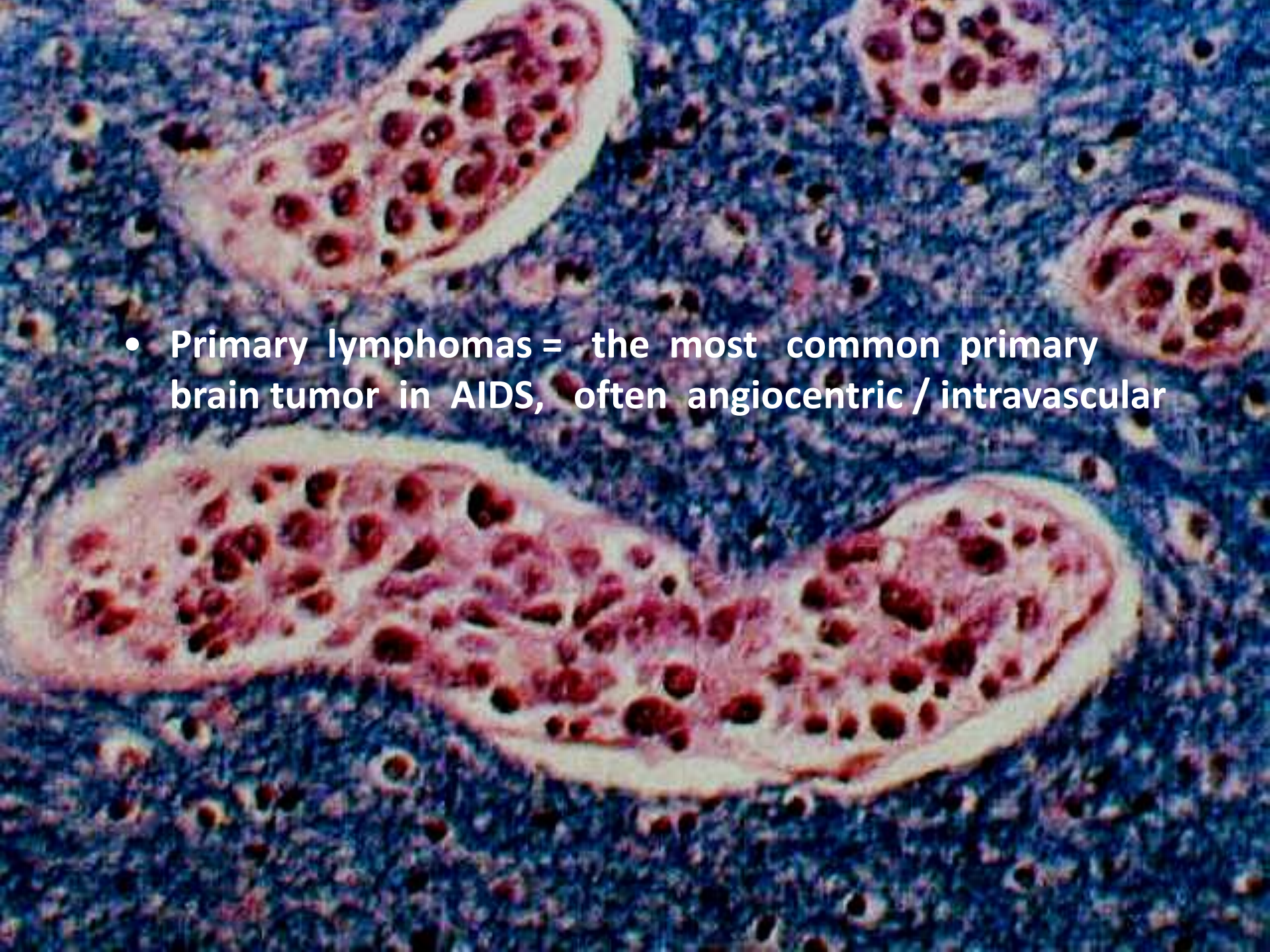




BBB



i.v. methotrexate

- 
- Primary lymphomas = the most common primary brain tumor in AIDS, often angiocentric / intravascular

CNS and HIV

```
graph TD; A[CNS and HIV] --> B[HIV]; A --> C[HIV-associated [opportunistic]]; A --> D[CNS Lymphoma]; B --> B1[HIV meningoencephalitis (subacute, giant cells)]; C --> C1[Papovavirus (J C Virus)]; C --> C2[Progressive multifocal leukoencephalopathy]; C --> C3[Toxoplasma Gondii]; C --> C4[Cryptococcus neoformans]; D --> D1[Intravascular B-cell Lymphoma];
```

HIV

HIV meningoencephalitis
(subacute, giant cells)

HIV-associated [opportunistic]

Papovavirus (J C Virus)
Progressive multifocal
leukoencephalopathy

Toxoplasma Gondii

Cryptococcus neoformans

CNS Lymphoma

Intravascular B-cell
Lymphoma

Demyelinating Disease

- Acquired conditions characterized by preferential damage to myelin with relative preservation of axons (eg multiple sclerosis)
- The clinical deficits are due to the effect myelin loss on transmission of electrical impulse along axons

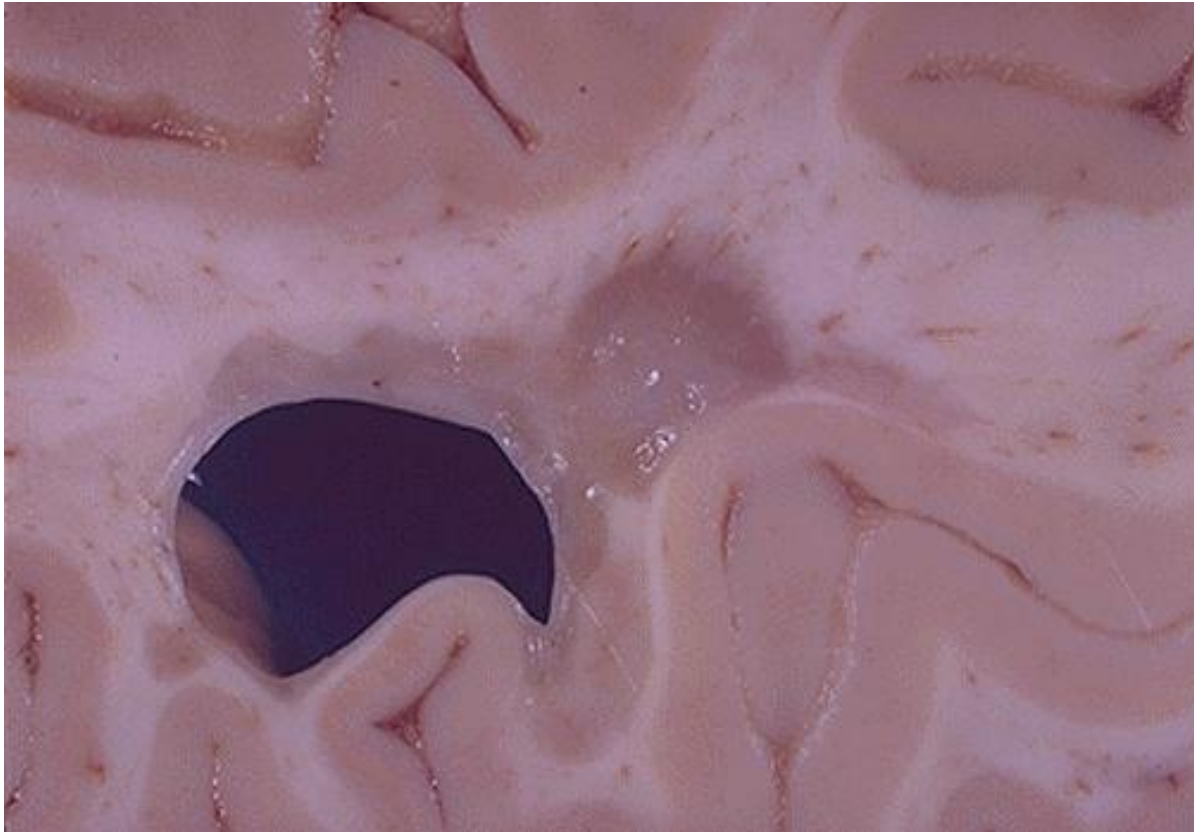
Multiple sclerosis

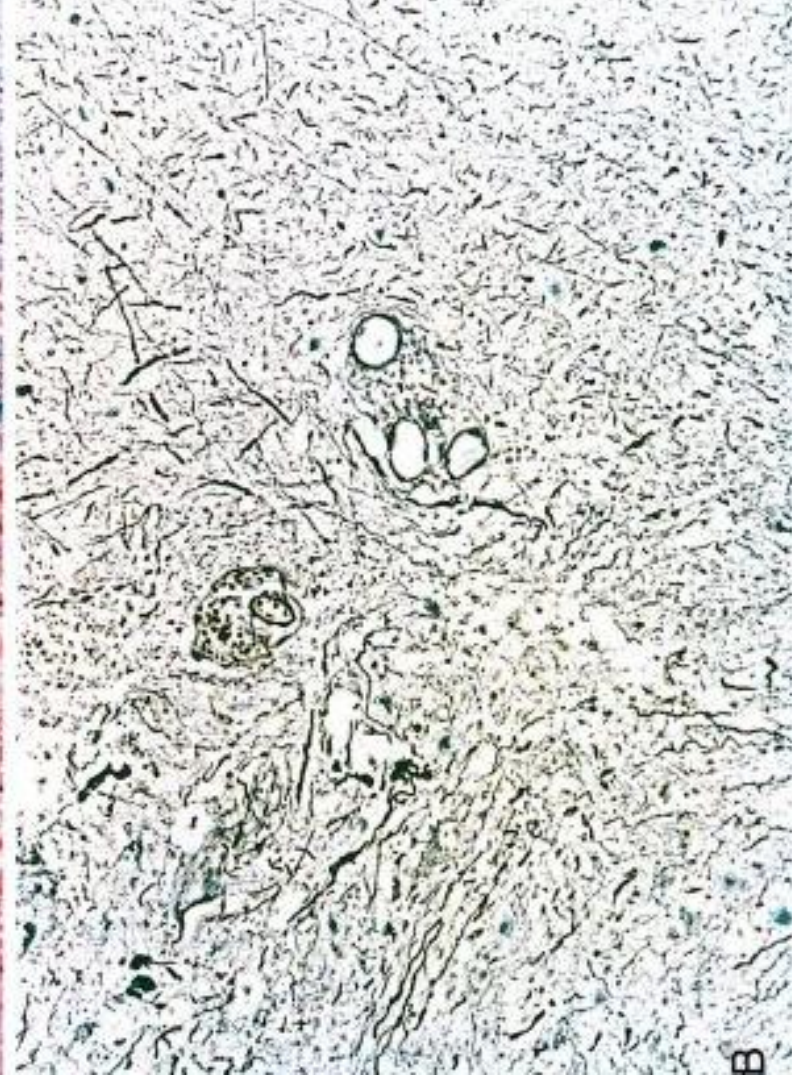
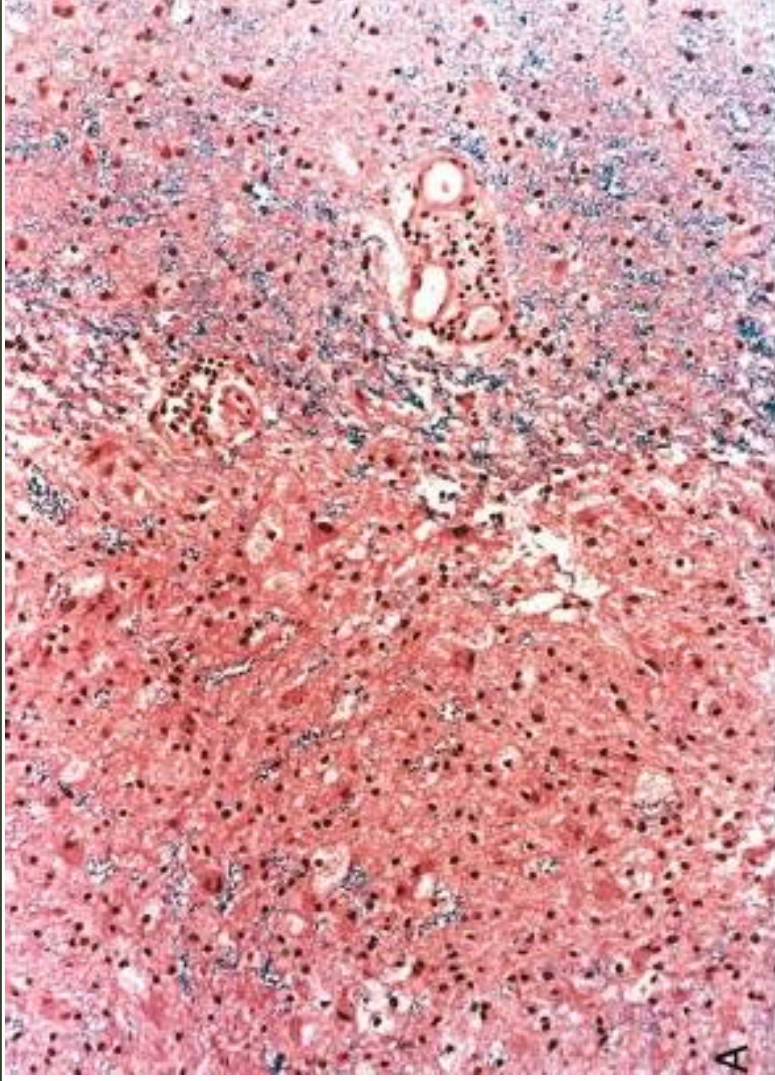
- It is the commonest demyelinating disorder affecting the CNS.
- Autoimmune demyelinating disorder characterized by distinct episodes of neurologic deficits, separated in time, attributable to white matter lesions that separated in space.

Clinical picture

- Most cases are b/n 20 & 40 years of age
- It is more common in females than in males
- The disease follows characteristics relapsing & remitting course. Initial manifestation include visual impairment(optic neuritis) limb weakness bladder dysfunction etc
- Progressive clinical deterioration occurs







Degenerative Diseases

- These are diseases of gray matter characterized by progressive selective loss of neurons associated with secondary changes in white matter.
- A common occurrence among neurodegenerative disorders is the development of protein aggregates that are resistant to normal cellular mechanism of degradation (eg A β in Alzheimer disease)
- These proteins are generally cytotoxic

Degenerative disease affecting the cerebral cortex

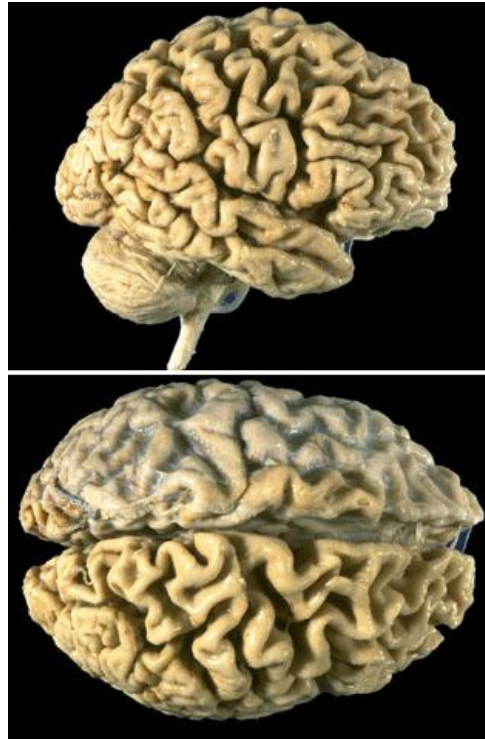
Alzheimer Disease

- The most common cause of dementia in the elderly
- Clinically, it manifests as insidious impairment of higher intellectual function with alteration in mood & behavior, later progressive disorientation , memory loss & aphasia indicate severe cortical dysfunction & eventually in 5 to 10 yrs , the patient becomes disabled, mute, & immobile.

Morphology

- Macroscopic examination of the brain shows variable degree of cortical atrophy with widening of the sulci
- The cortical atrophy leads to compensatory dilation of the cerebral ventricles known as "hydrocephalus ex vacuo".

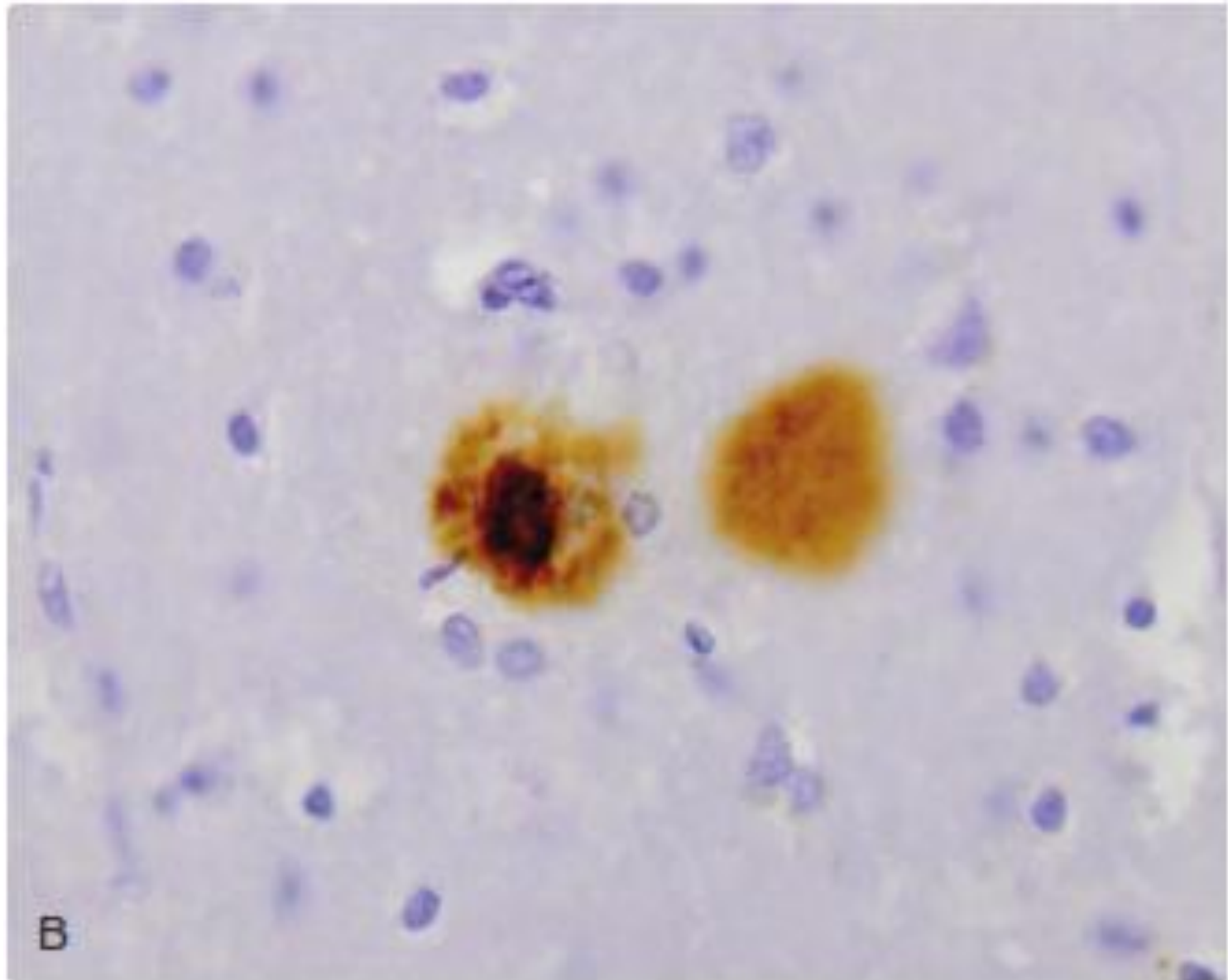


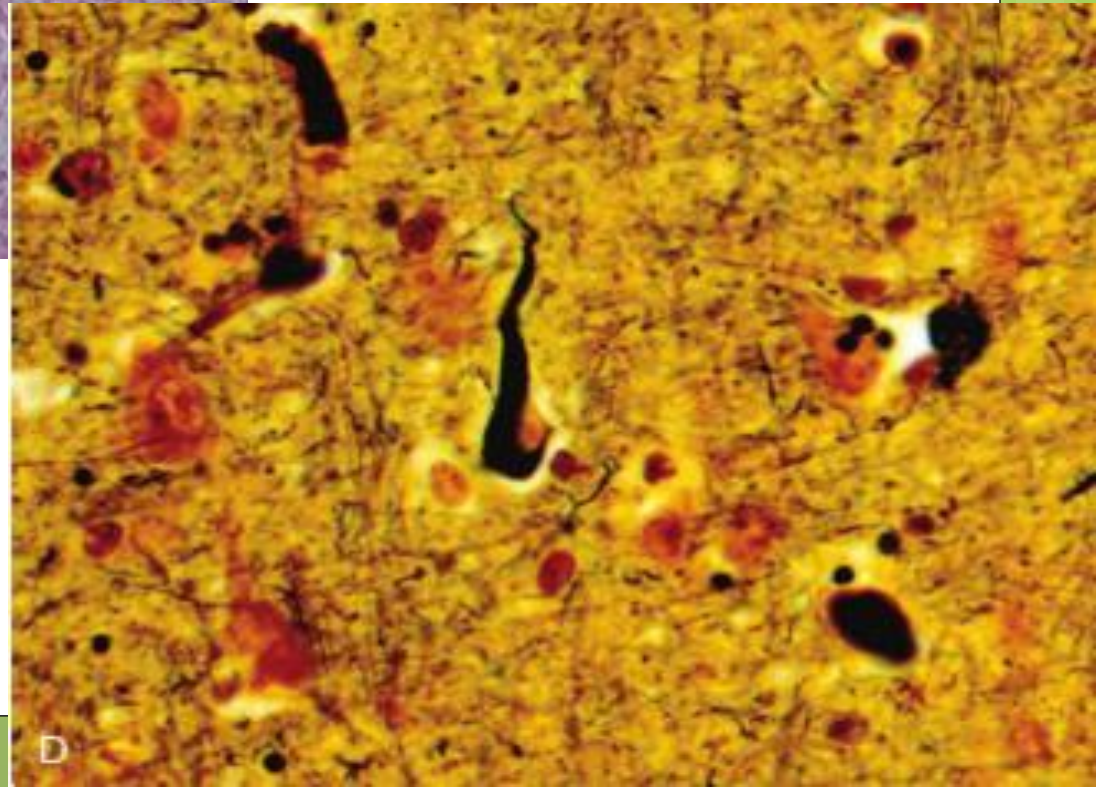
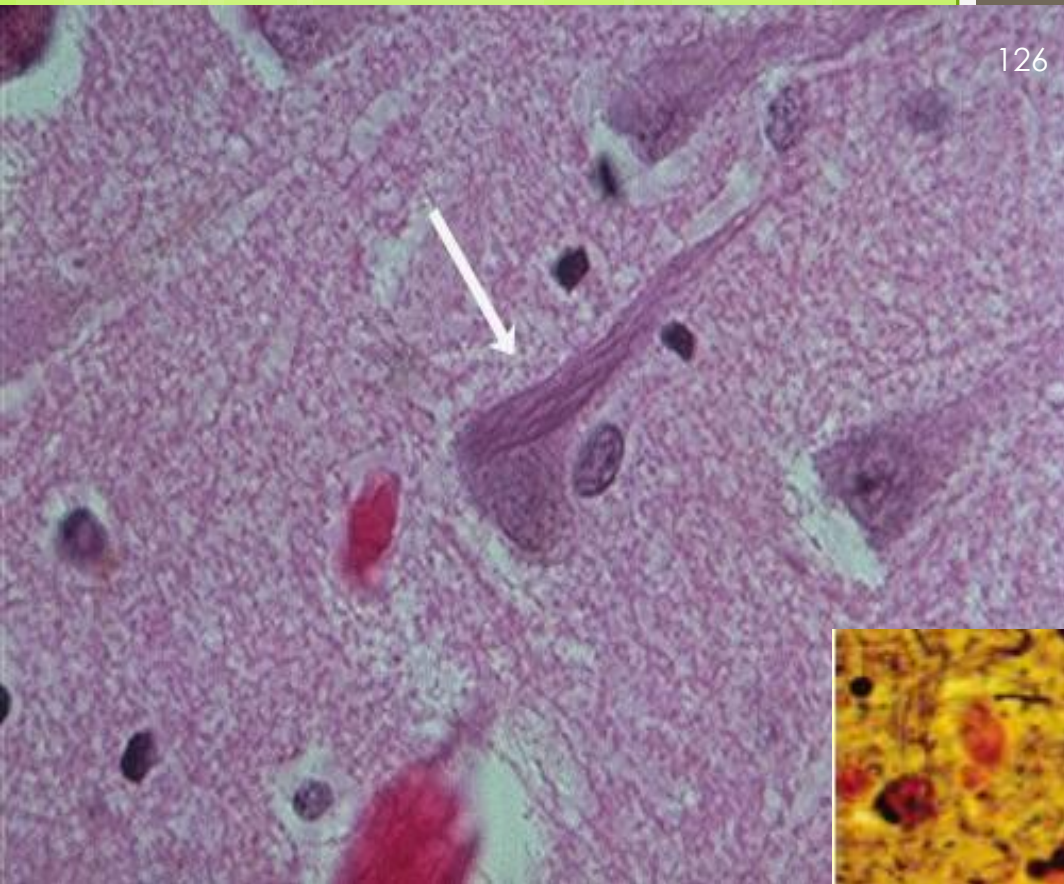


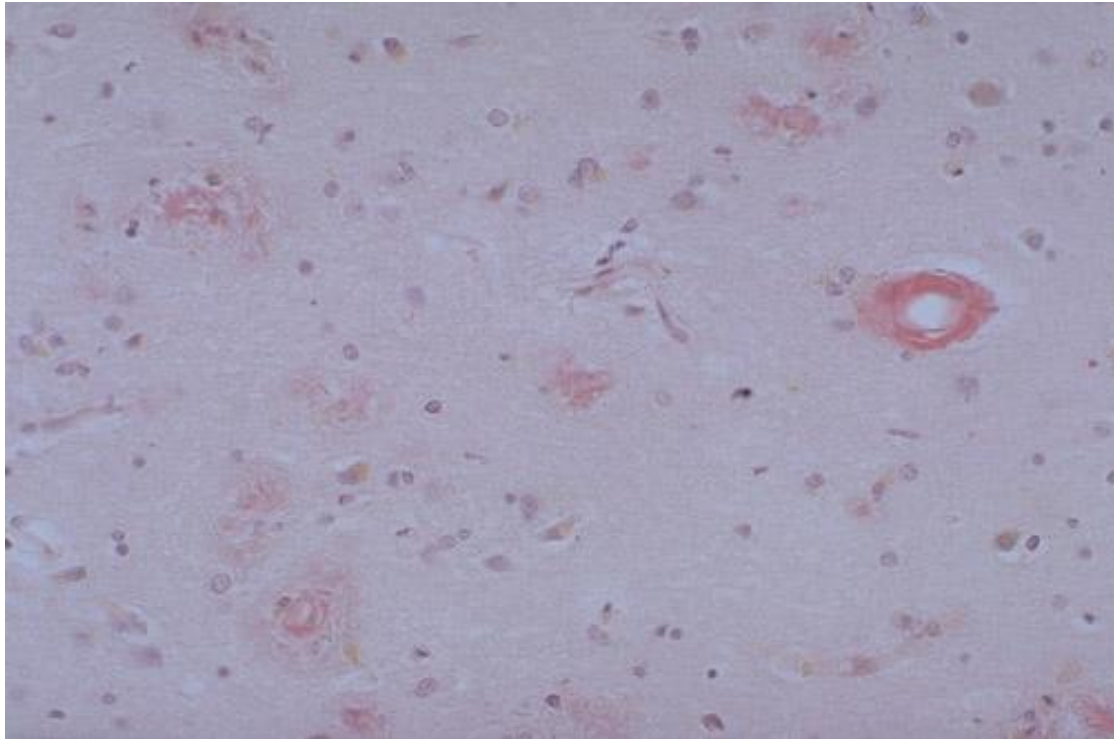


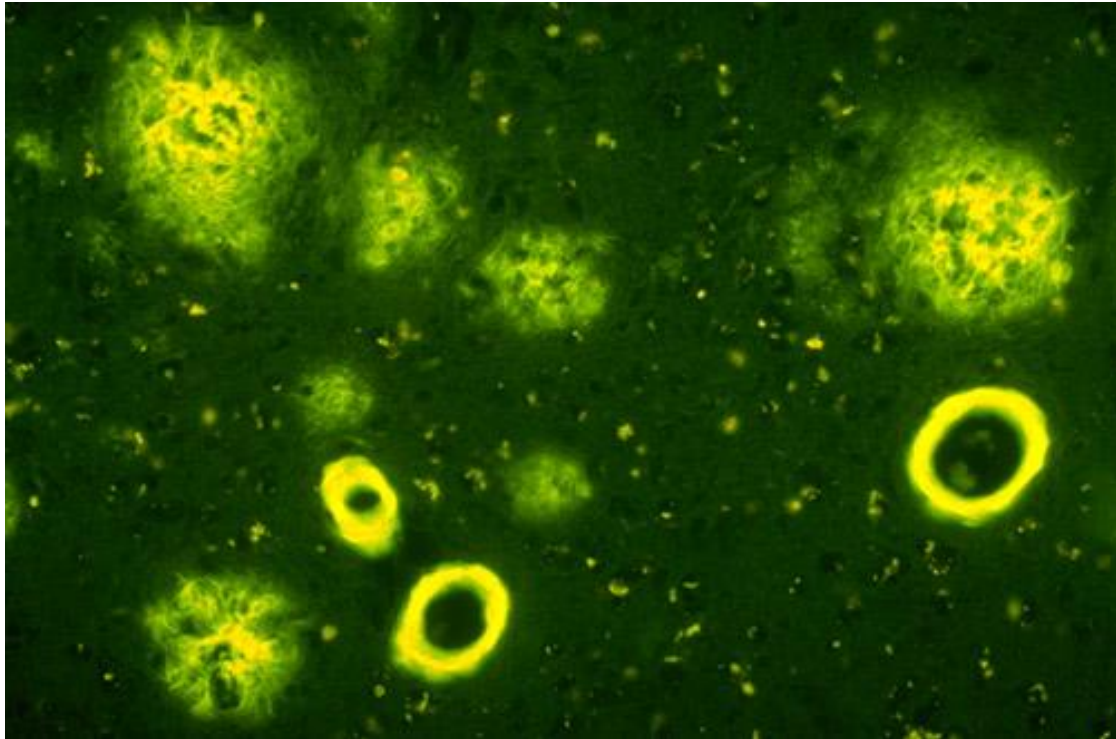
Microscopic abnormalities of Alzheimer disease include

- Neuritic (senile) plaques- collection of neuritic processes around central amyloid core composed of A β amyloid
- Neurofibrillary tangles – bundles of filaments in the cytoplasm of neurons(tau protein)
- Cerebral amyloid angiopathy – amyloid A β deposition in vascular wall









Degenerative disease of basal ganglia

- Diseases affecting these regions of the brain are associated with movement disorders , including rigidity, abnormal posturing & chorea.

Parkinsonism

- A clinical syndrome characterized by diminished facial expression, stooped posture , slowness of voluntary movement , festinating gait (progressively short & accelerated steps), rigidity & 'pill rolling' tremor .
- It occurs in conditions associated with damage to the nigrostriatal dopaminergic system.



Parkinson's disease

- Patients with progressive parkinsonism in the absence of a toxic or other known etiologies

Pathogenesis

- Degeneration of dopaminergic neurons of the substantia nigra

Pathogenesis ctd

- Genetic changes leading to accumulation of α -synuclein protein, mitochondrial dysfunction and enzymatic problems.
- Lead to accumulation of proteins(eg.- α -synuclein) which is toxic to the neuron.
- Lewy-Bodies-cytoplasmic eosinophilic inclusions composed of α -synuclein

Morphology

- The typical macroscopic finding is pallor of the substantia nigra.
- On microscopic examination, loss of melanin-containing neurons in the substantia nigra
- the remaining neurons in these areas, as well as in other subcortical regions, contain concentrically laminated, eosinophilic, intracytoplasmic inclusions known as **Lewy bodies**.

